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TECHNICAL SUPPLEMENT TO NEURAL BELLMAN OPERATORS

QIAN QI
Peking University

This supplement supplies the complete proofs for Neural Bellman Operators: feasible own-future construction; finite-horizon existence, uniqueness and stability; action and centered recursive-risk accounts; continuous-risk qualification; constrained original marginal regularity; actor/critic and smooth computational graphs; and the conditional Brownian recursive-utility bridge with its stopping boundaries. The current additions prove direct action and value stopping, rational multi-cut certification, continuous-action regret resolution, conditional economic stopping, and independently checked finite-design fit bounds. The retained results prove exact finite interpolation above all paid quadratic Bellman supports, rational residual certification, an exact action-information witness, and complete precision-work and terminal decision bounds. The prior memory and joint interpolation results are included with their full proofs and original scope. Fixed-information minimax statements retain their specified interfaces. Results for distinct objectives and historical experiments are not transferred by notation. The frozen original-economy service implements only the algorithms explicitly identified in its source protocol. Analytic extensions completed after that freeze do not modify its scientific code or measured outcomes. Statements introduced here are numbered S.*.

1. CURRENT PROOF ROADMAP

This supplement contains the complete mathematical chains needed for the current article: original operator construction, signed and centered risk qualification, original investment regularity, marginal learning and the all-state policy service, paid-support policy improvement, its precise information bound, same-incumbent gain propagation, and the declared simultaneous inference. Proofs specific to historical experiments and additional retained models appear in the development

Qian QI: qiqian@pku.edu.cn

companion. All original source files remain unchanged; exact byte views route shared statements and proofs without rewriting them.

2. SIGNED TRANSITION QUALIFICATION FOR RECURSIVE PREFERENCES

The transport account can use more information than an absolute error radius. A numerical transition may have a known direction of bias, and the implemented action may have a more precise upper error bound than the whole feasible menu. We retain these distinctions. The resulting qualification applies to the original recursive evaluation at every date, not only to expected cost with a fixed continuation. It does not change an expectation objective into a risk objective or identify a static tail-risk criterion with its nested counterpart.

Retain the augmented economic state, original feasible actions and measurable complete policies of Section 2. Suppose bounded original values satisfy backward recursion with

$$Q_t(v;x,a) = c_t(x,a) + \beta_t \rho_{t,x,a}(v), \qquad T_t v(x) = \inf_{a \in A_t(x)} Q_t(v;x,a),$$

where each original conditional evaluation is monotone and cash invariant: $\rho(v + k) = \rho(v) + k$ for constants k. Discounts $\beta_t \geq 0$ are deterministic, $B_0 = 1$, and $B_t = \prod_{s<t} \beta_s$. The state includes the preference variables required by the original model. The relevant bounded functions are in the evaluation's domain, and the original Bellman optimum and supplied feasible policies are well defined. For each common parameter θ, store bounded $f_{t,\theta}$, a complete feasible π_θ, and numerical action values $\widehat{Q}_{t,\theta}$ on the original feasible actions.

Put $\widehat{T}_{t,\theta}(x) = \inf_a \widehat{Q}_{t,\theta}(x,a)$ and define the signed discrepancy

$$D_{t,\theta}(x,a) = Q_t(f_{t+1,\theta};x,a) - \widehat{Q}_{t,\theta}(x,a).$$

Assume uniform, verified bounds

$$\begin{aligned} \ell_{t,\theta} \leq \widehat{T}_{t,\theta}(x) - f_{t,\theta}(x) \leq u_{t,\theta}, \\ 0 \leq \widehat{Q}_{t,\theta}(x, \pi_{t,\theta}(x)) - \widehat{T}_{t,\theta}(x) \leq \gamma_{t,\theta}, \\ b^-_{t,\theta} \leq D_{t,\theta}(x,a) \quad (a \in A_t(x)), \end{aligned}$$

$$D_{t,\theta}(x, \pi_{t,\theta}(x)) \leq b_{t,\theta}^{\pi,+}. \tag{1}$$

Here $b_{t,\theta}^{-}$ and $b_{t,\theta}^{\pi,+}$ may be signed. A bound confined to sampled actions cannot replace the all-action lower bound. Original-owner repairs, unsampled feasible actions, integration error and arithmetic error must be included in the relevant inequalities. Let $d_\theta^- \leq g - f_{T,\theta} \leq d_\theta^+$ throughout the terminal domain.

THEOREM S.1—Signed, recursive original-optimum qualification: *Under the stated conditions, every supplied complete policy satisfies, at every initial state,*

$$\begin{aligned} 0 \leq V_0^{\pi_\theta} - V_0^* \leq \mathcal{S}_\theta, \\ \mathcal{S}_\theta = \sum_{t<T} B_t \{u_{t,\theta} - \ell_{t,\theta} + \gamma_{t,\theta} + b_{t,\theta}^{\pi,+} - b_{t,\theta}^{-}\} \\ + B_T(d_\theta^+ - d_\theta^-). \end{aligned} \tag{2}$$

Consequently a class for which these inequalities hold uniformly and $\sup_\theta \mathcal{S}_\theta \leq \varepsilon$ *consists entirely of complete policies with original-recursion loss at most* ε*. Actions, attained continuations and endogenous occupations may change at every date. The assertion is pointwise in the augmented state and does not require independent deployment contexts or a continuous actor.*

The bound is invariant to arbitrary date-specific, state-independent shifts of the stored continuations when the numerical action values are shifted consistently. If the original-minus-numerical action value is a constant $k_{t,\theta}$ *independent of state and action, its contribution to* (2) *is zero. A bias common to all actions at a given state, but varying across states, does not in general cancel. The coefficient one on* $b^{\pi,+} - b^{-}$ *cannot be reduced uniformly, even for one decision date and deterministic transitions.*

The proof compares both the complete policy and the original optimum with the same stored continuation by recursive order. It does not propagate a linear occupation identity through nonlinear preferences. Setting $b^- = -b$ and $b^{\pi,+} = b$ recovers Theorem A.60; a verified signed enclosure can strictly improve it without a new fit or economic observation. With stationary bounded data and $0 \leq \beta < 1$, the

corresponding stationary bound is

$$V^{\pi} - V^{*} \leq \frac{u - \ell + \gamma + b^{\pi,+} - b^{-}}{1 - \beta}.$$

The proof and the necessary bias counterexamples are in Supplement 5.

2.1. *Explicit qualification of two nonlinear evaluations*

Wasserstein distance controls expectations of Lipschitz functions. For nonlinear evaluations its price must be derived, not borrowed from the expectation model. For a bounded K-Lipschitz continuation f and probability laws $P, \widehat{P}$, define the upper-tail average value at risk and entropic evaluation by

$$\begin{aligned} \rho_P^{\alpha}(f) &= \inf_{z \in \mathbb{R}} \left\{ z + \frac{P(f - z)_+}{1 - \alpha} \right\}, && 0 \leq \alpha < 1, \\ \rho_P^{\lambda}(f) &= \lambda^{-1} \log P \exp(\lambda f), && \lambda > 0. \end{aligned} \tag{3}$$

The limit at $\lambda = 0$ is expectation. The first definition includes atoms; it is not the conditional mean obtained by discarding all mass at a quantile. The variational representation and its treatment of general loss distributions are established results (Rockafellar and Uryasev, 2002). We use these evaluations only when they are the original one-step primitives of the recursive economic objective.

PROPOSITION S.2—Risk-specific transport allowances: *If* $W_1(P, \widehat{P}) \leq e$, *then*

$$\begin{aligned} |\rho_P^{\alpha}(f) - \rho_{\widehat{P}}^{\alpha}(f)| &\leq \frac{Ke}{1 - \alpha}, \\ |\rho_P^{\lambda}(f) - \rho_{\widehat{P}}^{\lambda}(f)| &\leq K \exp\{\lambda \operatorname{osc}(f)\}\, e. \end{aligned} \tag{4}$$

The entropic expression is invariant to constants added to f. *The factor* $(1 - \alpha)^{-1}$ *is necessary uniformly over bounded Lipschitz continuations and probability laws. If a coupling satisfies* $f(Y) \geq f(\widehat{Y})$ *almost surely, the original-minus-numerical risk difference is nonnegative for either evaluation.*

If the additional numerical action-value error is at most ξ_t in absolute value, a symmetric choice in (1) is

$$b_t^- = -\beta_t \kappa_t e_t - \xi_t, \qquad b_t^{\pi,+} = \beta_t \kappa_t e_t + \xi_t,$$

where $\kappa_t = K_{t+1}/(1-\bar{\alpha}_t)$ for a verified risk-level cap $\alpha \le \bar{\alpha}_t < 1$, or $\kappa_t = K_{t+1}\exp(\bar{\lambda}_t O_{t+1})$ for verified bounds on the entropic parameter and continuation oscillation. A monotone coupling instead permits $b_t^- = -\xi_t$ and the same upper endpoint. Approximate evaluation of the exponential requires an outward upper bound; machine output is not an exact modulus.

These local risk coefficients enter the dated budget additively:

$$\sum_{t<T} B_t \eta_t + 2\sum_{t<T} B_{t+1}\kappa_t e_t + 2\sum_{t<T} B_t \xi_t + B_T(d^+ - d^-) \le \varepsilon, \tag{5}$$

where η_t bounds residual span plus greediness. Monotonicity and cash invariance propagate constant error by β_t, not by a product of all later tail-risk multipliers. The continuation moduli themselves may grow with horizon or preference parameters and must still be established and paid for. Subject to this qualification, the coupled resource program of Section CX allocates residual and risk-adjusted transition precision without deleting descendant or representation costs.

Economic and strategic interpretation. A larger tail-risk level can require finer transition information for the same original-policy loss allowance. That requirement is part of the original preference primitive, not a favorable reweighting of measured work. When preference levels evolve endogenously, the bounds must cover the augmented state and actual feasible actions. For games and temporal selves, the theorem applies to each fixed-opponent or fixed-future best-response recursion under that decision maker's original objective. Applying nested average value at risk to an originally static terminal-risk problem would change that problem; no such substitution is made. General non-cash-invariant recursive aggregators retain Proposition A.33 and its separate continuation modulus. At a fixed state write

$$f_a(z) = V_1(F(x,a,z)), \qquad Q_x(a) = S(x) + h(a) + \beta\rho_\alpha(f_a).$$

The actual complete-policy cost is $J_x(a)$, with the same implemented last-date policy for every current action. Its retained uniform guarantee is

$$J_x(a) = Q_x(a) + \eta_x(a), \qquad 0 \le \eta_x(a) \le \delta, \qquad \delta = \beta\varepsilon_1, \quad \varepsilon_1 = 2^{-20}. \tag{6}$$

The function η_x is not assumed convex or differentiable.

2.2. *A comparison modulus from the original technology*

The following constant concerns the original optimal continuation, not a fitted critic or an empirical gradient bound. This addition is restricted to the original two-dimensional application.

THEOREM S.3—Original action comparison modulus: *For* $d = 2$, $\beta = 15/16$, $R = 1/32$, *and* $\alpha \in \{1/2, 3/4\}$, *put* $t = 1 - \alpha$ *and*

$$M_\alpha = \frac{1413}{128} + \beta\left\{\frac{127}{16} + \frac{19625\alpha}{20480t}\right\}. \tag{7}$$

The original V_1 *has an* M_α*-Lipschitz gradient on the state cube, with the continuous boundary extension. Consequently, for every original state and every two feasible actions,*

$$\mathrm{Lip}_z(f_b - f_a) \le E_{ab} := \frac{4}{5}M_\alpha|b - a|. \tag{8}$$

Both conclusions include switches of the constrained terminal optimizer and of its active risk region.

The proof normalizes terminal investment as $A(y)v$, $v \in [0,1]$, and bounds the state semiconcavity of that fixed-domain objective. It therefore does not differentiate a rounded actor or ignore the capacity multiplier. The earlier expected-cost smoothness result in Section FD is retained with its original criterion; the extra term in (7) pays for the movement of the original AVaR tail.

3. LEARNING THE MARGINAL CONTINUATION OF THE ORIGINAL BELLMAN OPERATOR

The preceding comparison methods certify an acquired actor. We now change what the critic learns. The target remains the original two-date investment problem

in Section 5, with its continuous innovation law, capacity, discount and nested AVaR preference. The construction first evaluates the original risk operator and then learns its action marginal. In particular, it does not replace the optimizing risk density by the density selected by a fitted terminal continuation.

Write $a = A(x)v$, $v \in [0,1]$, and define

$$\begin{aligned} R_\alpha(x,v) &= \beta\rho_\alpha\{V_1(F(x,A(x)v,Z))\}, \\ Q_x(A(x)v) &= S(x) + h(A(x)v) + R_\alpha(x,v), \\ H_\alpha(x,v) &= A(x)^{-1}\partial_v R_\alpha(x,v). \end{aligned} \tag{9}$$

Thus H_α is a marginal continuation cost per unit of the original action. The known stage cost is retained analytically. The original optimal last-date value $V_1 = T_1 g$ is the target of the paid acquisition calculation. Deployment still uses the same measurable last-date controller as the retained study. Its uniform guarantee gives

$$J_x(a) = Q_x(a) + \eta_x(a), \qquad 0 \le \eta_x(a) \le \delta := \beta 2^{-20}. \tag{10}$$

No differentiability or convexity of η_x is assumed.

Derivative-based economic dynamic programming, derivative supervision and decision-aware learning have established precedents (Cai and Judd, 2015, Czarnecki et al., 2017, Donti et al., 2017, Elmachtoub and Grigas, 2022). The action-null quotient principle is also retained from R52, and the constrained directional regret bound from Theorem A.7. R67 already supplies a certified fixed-feature readout and a dual gap. The additional construction here supplies regularity constants for the original recursive AVaR target, outward derivative labels made from paid finite-risk values, a finite-class training certificate, and a feature-aware extension from those labels to off-design decisions. These ingredients apply to conventional and neural representations with the same information and feature restrictions.

3.1. *Regularity of the original integrated risk marginal*

Throughout this section $d = 2$, $R = 1/32$, $\beta = 15/16$, $\alpha \in \{1/2, 3/4\}$, and $t = 1 - \alpha$. Retain the original state-gradient constant from Theorem S.3:

$$M_\alpha = \frac{1413}{128} + \beta\left\{\frac{127}{16} + \frac{19625\alpha}{20480t}\right\},$$
$$m_z = \frac{181}{64}, \qquad C_\alpha = \frac{4}{5}M_\alpha,$$
$$D_\alpha = \frac{5M_\alpha}{16} + \frac{C_\alpha^2\alpha}{m_z t}, \qquad L_\alpha = 5 + \beta D_\alpha. \tag{11}$$

All these constants are derived from the original primitives. They are not fitted from the acquisition or evaluation sample.

THEOREM S.4—An original-risk marginal with explicit boundary control: *The function H_α in (9) is continuous on $\Omega = [0,1]^2 \times [0,1]$, with the inward derivative convention at the action endpoints. For each state, Q_x is continuously differentiable and 2-strongly convex on $[0, A(x)]$, and*

$$|Q_x'(a) - Q_x'(b)| \le L_\alpha |a - b|.$$

Consequently its normalized action curvature lies between $2A(x)^2$ and $L_\alpha A(x)^2$. For $u = (x, v)$ and $u' = (x', v')$,

$$|H_\alpha(u) - H_\alpha(u')| \le \Lambda_\alpha \|u - u'\|_2, \tag{12}$$
$$\Lambda_\alpha = \beta\left\{\frac{441M_\alpha}{1024} + \frac{147M_\alpha^2\alpha}{160m_z t}\right\}.$$

These statements include a minimizing central risk interval at either endpoint of its feasible set and a binding economic capacity.

The proof uses the original M_α-Lipschitz gradient of V_1 and the m_z-strong convexity of $z \mapsto V_1(F(x, a, z))$. A central interval of length $2R\alpha$ has a unique minimizing location. Its constrained first-order inequalities, rather than an assumed interior root, control changes in the adverse tail. That step is essential: controlling a fitted

terminal continuation in one action direction alone need not control which risks receive AVaR weight.

3.2. *Paid derivative labels and finite-class training*

At a declared acquisition site (x, v), set $a = A(x)v$. Suppose an original-Q_x oracle returns outward value intervals at a and the available inward probes $a \pm d$, where $d > 0$ is fixed before their outcomes and each queried action is feasible. Write $\mathcal{I}_0, \mathcal{I}_+, \mathcal{I}_-$ for these intervals, and use outward interval subtraction and division to form

$$\mathcal{S}_+ = (\mathcal{I}_+ - \mathcal{I}_0)/d, \qquad \mathcal{S}_- = (\mathcal{I}_0 - \mathcal{I}_-)/d, \qquad \mathcal{S}_c = (\mathcal{I}_+ - \mathcal{I}_-)/(2d).$$

PROPOSITION S.5—Finite-risk secants enclose the original marginal: *Every available interval in the following list contains $Q'_x(a)$:*

$$\begin{aligned} \mathcal{G}_+ &= \mathcal{S}_+ + [-L_\alpha d/2, -d], \\ \mathcal{G}_- &= \mathcal{S}_- + [d, L_\alpha d/2], \\ \mathcal{G}_c &= \mathcal{S}_c + [-(L_\alpha - 2)d/4, (L_\alpha - 2)d/4]. \end{aligned} \tag{13}$$

The central interval requires both probes. The forward interval alone is valid at $a = 0$, and the backward interval alone at $a = A(x)$. Their outward intersection, followed by subtraction of the known $h'(a)$, gives a valid label $[\ell_i, u_i]$ for $H_\alpha(x, v)$.

The experiment uses the prospectively specified normalized probe 2^{-12}, hence $d = A(x)2^{-12}$. Its five main action sites are $v \in \{0, 1/4, 1/2, 3/4, 1\}$. The three interior sites buy three original-Q values each, and the endpoints buy two each: there are thirteen paid contexts at each state. The nested state meshes have 5^2 and 9^2 sites, giving 325 and 1053 contexts per risk level and standalone acquisition budget. The protocol specifies how each continuous-risk interval is constructed and charges all optimal-continuation solves, probes and enclosure work. The allowance δ in (10) does not enter an optimal-Q derivative label; it enters the subsequent implemented-policy statement once.

For a declared family of continuous piecewise-polynomial features, use the anchored critic

$$\widehat{R}_\theta(x,v) = \widehat{R}_0(x) + \sum_{j=1}^{p} \theta_j\{\phi_j(x,v) - \phi_j(x,0)\}. \tag{14}$$

The implementation stores an approximation $\widehat{Q}_0(x)$ to $Q_x(0)$. In (14) this means $\widehat{R}_0(x) = \widehat{Q}_0(x) - S(x)$, so the full action critic is $\widehat{Q}_0(x) + h(A(x)v)$ plus the learned anchored difference. The state-only anchor affects a displayed value level and cancels from every current action comparison. At a label site, let $A_{ij} = A(x_i)^{-1} D_v \phi_j(x_i, v_i)$. Include both feasible one-sided rows at an interior kink and only the inward row at $v = 0$ or $v = 1$. The same label is attached to each row. Let the coefficient box $\Theta = \prod_j [\underline{\theta}_j, \overline{\theta}_j]$ be fixed before acquisition. The action-marginal training objective is

$$T(\theta) = \max_i \{A_i\theta - \ell_i,\ u_i - A_i\theta\}, \qquad T_* = \min_{\theta\in\Theta} T(\theta). \tag{15}$$

It is the largest possible node error over the retained outward labels, in original-action units. The value-loss control uses the same box, features and complete paid bundle, and fits the anchored values by bounded least squares. A smaller value loss is not presumed to minimize (15).

PROPOSITION S.6—A checkable global training gap in the declared box: *The minimum T_* exists. Any exactly checked $\widehat{\theta} \in \Theta$ gives the primal upper bound $P = T(\widehat{\theta})$. Let $y_i^+, y_i^- \geq 0$ be rational weights whose combined sum is one, and set*

$$r = \sum_i (y_i^+ - y_i^-) A_i, \qquad B_y = -\sum_i \ell_i y_i^+ + \sum_i u_i y_i^- + \sum_j \min\{\underline{\theta}_j r_j, \overline{\theta}_j r_j\}.$$

Then

$$\max\{0, B_y\} \leq T_* \leq P, \qquad 0 \leq P - T_* \leq P - \max\{0, B_y\}. \tag{16}$$

In particular, no exact dual stationarity equations or solver status are needed for the lower bound. Exact verification must bind the actual stored coefficients, feature rows, labels and box.

The same independent rational audit checks the value-loss control's optimization gap by minimizing the affine gradient support of its convex sum of squares over the declared box. Proposition S.17 supplies that lower bound and its proof. Thus neither training objective is declared globally solved from a floating optimizer's success flag.

The declared box is $[-128, 128]$ except for the nonnegative quadratic curvature coefficient and the nonnegative nonlinear ReLU or matched-hinge coefficients. A signed coordinate contributes $-128|r_j|$ to B_y; a nonnegative one contributes $\min\{0, 128 r_j\}$. The quadratic, polynomial, spline, tensor-grid and matched-hinge controls face the same acquisition interface. A coordinate-loop hinge carrying the same features and coefficients represents the same function as its ReLU encoding. This experiment can identify the consequence of a learning loss or encoding cost, not a neural-specific approximation advantage from that equality.

3.3. *From finite labels to an off-design policy certificate*

A node training certificate does not supply uniform derivative accuracy. To make the remaining calculation explicit, take a finite cover $\mathcal{C}$ of Ω. For each cell C, construct intervals

$$d_j(C) \supseteq \{A(x)^{-1} D_v \phi_j(x, v) : (x, v) \in C\},$$

including every relevant one-sided value. Let $[\underline{h}_\theta(C), \overline{h}_\theta(C)]$ enclose $\sum_j \theta_j d_j(C)$ by interval arithmetic, and let $r_i(C) \geq \sup_{u \in C} \|u - \xi_i\|_2$, where $\xi_i = (x_i, v_i)$ is the parameter node. Define

$$\begin{aligned} \underline{H}(C) &= \max_i \{\ell_i - \Lambda_\alpha r_i(C)\}, \\ \overline{H}(C) &= \min_i \{u_i + \Lambda_\alpha r_i(C)\}, \\ \epsilon(C) &= \max\{0, \overline{h}_\theta(C) - \underline{H}(C), \overline{H}(C) - \underline{h}_\theta(C)\}. \end{aligned} \tag{17}$$

Using only a declared subset of label sites in the maximum and minimum remains valid. Empty target intersections indicate a failed numerical enclosure and do not authorize selection of a favorable endpoint.

THEOREM S.7—Feature-aware continuation and decision control: *Suppose the label and feature-range inequalities above are valid. For each cell C, the original-action marginal error of (14) is at most $\epsilon(C)$, including one-sided limits. Fix a state x, and let $\epsilon_x(v)$ be any measurable choice of a valid cell bound at (x,v). For feasible original actions a,b, put*

$$\mathcal{D}_x(a,b) = A(x) \int_{\min(a,b)/A(x)}^{\max(a,b)/A(x)} \epsilon_x(v)\, dv,$$

$$\widehat{q}_x(a) = S(x) + h(a) + \widehat{R}_\theta(x, a/A(x)). \tag{18}$$

Then

$$|Q_x(b) - Q_x(a) - \{\widehat{q}_x(b) - \widehat{q}_x(a)\}| \le \mathcal{D}_x(a,b), \tag{19}$$

$$J_x(b) - J_x(a) \le \widehat{q}_x(b) - \widehat{q}_x(a) + \mathcal{D}_x(a,b) + \delta,$$

$$0 \le J_x(a) - V_0(x) \le \delta + \sup_{0 \le b \le A(x)} \{\widehat{q}_x(a) - \widehat{q}_x(b) + \mathcal{D}_x(a,b)\}. \tag{20}$$

Any certified upper bound on the final supremum remains valid. These conclusions do not require a convex fitted critic.

The finite cover pays for the actual feature geometry. An anchored ReLU feature has derivative zero or its declared action coefficient; both sides enter a crossing cell. Polynomial and spline pieces have computable polynomial derivative ranges. Tensor hats have one-sided action derivatives at their knots. Dividing their numerator bounds by the positive capacity interval is valid because $A(x) \ge 1/8$. The supplement gives constructive range rules for every declared family. Consequently (17) does not infer a smooth fitted derivative from smoothness of the original target.

The bound can be conservative at the fixed training meshes. Unresolved cells, paid refinements and fallback original-risk qualification remain part of the numerical record. No off-design certificate is credited merely because a finite-dimensional fit has a small node loss.

PROPOSITION S.8—Capacity-sensitive actor extraction: *Suppose a feasible returned action a has an outward original derivative interval $[g_-, g_+]$. Define*

$$\mathcal{K}_x(a;g) = -\min_{d\in[-a,A(x)-a]}\{gd + d^2\},$$
$$K_x(a) = \max\{\mathcal{K}_x(a;g_-), \mathcal{K}_x(a;g_+)\}. \tag{21}$$

Then

$$Q_x(a) - V_0(x) \le K_x(a), \qquad J_x(a) - V_0(x) \le K_x(a) + \delta, \qquad |a - a_0^*(x)|^2 \le K_x(a).$$

The scalar minimum is attained at $\mathrm{proj}_{[-a,A(x)-a]}(-g/2)$. *At $a = 0$, $g_- \ge 0$ implies $K_x(a) = 0$; at $a = A(x)$, $g_+ \le 0$ implies $K_x(a) = 0$. If the fitted actor has a checked constrained stationarity residual at most r, and every required one-sided marginal error is at most ϵ, the interval supplied by that error bound can be chosen so that*

$$K_x(a) \le (r + \epsilon)^2/4.$$

This is the retained directional strong-convexity bound applied to a newly acquired and extended marginal label. It makes a capacity action with the correct marginal sign economical to certify. A numerical polynomial root or an optimizer status alone does not supply the stated residual; the returned action, piecewise derivatives and rounding must be checked or covered by the original certificate service. Similarly, neither a training-gap certificate nor a certified action identifies a computational adoption region without the full acquisition and decision-service costs and the common-information comparisons.

3.4. *The anchor that a recursive Bellman operator must retain*

An action-independent level cancels at its current state. It need not cancel when that learned value is exported as the next continuation. The original investment transition gives an exact example. If $W_1(y) = V_1(y) + \gamma(y_1 + y_2)$, the zero-sum shock and cash invariance imply

$$Q_x^{W_1}(a) - Q_x(a) = \beta\gamma\{F_1(x,a,0) + F_2(x,a,0)\}.$$

The added root marginal is $3\beta\gamma/4$. The same state-dependent level could leave a last-date action minimization unchanged. Thus discarding that anchor would change the recursive decision problem.

The complete-value statement in Theorem S.18 supplies the required account. Let W_{t+1} be an exported continuation with signed uniform errors $E^-_{t+1} \le W_{t+1} - V_{t+1} \le E^+_{t+1}$. Suppose the acquired action critic is compared with the original Bellman objective using that W_{t+1}, and its retained anchor and action-contrast errors give full-level error bounds $d^-_t \le \widehat{Q}_t - Q_t^{W_{t+1}} \le d^+_t$. If the exported value has an additional checked minimization error at most ζ_t in absolute value, then a monetary recursive risk map gives

$$\begin{aligned} E^-_t &= d^-_t + \beta_t E^-_{t+1} - \zeta_t, \\ E^+_t &= d^+_t + \beta_t E^+_{t+1} + \zeta_t. \end{aligned} \tag{22}$$

A local certified action gap g_t under $Q_t^{W_{t+1}}$ and future policy bound P_{t+1} give

$$P_t \le g_t + \beta_t (E^+_{t+1} - E^-_{t+1}) + \beta_t P_{t+1}.$$

In the present paid two-date construction the root target uses the original V_1, so its continuation approximation term is zero; the unchanged terminal controller contributes $\beta\varepsilon_1 = \delta$. The supplementary theorem separately states the bound for a noncash-invariant recursive preference under an explicit continuation Lipschitz assumption. Such a preference does not inherit the monetary shift identity by notation.

4. PROOFS OF THE INHERITED NBO RESULTS

These proofs retain the original statements and arguments. Their placement in the supplement changes the reading order, not their hypotheses.

4.1. *Proof of the result 1*

PROOF: Backward comparison and cash invariance give

$$V^*_t \ge f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J^\pi_t \le f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3). A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t. The terminal width is unchanged. *Q.E.D.*

4.2. *Proof of the result 2*

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that every cone lies above it minus e_t. At a nearest node its cone exceeds the infimum by at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \le y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \mathrm{ReLU}(z) + \mathrm{ReLU}(-z)$ and $\min(u, v) = u - \mathrm{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. *Q.E.D.*

5. SIGNED QUALIFICATION AND NONLINEAR TRANSPORT: PROOFS

5.1. *Proof of Theorem S.1*

Fix θ and omit it from the notation. Because b_t^- is valid for every original feasible action, taking infima gives

$$T_t f_{t+1} - f_t \ge \widehat{T}_t - f_t + b_t^- \ge \ell_t + b_t^-.$$

For the actual feasible action, the numerical greediness and actor-specific upper bounds give

$$T_t^\pi f_{t+1} - f_t \le \widehat{Q}_t(\cdot, \pi_t) - f_t + b_t^{\pi,+} \le u_t + \gamma_t + b_t^{\pi,+}.$$

Define real constants backwards by

$$a_T = d^-, \quad c_T = d^+, \qquad a_t = \ell_t + b_t^- + \beta_t a_{t+1}, \quad c_t = u_t + \gamma_t + b_t^{\pi,+} + \beta_t c_{t+1}.$$

At the terminal date $f_T + a_T \le g \le f_T + c_T$. If $V_{t+1}^* \ge f_{t+1} + a_{t+1}$, monotonicity and cash invariance, including under the infimum, imply

$$V_t^* = T_t V_{t+1}^* \ge T_t f_{t+1} + \beta_t a_{t+1} \ge f_t + a_t.$$

Similarly, $V_{t+1}^\pi \le f_{t+1} + c_{t+1}$ implies

$$V_t^\pi = T_t^\pi V_{t+1}^\pi \le T_t^\pi f_{t+1} + \beta_t c_{t+1} \le f_t + c_t.$$

Induction proves both comparisons. Subtracting at date zero gives $V_0^\pi - V_0^* \le c_0 - a_0$, which expands to (2). Feasibility and optimality give its lower bound zero. Infima need not be attained: the order comparison holds directly under infima, provided the stipulated original value recursion and feasible policy are defined. Every parameter is treated by the same argument; a supremum of valid bounds provides the stated classwise conclusion. No union over parameter vectors and no reference-policy occupation law is introduced.

For shifts $f_t' = f_t + k_t$, set $\widehat{Q}_t' = \widehat{Q}_t + \beta_t k_{t+1}$. Both numerical residual endpoints shift by $\beta_t k_{t+1} - k_t$; their difference is unchanged. Numerical action differences and original-minus-numerical action values are unchanged by cash invariance. Both terminal endpoints shift by $-k_T$. Thus every term in (2) is invariant. If the action-value discrepancy itself is a constant k_t, choose $b_t^- = b_t^{\pi,+} = k_t$ and its charge vanishes.

For the stationary assertion, the Bellman and fixed-policy maps are sup-norm contractions with modulus β: monotonicity and cash invariance imply sup-norm nonexpansiveness of each one-step evaluation. The finite-horizon argument with a bounded terminal comparison then has a terminal remainder of order β^T. Letting T tend to infinity proves the displayed stationary bound. The asserted constant-shift invariance does not justify shifting an original preference primitive or an action-dependent immediate cost.

5.2. *Sharpness and the distinction between constant and statewise bias*

For $h > 0$, take one decision date, terminal cost $g(y) = y$, discount one and zero immediate costs. The original successors of actions zero and one are respectively h and zero; the numerical successors are both zero. Store $f_0 = 0$ and $f_1 = g$, and let the numerical tie convention select action zero. All numerical residual and greediness terms vanish. The original-minus-numerical action values lie in $[0, h]$, the selected-action upper endpoint is h, and the implemented policy loses exactly h. Thus the signed charge h is attained. The symmetric allowance $2h$ remains valid but is strictly larger. The example works for any monotone cash-invariant normalized risk evaluation because its successors are deterministic. Both laws can be singular, so no total-variation convergence is being used.

A statewise common-action bias is different. Let a root action choose state L or R at the next decision date. At that date there is only one feasible action, with original cost $2h$ in L and zero in R. Let every numerical cost, stored continuation and terminal cost be zero, with discount one. Break the root numerical tie toward L. All numerical residual spans and greediness gaps vanish. At the second date the discrepancy is common to every action at its state, but equals $2h$ or zero depending on that state. The implemented policy loses $2h$. Cancelling the discrepancy merely because it is action independent at each state would falsely report zero loss. The signed uniform range correctly charges $2h$. This is precisely the earlier action's effect on future states; fixing the reference occupation would also lose it.

5.3. *Proof of Proposition S.2*

Let $(Y, \widehat{Y})$ have any coupling of the two successor laws. For every real threshold z, the map $r \mapsto (r - z)_+$ is one-Lipschitz. Hence

$$\left| P(f - z)_+ - \widehat{P}(f - z)_+ \right| \leq \mathbb{E}|f(Y) - f(\widehat{Y})| \leq K\mathbb{E}d(Y, \widehat{Y}).$$

The same bound, divided by $1 - \alpha$, is uniform in z for the two functions minimized in (3). Infima of uniformly close functions are close. Taking the infimum over couplings proves the first bound, with approximating couplings sufficient when an optimum is not attained. At $\alpha = 0$ the variational expression equals expectation:

the expression is at least Pf and equality holds for any threshold no larger than the bounded loss's infimum.

To verify that the tail factor cannot be uniformly reduced, take $f(y) = y$ on $[0,1]$, $\widehat{P} = \delta_0$ and $P = (1-h)\delta_0 + h\delta_1$, with $0 < h \leq 1$ and $\alpha = 1 - h$. Then $K = 1$, $W_1(P, \widehat{P}) = h$, $\rho_P^\alpha(f) = 1$ and $\rho_{\widehat{P}}^\alpha(f) = 0$. The bound $Ke/(1-\alpha) = 1$ is attained. In particular, a bound uniform up to risk level one cannot be deduced from Wasserstein distance alone with a fixed coefficient.

For the entropic assertion, choose $m \leq f \leq M$. For positive λ, the mean value theorem gives

$$|\exp(\lambda f(y)) - \exp(\lambda f(\widehat{y}))| \leq \lambda \exp(\lambda M) K d(y, \widehat{y}).$$

Both exponential integrals are at least $\exp(\lambda m)$, and $|\log s - \log t| \leq |s - t| / \min(s,t)$ for positive s, t. Combining these facts, dividing by λ and taking the infimum over couplings yields $K \exp\{\lambda(M-m)\} e$. Letting m and M approach the continuation's infimum and supremum proves (4). Translation invariance follows from the oscillation rather than the absolute level. The expectation limit follows from boundedness and expansion of the exponential integral; its displayed coefficient tends to K.

If $f(Y) \geq f(\widehat{Y})$ almost surely, every positive-part integral is weakly larger and so is its minimized variational expression. Exponentiation and logarithm preserve the same order for the entropic evaluation. This gives the signed lower endpoint zero. If $|\widehat{Q} - c - \beta \rho_{\widehat{P}}(f)| \leq \xi$, subtracting the numerical approximation places the action-value discrepancy in $[-\xi, \beta\kappa e + \xi]$ under that ordering, or in $[-\beta\kappa e - \xi, \beta\kappa e + \xi]$ without it. These are bounds for the original risk primitive; no statement is made about replacing one primitive by another.

5.4. *Nested objectives and the dated resource account*

Nested one-step evaluations and a static terminal evaluation differ even without decisions. With four equiprobable terminal losses $0, 0, 0, 4$, group the first two and last two under two equiprobable intermediate states. At risk level one half, the intermediate evaluations are zero and four, and the nested value is four. Static

average value at risk of the four terminal outcomes is two. Consequently a static objective needs its own state augmentation or a different analysis; applying the preceding nested recursion without that step changes the objective.

For an originally nested criterion, the preceding induction propagates a state-independent continuation allowance a as $\beta_t a$. The local Wasserstein-to-risk coefficient κ_t enters only the current discrepancy. Substituting the symmetric signed interval into (2) gives (5); monotone-coupling information replaces the factor two on that transition component by one. This establishes the budget used for dated allocation. It does not bound the cost of obtaining K_{t+1} or the number of labels, innovations or feasible-action searches. In particular, a growth of the verified continuation modulus with horizon is retained in κ_t and cannot be removed by this algebra.

For a fixed-opponent best-response problem the identical induction compares a player's implemented complete strategy with the original best-response optimum. Applying it separately in every declared subgame gives the corresponding unilateral-deviation certificate. Each temporal self likewise retains its own original criterion and fixed future strategies. Monotone but non-cash-invariant aggregators do not satisfy the constant-shift step above and must instead use their certified absolute-error recursion. This establishes the advertised scope without conflating expectation, nested risk, static risk and strategic evaluation.

6. PROOFS FOR REFERENCE-CENTERED RISK QUALIFICATION

6.1. *Centering and the original Bellman optimum*

For any bounded z, substituting $f_{t+1} + z$ in (8) and subtracting the finite number $\rho(f_{t+1})$ proves (10). Each dual objective at f_{t+1} is at most its supremum, giving $k^f \geq 0$. Approximating that supremum gives $\inf_q k^f(q) = 0$ without an attainment assumption. Since each q is a probability, adding a constant b to z adds b to $\mathcal{E}^f(z)$. If $z \leq w$, every dual objective at z is no larger than its counterpart at w, proving monotonicity. The inequalities $w - \|z - w\|_\infty \leq z \leq w + \|z - w\|_\infty$ prove nonexpansiveness.

PROOF OF THEOREM 3: Let $d_t^\pi = V_t^\pi - f_t$ and $d_t^* = V_t^* - f_t$. The original recursions and (10) give the exact identities

$$d_t^\pi(x) = r_t(x, \pi_t(x)) + \beta_t \mathcal{E}^f_{t,x,\pi_t(x)}(d_{t+1}^\pi),$$
$$d_t^*(x) = \inf_{a \in A_t(x)} \{r_t(x,a) + \beta_t \mathcal{E}^f_{t,x,a}(d_{t+1}^*)\}. \tag{23}$$

Both have terminal value $g - f_T$. Suppose $d_{t+1}^\pi \le U_{t+1}$. Monotonicity, nonnegative β_t, the upper residual inequality and (11) imply $d_t^\pi \le U_t$. Similarly $L_{t+1} \le d_{t+1}^*$ implies $L_t \le d_t^*$. Backward induction establishes both statements. Feasibility of the supplied complete policy implies $V_t^* \le V_t^\pi$, giving (12). The argument takes place at each fixed parameter vector; applying it to every vector satisfying the premises proves the class statement without assembling incompatible continuations.

Exact equality recursions coincide with (23), proving the exactness assertion by induction. For the scalar comparison, set $a_t = \ell_t + b_t^-$ and $b_t = u_t + \gamma_t + b_t^{\pi,+}$ from the signed assumptions. Every original action has residual at least a_t, since its numerical action value is at least the numerical Bellman infimum. The implemented action has residual at most b_t. Define $\underline{h}_T = d^-$, $\overline{h}_T = d^+$, and

$$\underline{h}_t = a_t + \beta_t \underline{h}_{t+1}, \qquad \overline{h}_t = b_t + \beta_t \overline{h}_{t+1}.$$

Since $\mathcal{E}^f(b) = b$ for constants, these are feasible lower and upper certificates. Exact backward evaluation is order preserving, so $L_t \ge \underline{h}_t$ and $U_t \le \overline{h}_t$ for exact residuals and terminal discrepancies. Their date-zero difference is at most (2).

For two policy supersolutions, their pointwise minimum remains a supersolution: the operator at the minimum of their successor bounds is at most its value at either bound. The corresponding pointwise maximum of two optimum subsolutions remains a subsolution by the reverse argument. Finite combinations therefore preserve an already valid scalar guarantee. No claim about improved bounds from an uncontrolled numerical optimizer is needed. *Q.E.D.*

6.2. *Risk occupations and certificate orientation*

PROOF OF PROPOSITION A.42: For fixed π, (23) is the Bellman equation for a finite-horizon maximizer selecting a risk index j after observing the current state. Its

reward is $r_{ts} - \beta_t k_{tsj}$, transition vector is q_{tsj}, and terminal reward is $g - f_T$. Finite maxima have measurable maximizers. Backward induction establishes a Markov maximizing choice at each date and state.

Any nonanticipating choice rule, including a history-dependent one, induces occupation masses satisfying (A.97). Conversely, for a feasible flow, choose j at (t,s) with probabilities y_{tsj}/m_{ts} when $m_{ts} > 0$; at zero-mass states choose an arbitrary feasible index. Induction on t recovers the declared masses. The corresponding expected sum of rewards is exactly the objective in (A.98). Thus the flow maximum equals the centered fixed-policy value. This representation is under the adversarial risk kernels and must not be called the physical occupation law.

Multiply each dual inequality by $B_t y_{tsj}$ and sum. The flow equalities cancel intermediate potential terms, leaving the primal objective at most $\mu^\mathsf{T} U_0$. The argument allows a zero discount: it uses $B_{t+1} = B_t \beta_t$ but never divides by either factor. Feasible primal and dual witnesses with equal objectives certify the maximum. For an optimum subsolution, a separate feasible risk witness for every original action supplies a lower bound on that action's supremum; a minimization over a restricted sampled action set does not supply the required all-action inequality. *Q.E.D.*

6.3. *Kernel exclusion and its sharp boundary*

For any $z \in \mathcal{D}$, the difference between the affine pieces indexed by q and q_0 is

$$(q - q_0)^\mathsf{T} z - k^f(q) + k^f(q_0).$$

A strict negative upper bound excludes q uniformly. This proves (A.99). Maximizing coordinate by coordinate gives the box formula. For a span ball, the coefficients sum to zero; adding a constant to z changes nothing. Normalize $\min_i z_i = 0$ and assign s to the positive coefficients and zero to the negative ones. The maximum is $s \sum_i (q_i - q_{0i})_+ = s\|q - q_0\|_1/2$.

As a sharp example, take two equally likely outcomes and average value at risk at level $1/2$. Then $\rho(v) = \max(v_1, v_2)$ and the dual vertices are e_1, e_2. At $f = (0,1)$ their centered penalties are $1, 0$. For $z = (2,0)$, the centered value is 1. Keeping only the maximizing kernel e_2 from the reference gives 0, which is invalid as an upper

bound; dropping penalties gives 2, which is conservative but not exact. On the span ball of radius $1/2$, the first kernel is safely excluded. At radius 1, $z = (1, 0)$ makes the two pieces equal, so the strict exclusion must retain the boundary. This example proves both the necessity of allowing the risk kernel to change and sharpness of the strict threshold convention.

6.4. *Entropic duality without changing the objective*

Let p be a probability measure, f, z bounded, and $\lambda > 0$. Boundedness makes $Z_f = \int e^{\lambda f} dp$ strictly positive and finite, and makes p_f equivalent to p on its support. For $q \ll p$ with finite relative entropy,

$$\lambda^{-1}\mathrm{KL}(q\|p_f) = \lambda^{-1}\mathrm{KL}(q\|p) - \int f\, dq + \lambda^{-1}\log Z_f.$$

If q places mass outside the support of p, its penalty is infinite. For any bounded v, define $p_v = e^{\lambda v} p / Z_v$. Nonnegativity of $\mathrm{KL}(q\|p_v)$ gives

$$\int v\, dq - \lambda^{-1}\mathrm{KL}(q\|p) \le \lambda^{-1}\log Z_v,$$

with equality at $q = p_v$. This proves the variational representation, its centered penalty, and (A.100). It also shows exactly why using the fixed tilt p_f as an expectation is not generally exact: the centered evaluation remains entropic under p_f, rather than linear. For $\lambda \downarrow 0$, bounded convergence gives the expectation limit. Exponential and logarithmic evaluations in a numerical certificate require outward arithmetic; the rational average-value-at-risk experiments are not evidence of exact floating-point entropic computation.

6.5. *Occupation-sensitive precision and a strict improvement*

For fixed penalties and transitions, write the flow objective as $A(y)$ and the additional nonnegative residual contribution as $E(y) = \sum_{t,s} B_t e_{ts} m_{ts}$. If $H = \max_y A(y)$ and $H_e = \max_y \{A(y) + E(y)\}$, then

$$0 \le H_e - H \le \max_y E(y).$$

The upper inequality follows from $A(y) \leq H$ for every feasible y; the lower follows from $E(y) \geq 0$. Since each occupation layer has total mass one, $\max_y E(y) \leq \sum_t B_t \max_s e_{ts}$. This establishes (A.101). The argument assumes the penalties are held fixed while residual uncertainty is enlarged. Altering the underlying stored continuation changes the penalties and requires requalification.

For a strict horizon example, let two states be absorbing under both actions, let action zero cost zero and action one cost one in both states, let $g = 0$, and let all discounts equal one. Every normalized monetary risk evaluation of these deterministic transitions agrees with evaluation at the absorbing state. Put $f_t(s) = (T - t)s$ for $s \in \{0, 1\}$ and choose action zero everywhere. Each date's implemented and minimum residuals are $(0, -1)$; the signed scalar account charges one at each date, yielding T. The exact state-dependent lower and upper recursions both equal $-(T - t)s$, giving zero policy loss at every state. There are two feasible actions and every action is checked. No state-dependent bias was cancelled without propagation: its entire state dependence was carried through the original transition at every date. This is consistent with, and sharpens rather than contradicts, the statewise-bias warning in Section 2.

Finally, the finite backward recursion evaluates each stored transition support once for each retained risk piece at each date and state-action pair. Summing these support sizes yields the stated arithmetic count. A direct descendant recursion repeatedly evaluates the same subproblem; removing that repetition does not establish a polynomial bound in the number of continuous states, risk pieces, acquisition cells or rational bit length. The comparison is a representation-conditioned resource statement, and the full construction and support-qualification work remains separately charged.

7. CONTINUOUS INVESTMENT: PRIMITIVE PROOFS

7.1. *Curvature, the action constraint, and state moduli*

PROOF OF PROPOSITION 4: For $x \in X$, $0 \leq a \leq 1/4$ and $|z| \leq 1/32$, the transition remains in X. Indeed, the state part is increasing in both of its arguments, its maximum is $1/16 + 1/2 + 1/8$, and $b_j a \leq 1/8$. Its minimum after the shock is at

least $1/32$, and its maximum is at most $27/32$. Thus every derivative bound below is used on its stated domain.

Both S and g are convex and continuously differentiable; their squared hinges have positive semidefinite second derivative on each smooth region. Direct differentiation gives

$$|\partial_j g| \le \frac{10}{d}, \qquad |\partial_j S| \le \frac{15}{2d}, \qquad \nabla^2 S \succeq \frac{4}{d} I.$$

The first bound adds $5/d$ from the target term, $1/d$ from the cyclic difference term, and $4/d$ from the hinge. Only $-x_j x_{j+1}/16$ in F_j has nonzero second derivative. For a joint direction (v, w), the possibly negative part of the second derivative of $g(F)$ is

$$-\frac{1}{8}\sum_j \partial_j g(F) v_j v_{j+1} \;\ge\; -\frac{5}{4d}\|v\|_2^2,$$

since $\sum_j |v_j v_{j+1}| \le \|v\|_2^2$. The remaining term is the quadratic form of the positive semidefinite Hessian of g applied to the full state-action Jacobian. In particular, it includes the mixed state-action terms and is not discarded selectively. Since $h''(a) = 2 + 48a^2 \ge 2$, every fixed-shock objective satisfies (15). Integrating against a fixed feasible risk density preserves this bound. Taking the supremum over those densities preserves the same strong convexity. The bound extends across the hinge by continuity of the first derivative, or by integrating the almost-everywhere second-derivative inequality along a segment.

The graph $\{(x, a) : x \in X, 0 \le a \le A(x)\}$ is convex because A is affine. Partial minimization of the jointly convex objective over this graph proves convexity of V_1. Compactness and continuity supply an attained minimum. The positive capacity permits constraint qualification in the action variable.

For a fixed feasible density, the supporting strong-convexity inequality gives, for every (y, u),

$$Q_1(y, u) \ge W + p_x^\top (y - x_0) + p_a(u - a_0) + (u - a_0)^2.$$

For feasible u, adding $\lambda(u - A(y)) - \nu u \le 0$ to the right side preserves a lower bound. Minimize this augmented quadratic over all real u, an enlargement of the feasible interval. Completing the square gives (17). This proof uses neither equality

in the risk supremum nor exact stationarity. Interval implementations bound the complete expression outwards, including its residual square and its slope.

For completeness we bound true value subgradients, rather than the potentially large slopes of arbitrary poor lower witnesses. A transition Jacobian column has nonnegative entries summing to at most $11/16$. Therefore any state subgradient of a risk-maximizing one-date objective has coordinate magnitude at most

$$\frac{1}{d}\left(\frac{15}{2}+\beta\frac{55}{8}\right).$$

At an upper-bound action, the multiplier can be chosen with $0 \le \lambda \le \beta(10/d)\sum_j b_j = 15\beta/4$: the nonnegative derivative of h cannot increase the negative derivative which this multiplier offsets. At a lower-bound action take $\lambda = 0$. At an interior minimizer both action-bound multipliers vanish. Envelope subgradients thus have coordinates bounded by

$$\frac{1}{d}\left\{\frac{15}{2}+\beta\left(\frac{55}{8}+\frac{15}{32}\right)\right\} < \frac{15}{d}.$$

Where the risk function is nonsmooth, the same bound holds for convex combinations of active risk gradients and valid KKT multipliers. Integrating bounded subgradients along a segment, with the one-sided limit at the boundary of X, proves (16). *Q.E.D.*

7.2. *Exact one-date risk and outward construction*

Put $m = F(x,a,0)$, $R = 1/32$ and $D = 10d^{-1}\sum_j m_j s_j$. Since $\sum_j s_j = 0$, the hinge in $g(m+sz)$ is constant in z. The target and cyclic terms give exactly

$$g(m+sz) = g(m) + Dz + 5z^2. \tag{24}$$

Let $L = 2R\alpha$ and $h = \max\{-R, \min\{R-L, -D/10 - L/2\}\}$. The interval $[h, h+L]$ is a lowest-value interval of length L for the convex quadratic. The maximizing AVaR density is $(1-\alpha)^{-1}$ on its complement and zero on that interval. The two tail moments are

$$M_1 = -\frac{L(2h+L)}{2(2R-L)}, \qquad M_2 = \frac{2R^3 - L(3h^2+3hL+L^2)}{3(2R-L)}.$$

Hence the risk equals $g(m) + DM_1 + 5M_2$. This also follows from the threshold representation of AVaR, with the threshold chosen at the boundary of the retained lower interval. Endpoint minima and equal boundary values are included by the clipping rule.

For a lower witness, any dyadic $\widehat{h} \in [-R, R - L]$ defines an exactly feasible excluded interval and a feasible tail density. Differentiating its objective holds $\widehat{h}$ fixed. This distinction is essential: a derivative of a changing approximate tail need not be the derivative of a supporting objective. Upper values use an outward enclosure of the maximizing h, while lower witnesses use the separately retained feasible $\widehat{h}$. The lower witness therefore remains valid when the two differ.

At a fixed initial state, $F(x, u, z)$ is affine in u and z. A convex continuation has a global affine minorant at every successor anchor. For a density constant on a bin, integration of this affine minorant equals its value at the midpoint. Its action dependence is affine. Adding the strong-convexity minorant of h gives (18). A fitted continuation is used only where its nonnegative convex readout makes this argument valid; no analogous assertion is made for an unrestricted nonconvex fit.

For a value upper bound, convexity places a fitted continuation below the larger endpoint value on a bin. For V_1, the midpoint upper value plus $15R/n$ bounds the whole bin by (16). If $l_i \le v(z) \le u_i$ on each bin, monotonicity of the original risk gives

$$\rho_\alpha\left(\sum_i l_i 1_{B_i}\right) \le \rho_\alpha(v) \le \rho_\alpha\left(\sum_i u_i 1_{B_i}\right).$$

The step-function risks are finite upper-tail means because the bin probabilities are exactly equal and $n(1-\alpha)$ is an integer. Their computation does not assume that v is constant on the bins.

7.3. *Complete policy and centered comparison*

PROOF OF THEOREM 5: Every fixed density is feasible for the original risk supremum. Every continuation support is below V_1 at every successor. Accordingly its all-action quadratic minimum is below V_0, irrespective of which feasible proposal supplied the anchor. The binwise upper calculation dominates the value of the

entire terminal controller; monotonicity and cash invariance charge at most $\beta\varepsilon_1$ for its terminal optimality allowance. It follows that $\underline{V}_0 \le V_0 \le J_0^{\widehat{\pi}} \le \overline{J}_0^{\widehat{\pi}}$. Subtracting proves (19).

For the centered statement take any lower and upper continuation functions ℓ, u furnished by the preceding supports and bounds. Their differences from f give admissible error functions. The exact centered identity cancels the common $\rho_\alpha(f)$ before any numerical enclosure. The preceding lower witness is a minorant of the resulting all-action expression; the preceding policy upper envelope is a majorant of the resulting policy expression. They are sufficient subsolution and supersolution inequalities in the general theorem. Computing them jointly is an implementation of that comparison, not a new objective. Loose global residual constants remain a valid alternative, but need not replace these sharper local witnesses. *Q.E.D.*

Whole-cell residual interpolation. On a state cell with vertices v_i, choose its multilinear convex weights $\omega_i(y)$. If f and V_1 are convex, their vertex upper enclosures interpolate to upper bounds. A fixed affine support L_V of V_1 satisfies

$$V_1(y) - f(y) \ge L_V(y) - \sum_i \omega_i(y)\overline{f}(v_i) = \sum_i \omega_i(y)\{L_V(v_i) - \overline{f}(v_i)\}.$$

Its right side is at least the smallest vertex difference. Maximize that lower bound over the stored supports. Reversing V_1 and f, using upper values of V_1 and lower supports of f, gives the upper residual bound. The finite maximum or minimum over anchors and cells is taken after these whole-cell inequalities. No interpolation equality for the unknown function is asserted.

Strict tail exclusion. Suppose $u_i < \tau$ and the union H of $k = n(1-\alpha)$ other bins has $v(z) \ge \tau$ throughout. It has probability at least $1-\alpha$. If a feasible maximizing density assigned positive mass to B_i, some capacity $q < (1-\alpha)^{-1}$ would remain on H. Moving equal mass from B_i to H strictly increases its risk objective, a contradiction. Equality yields only weak improvement and cannot support exclusion. Bin bounds used to construct H must come from the unpruned calculation. The argument covers a continuum of densities without enumerating all of them.

8. DECISION-SCALE QUALIFICATION: PROOFS

PROOF OF PROPOSITION A.34: For every real threshold η, the function $\phi_\eta = (v - \eta)_+$ is convex and K-Lipschitz. Conditional Jensen and the chord inequality give $M_n\phi_\eta \le P\phi_\eta \le E_n\phi_\eta$. Apply the AVaR threshold formula $\inf_\eta\{\eta + (1-\alpha)^{-1}\mu\phi_\eta\}$ for each of the three probability measures. Taking infima preserves the two orders and proves the enclosure. In particular, no interchange of an action infimum and a risk supremum occurs.

Here is a uniform bound which remains valid at kinks. Let $h = 2R/n$. Choose one-sided slopes of ϕ_η at each bin's outer endpoints. The slope increases across each bin; its total increase over the whole interval is at most $2K$. The difference between the endpoint average and midpoint value on bin i is no greater than $h/4$ times the slope increase across that bin. This follows by integrating the left and right slopes over the two half-bins. Choose the left endpoint right derivative and right endpoint left derivative; jumps at shared endpoints may be omitted, which only decreases the sum of within-bin slope increases. Therefore

$$0 \le E_n\phi_\eta - M_n\phi_\eta \le \frac{h}{4n}(2K) = \frac{KR}{n^2}.$$

For completeness, the half-bin expression equals one half the integral of the difference between the right and left slopes at equal distances from the midpoint, so its upper bound also follows directly without assuming a classical second derivative. The preceding uniform bound passes through the two infima and gives (A.86). Monotonicity and cash invariance add $e_- + e_+$ for endpoint error. They also charge the actual terminal controller's one-sided error only on the upper side.

For a product law, apply the one-coordinate midpoint–endpoint comparison successively, holding the other coordinates fixed. Products of positive averaging operators preserve inequalities and telescope. Each partially averaged positive-part function remains convex and has the same Lipschitz modulus in each unintegrated coordinate. Summing the one-coordinate error bounds and then taking the threshold infimum proves the product statement. This argument uses neither a discrete economic transition assumption nor differentiability of v. *Q.E.D.*

PROOF OF THEOREM A.36: Every service's regret belongs to $[0, \tau]$ at every possible initial state. The difference of two complete-policy values therefore belongs to $[-\tau, \tau]$, because the common original optimal value cancels. Each interval intersection retains the true paired difference. A difference of two intervals of width at most w has width at most $2w$. Intersecting with a valid support cannot increase that width.

Hoeffding's inequality for a variable of range length 2τ gives

$$\Pr\{|\overline{D} - \mathbb{E}D| > r\} \leq 2\exp\{-Nr^2/(2\tau^2)\}.$$

A union bound across the H fixed contrasts gives the stated coverage. Pairwise contrasts need not be mutually independent; only the initial states within each contrast must be independent, conditional on the frozen preparation. Replacing the unobserved sample mean by its numerical lower and upper bounds yields (A.87). Their difference is at most $2(r + w)$. This is an application of the established concentration inequality (Hoeffding, 1963), not a claim to a new concentration principle.

For the chosen rational parameters, $Nr^2/(2\tau^2) = 8$. The inequality $24e^{-8} < 1/20$ can be checked entirely with rational arithmetic: the partial sum $\sum_{j=0}^{11} 8^j/j!$ is greater than 480. Equation (A.88) is an exact rational comparison. In numerical classification the margin is rounded inward and the reported confidence endpoints outward. No observed range or variance is used to choose the radius. *Q.E.D.*

Totalization of the admitted service. The first-date objective with the original V_1 has action Lipschitz constant less than 7: the primitive action derivative is at most $3/4$, and the discounted continuation contribution is at most $(15/16)(15/d)(3d/8)$. The feasible interval has length at most $1/4$. A uniform net with $m = 1024$ intervals therefore gives an all-action lower value by subtracting $7/(8m)$ from the smallest grid cost lower bound. The grid point with the smallest upper value supplies a feasible policy upper value.

The exact fallback uses $n = 256$ innovation bins and the preceding convex-order rule. Exact one-date rational supports have gaps at most 2^{-30}; the terminal controller contributes 2^{-20}. The resulting quadrature width, including these allowances, is

below $1/8192$ for both risk levels. Adding $7/(8m)$ leaves the original-optimum bound below $1/256$. Downward rounding of the first action to 40 bits costs at most 72^{-40} on each side and fits strictly inside both remaining allowances. All values, risk weights, feasible actions and endpoint computations in this final construction are rational, and the terminal strong-convexity bisection reaches its stated gap in finitely many steps. Thus the fallback defines a finite valid action at every dyadic initial observation in the target distribution. For arbitrary real initial states, the separate acquisition transfer remains required; the confirmation theorem here concerns the declared dyadic initial law.

The floating implementation may reach the two targets much sooner. That does not permit removing the exact fallback from the policy definition, treating rejected proposals as missing observations, or asserting that the fallback was exercised when the actual counter is zero. Hardware or resource failure aborts the experiment and is recorded as a failure, not a change in its conditional sampling law.

PROOF OF PROPOSITION A.37: Subtract the two complete acquisition-and-deployment accounts. Using the upper policy, acquisition and fee bounds and the lower computational saving bounds above (A.89) by $m(d^+ - \lambda s^-) + \lambda k^+ + F^+$. Nonpositivity gives (A.90). Solving this scalar inequality proves the integer threshold and its sign qualifications. The separate quality inequality depends on policy cost rather than price and cannot be deduced by increasing λ. A common audit cost cancels only when it is actually common and purchased in both alternatives; otherwise its incremental charge remains in k^+ or F^+. *Q.E.D.*

PROOF OF COROLLARY A.35: Write $h = 2R/n$ and $k = n(1-\alpha)$. The two roots of $v_n - \eta_n$, where $\eta_n = K\alpha R$, are $c_n - \alpha R$ and $c_n + \alpha R$. They lie strictly inside the domain and at two distinct bin midpoints, since αn is an even integer. Under M_n the mass strictly above η_n is $(k-1)/n$ and the mass at η_n is $2/n$. Under E_n there is no mass at η_n, and the mass above it is exactly k/n. Hence η_n minimizes the AVaR threshold objective under both measures, by its left and right slope conditions.

The hinge $\phi = (v_n - \eta_n)_+$ is affine on every bin except the two whose midpoints are its roots. On either exceptional bin, the endpoint average minus the midpoint value is $Kh/4$. Each bin has weight $1/n$. The two threshold minima therefore differ

by

$$(1-\alpha)^{-1}(E_n\phi - M_n\phi) = \frac{1}{1-\alpha}\frac{2}{n}\frac{Kh}{4} = \frac{KR}{(1-\alpha)n^2}.$$

All equalities use the endpoint mixture's half weights at the outer boundaries. This explicit sequence establishes sharpness of the stated enclosure on its function class; it does not assert that the investment value itself equals v_n. *Q.E.D.*

9. A CONTROLLED RECURSIVE-UTILITY BSDE VERIFICATION BRIDGE

The connection between recursive utility and backward stochastic differential equations has a classical foundation (Pardoux and Peng, 1990, El Karoui et al., 1997). For an NDU/FBSDE application, the preference-state dynamics, information filtration and recursive objective must first be specified. The result below supplies a precise Brownian Markov subcase with an augmented but decoupled forward state. It does not identify a general coupled, general-martingale NDU system with the investment model's nested AVaR recursion. The two-date investment experiment supplies no observations for this different continuous-time criterion.

Fix a finite horizon T, deterministic initial state $X_0 = x$, a Brownian natural filtration with its usual augmentation, and an admissible progressively measurable action process a_s valued in a compact set. The augmented economic and preference state solves

$$dX_s = b(s, X_s, a_s)\,ds + \sigma(s, X_s, a_s)\,dW_s.$$

Assume bounded coefficients, uniform Lipschitz continuity in the state, and the usual measurability. A chosen feedback policy must additionally define an admissible, well-posed state process; arbitrary discontinuous feedback is not licensed by these assumptions on open-loop controls. The state may include preference coordinates, but there is no dependence of b, σ on the backward pair in this subcase. Let g and $f(s, x, 0, 0, a)$ be bounded. The scalar cost driver satisfies

$$|f(s,x,y,z,a) - f(s,x,y',z',a)| \le L_y|y-y'| + L_z\|z-z'\|.$$

The original policy criterion is Y_0^a, where

$$Y_s^a = g(X_T) + \int_s^T f(v, X_v, Y_v^a, Z_v^a, a_v)\, dv - \int_s^T Z_v^a\, dW_v. \tag{25}$$

Converting a utility into a cost requires the corresponding sign changes in both the driver and terminal condition. Merely changing the name of the output does not identify different utility functionals.

For a bounded $u \in C^{1,2}$ with the bounded derivatives needed below, set

$$z_u(s,x,a) = \sigma(s,x,a)^{\mathsf{T}} \nabla u(s,x),$$
$$\mathcal{H}^a u = \partial_s u + b(s,x,a)^{\mathsf{T}} \nabla u + \tfrac{1}{2} \operatorname{tr}(\sigma\sigma^{\mathsf{T}} D^2 u) + f(s,x,u,z_u,a).$$

The whole-space formulation introduces no artificial boundary. A bounded-domain version requires an invariant domain or a specified stopped process with its stopping-boundary error included in the terminal account. Deleting that term is not justified by a small interior residual.

PROPOSITION S.9—Original-recursion residual certificate: *For each admissible action process the forward equation and* (25) *have a unique square-integrable adapted solution. Suppose uniformly over the declared state-action domain*

$$\mathcal{H}^a u \geq -e_-(s) \quad \textit{for every admissible } a, \qquad \mathcal{H}^\pi u \leq e_+(s),$$
$$-\eta_- \leq g - u(T,\cdot) \leq \eta_+,$$

where $e_\pm$ are nonnegative integrable deterministic bounds and $\eta_\pm \geq 0$. Then the implemented admissible policy satisfies

$$0 \leq Y_0^\pi - \inf_a Y_0^a \leq e^{L_y T}(\eta_- + \eta_+) + \int_0^T e^{L_y s}\{e_-(s) + e_+(s)\}\, ds. \tag{26}$$

BSDE solvability alone does not imply the existence of an optimizing control. The infimum is sufficient for this certificate; compactness and continuity of the control-to-solution map are separate optimal-control assumptions.

PROOF: For the forward process, the Brownian isometry and maximal inequality bound squared maximal differences of successive Picard iterates on a short interval

by a fixed Lipschitz constant times the interval length and the preceding squared difference. Choose the interval so that this coefficient is smaller than one. Completeness yields a strong adapted fixed point, and the identical difference estimate proves uniqueness. Concatenating finitely many intervals covers $[0, T]$. Bounded coefficients give finite second moments. The argument uses the stipulated admissible control process; it does not establish well-posedness for a feedback whose closed-loop coefficients violate those conditions.

Fix this forward process. For a trial pair (y, z) in the weighted square-integrable space, Brownian martingale representation of $g(X_T) + \int_0^T f(s, X_s, y_s, z_s, a_s)\, ds$ defines the linear backward solution (Y, Z) with that prescribed driver. For differences between the outputs of two trial pairs, Itô's formula to $e^{ks}|\Delta Y_s|^2$, the common terminal value and expectation give

$$\mathbb{E}\int_0^T e^{ks}\{k|\Delta Y_s|^2 + \|\Delta Z_s\|^2\}\, ds \leq 2\mathbb{E}\int_0^T e^{ks}|\Delta Y_s|(L_y|\Delta y_s| + L_z\|\Delta z_s\|)\, ds.$$

Young's inequality bounds the integrand on the right by $(k/2)|\Delta Y_s|^2 + (4L_y^2/k)|\Delta y_s|^2 + (4L_z^2/k)\|\Delta z_s\|^2$. In the squared norm $\mathbb{E}\int_0^T e^{ks}\{(k/2)|Y_s|^2 + \|Z_s\|^2\}\, ds$, the map is a contraction for sufficiently large k, with squared-norm factor at most $\max\{8L_y^2/k^2, 4L_z^2/k\} < 1$. This complete weighted space has a norm equivalent to the usual finite-horizon one. Banach iteration gives existence and uniqueness. Applying Brownian maximal estimates to the backward equation gives $\mathbb{E}\sup_{s\leq T}|Y_s|^2 < \infty$. No general-martingale representation or fully coupled forward equation has been inserted into this argument.

For verification, fix any admissible action process and apply Itô's formula to $u(s, X_s)$. Put $\Delta Y = Y^a - u(s, X_s)$ and $\Delta Z = Z^a - z_u(s, X_s, a_s)$. Subtracting the equations yields a linear BSDE with terminal error $g(X_T) - u(T, X_T)$ and driver

$$A_s\Delta Y_s + \vartheta_s^{\mathsf{T}}\Delta Z_s + \mathcal{H}^{a_s}u(s, X_s), \qquad |A_s| \leq L_y, \quad \|\vartheta_s\| \leq L_z.$$

Measurable divided differences first change y at fixed z, then change z; set the relevant coefficient to zero when a denominator vanishes. The Lipschitz bounds

imply the displayed coefficient bounds. Bounded ϑ satisfies Novikov's condition. Under the measure whose density is $\exp\{\int_0^T \vartheta\, dW - \frac{1}{2}\int_0^T \|\vartheta\|^2 ds\}$, $W - \int \vartheta ds$ is Brownian. Multiplication by $\exp(\int_0^s A_v dv)$ and expectation give

$$\Delta Y_0 = \mathbb{E}^{\vartheta}\Big[e^{\int_0^T A_v dv}(g - u(T))(X_T) + \int_0^T e^{\int_0^s A_v dv}\mathcal{H}^{a_s}u(s, X_s)\, ds\Big].$$

Here is the integrability step. The bounded drift has an exponential density with all finite moments. Cauchy–Schwarz transfers the $L^2(P)$ supremum of ΔY and the $L^2(P)$ square-root quadratic variation of ΔZ to $L^1(P^{\vartheta})$. After multiplication by the bounded finite-horizon factor, the localized stochastic integral is therefore an $H^1(P^{\vartheta})$ martingale. The terminal and residual terms are dominated by their bounded or integrable deterministic envelopes. The localized identity consequently passes to expectation and the limit, rather than assuming the stochastic integral has mean zero without an integrability check.

The multiplier is positive and at most $e^{L_y s}$. The all-action lower residual and terminal lower error give

$$Y_0^a \geq u(0,x) - e^{L_y T}\eta_- - \int_0^T e^{L_y s} e_-(s)\, ds$$

for every admissible control. The policy-specific upper assumptions give the analogous upper bound for Y_0^{π}. Take the infimum in the first inequality and subtract. This proves (26); admissibility of π proves nonnegative loss. *Q.E.D.*

Research interface, not objective substitution. An NBO critic and actor for this declared problem must be checked against the original driver, the full graph $z_u = \sigma^{\mathsf{T}}\nabla u$, the all-action versus implemented-action residuals, and every terminal or stopping-boundary discrepancy. The proposition does not prove uniform approximation by a neural architecture, convergence of a training optimizer, discretization consistency, or a contraction for a fully coupled NDU FBSDE. A latent-preference application would first establish its Markov augmentation, information filtration, driver regularity and admissible policy class. The same economic error accounting

can then be developed for that problem; investment-catalogue results cannot be transferred to it by renaming a value function.

10. WELL-POSED ECONOMIC OPERATORS AND IMPLEMENTED POLICIES

This section supplies sufficient conditions and proofs for the existence, measurability, uniqueness, and stability obligations used in the paper. These are standard dynamic-programming foundations, stated explicitly rather than claimed as a new numerical contribution (Ruszczyński, 2010, 2014). They apply directly to the original investment primitives below. A different recursive utility, information filtration, or noncompact state model requires its own corresponding hypotheses; it does not inherit these properties by being described as a Bellman problem.

10.1. *A sufficient finite-horizon model*

Let X and A be compact metric spaces. At each finite date $j < T$, let $A_j : X \rightrightarrows \mathsf{A}$ have nonempty compact values and be both upper and lower hemicontinuous. Its graph is denoted G_j. Let the innovation space Z_j be compact metric with fixed Borel probability P_j. The original transition $F_j : G_j \times \mathsf{Z}_j \to X$ is jointly continuous and invariant in X. The cost $c_j : G_j \to \mathbb{R}$ and terminal payoff $g : X \to \mathbb{R}$ are continuous. The finite discounts $\beta_j \geq 0$ need not be smaller than one.

Let $\rho_j : L^\infty(P_j) \to \mathbb{R}$ be finite, monotone, cash invariant, and normalized by $\rho_j(0) = 0$. Without normalization, the identical original operator is obtained using $\bar{\rho}_j = \rho_j - \rho_j(0)$ and $\bar{c}_j = c_j + \beta_j \rho_j(0)$; thus the finite deterministic constant remains in the original stage account. For implemented Borel policies we also require the following parameter-measurability property:

$$(s,z) \mapsto w(s,z) \text{ is bounded and Borel} \quad \Longrightarrow \quad s \mapsto \rho_j(w(s,\cdot)) \text{ is Borel} \tag{27}$$

for every standard Borel parameter space and bounded Borel w. This property is verified below for the original expectation and AVaR primitives; it is not deduced from cash invariance alone.

The filtration is generated by the successive original innovations, with their stipulated fixed conditional laws, and the initial state is observed. An admissible history policy is a nonanticipative Borel action rule on this history, feasible at its

current original state. Its criterion is the *specified conditional nested recursion*: at a history, apply ρ_j to the next-innovation section of its continuation. This specifies the time-consistent original problem. Static AVaR of total cost and a randomization device with a different observation convention are different criteria and are not inserted into this policy class without being declared.

PROPOSITION S.10—Finite-horizon existence, selection, and validation: *Under these conditions, the original Bellman recursion has one bounded continuous value sequence* $V_T = g$,

$$V_j(x) = \min_{a \in A_j(x)} \{c_j(x,a) + \beta_j \rho_j(V_{j+1}(F_j(x,a,\cdot)))\}.$$

Each minimum is attained and admits a Borel selector. The resulting Markov policy is optimal among all admissible history policies for the stipulated recursion. Every implemented Borel feasible policy has a well-defined bounded Borel value sequence; its value is at least V_j *at every date and original history with state* x.

Uniqueness here concerns the value sequence. An optimizer need not be unique. If each action objective has a unique minimizer, that minimizer is continuous in x. *In particular, a strongly convex objective on a compact convex feasible action set has a unique minimizer.*

For the same primitives, two bounded continuous terminal continuations satisfy

$$\|V_j - \widetilde{V}_j\|_\infty \le \left(\prod_{k=j}^{T-1} \beta_k\right) \|g - \widetilde{g}\|_\infty. \tag{28}$$

For a fixed implemented policy the same bound holds for bounded Borel terminal continuations, whose recursions are well-defined by (27). Additive stage-cost discrepancies of at most e_k *add their dated discounted sum to the right side. Changes in a law, transition, constraints, or preference require their own operator discrepancy; they are not covered by this same-primitives statement.*

PROOF: If $\|v - w\|_\infty \le h$, then $w - h \le v \le w + h$. Monotonicity and cash invariance give

$$|\rho_j(v) - \rho_j(w)| \le h, \qquad |\rho_j(v)| \le \|v\|_\infty.$$

This also applies to equivalence classes under P_j; a stronger pointwise bound is sufficient. The feasible graph is closed in a compact product and hence compact. For continuous v, joint continuity on $G_j \times \mathsf{Z}_j$ implies uniform continuity. Consequently $(x, a) \mapsto v(F_j(x, a, \cdot))$ is continuous in the supremum norm. The displayed nonexpansiveness proves continuity of its risk evaluation, and thus of the full action objective.

Compactness of $A_j(x)$ gives an attained minimum. To prove value continuity without hiding a selection assumption, let $x_n \to x$. Along a subsequence attaining the lower limit of the values, compactness gives a convergent subsequence of minimizers $a_n \to a$. Upper hemicontinuity gives $a \in A_j(x)$, and continuity of the objective implies that the lower limit is at least $V_j(x)$. Conversely, choose a minimizer a at x. Lower hemicontinuity provides $a_n \in A_j(x_n)$ with $a_n \to a$; evaluating these feasible actions shows that the upper limit is at most $V_j(x)$. Thus the value is continuous. Backward induction from g proves existence and continuity of the complete value sequence; the finite recursion also proves its uniqueness. The norm estimate $\|V_j\|_\infty \leq \|c_j\|_\infty + \beta_j \|V_{j+1}\|_\infty$ proves boundedness.

Here is an explicit measurable-selector argument. The argmin graph is closed, because both the objective and value are continuous, and its fibers are nonempty compact. For each n, choose a finite cover of A by closed balls of radius 2^{-n} with fixed centers. In a fixed ordering choose the first ball meeting the argmin fiber, and at each successive level choose the first ball meeting its intersection with the previously chosen balls. All these intersections remain nonempty compact and are nested. The event that the argmin fiber meets a fixed finite intersection of closed balls is closed: it is the projection onto compact X of a closed subset of $X \times \mathsf{A}$. Induction over the finite-valued previous choices therefore makes each new chosen center a Borel function of x. The nested sets have diameters at most 2^{1-n}, so their intersection contains exactly one action. The chosen centers converge to that action; their pointwise limit is Borel and belongs to the argmin fiber. This constructs the required selector. When the argmin is already unique, the same subsequence argument used for value continuity shows continuity of the selector.

For an arbitrary implemented history policy, composition of Borel histories, actions, and the original transition gives a bounded Borel next-history kernel. Prop-

erty (27) then constructs its bounded Borel value backward from g. Suppose at the next date that the policy continuation is at least V_{j+1} at its next original state. Applying the same conditional risk at the current history preserves this inequality. The resulting current cost is at least the action objective at its feasible chosen action, hence at least V_j. Backward induction proves policy domination, without restricting the comparison to Markov policies. Using the Borel minimizing selector at every date makes every inequality an equality, which proves its optimality.

Finally, the preceding nonexpansiveness yields $\|\mathcal{T}_j v - \mathcal{T}_j w\|_\infty \leq \beta_j \|v - w\|_\infty$, by comparing infima over the same actions. A fixed policy has the identical inequality without the infimum. Backward induction proves (28). When the costs differ by at most e_j, the one-step estimate has the extra term e_j; iterating it gives the dated sum. *Q.E.D.*

The exact continuity and measurability obligations. Compact innovations can be replaced by essential-uniform continuity of $v(F_j(x,a,\cdot))$ on the declared continuation class. Pointwise continuity alone is insufficient for an arbitrary monetary risk. For example, let Z have an unbounded exponential law, $F(x,z) = 1 - e^{-xz}$ for $x \in [0,1]$, terminal value $g(y) = y$, and $\rho = \operatorname{ess\,sup}$. All these primitives are continuous and bounded where required, but $\rho(g(F(0,Z))) = 0$ and $\rho(g(F(x,Z))) = 1$ for every $x > 0$. Expectation has stronger dominated-continuity properties, but those cannot be assigned to every monotone cash-invariant risk.

For the original fixed-law AVaR and bounded jointly Borel w, the formula

$$\rho_\alpha(w(s,\cdot)) = \inf_{\eta \in \mathbb{Q}} \left\{ \eta + \frac{1}{1-\alpha} \int (w(s,z) - \eta)_+ \, P(dz) \right\}$$

proves (27). Each integral is Borel: the claim holds on measurable rectangles, extends to indicators of product-Borel sets by a monotone-class argument, and then to nonnegative Borel functions by monotone simple approximation. The infimum is countable and equals the real-threshold infimum because the displayed expression is continuous in η. Expectation is the same measurability argument without an infimum. Thus the needed property is established for both original preference cases, while remaining explicit for a different risk primitive.

10.2. *The original investment domain and unique constrained actions*

PROPOSITION S.11—Original primitive closure and actor regularity: *For the original even-dimensional investment primitives, every feasible state-action-innovation triple satisfies*

$$F(x,a,z) \in [1/32, 27/32]^d \subset [0,1]^d. \tag{29}$$

The capacity correspondence $[0, A(x)]$ is continuous, compact, and nonempty, with $1/8 \leq A(x) \leq 1/4$. Costs are continuous and nonnegative, with valid uniform bounds $S \leq 49/32$, $g \leq 37/16$, and $S + h \leq 103/64$. Hence the finite-horizon existence and policy theorem applies for the original expectation and nested AVaR criteria.

For the two-date AVaR model with $\beta = 15/16$, V_1 is strongly convex in state with modulus $(4 - 5\beta/4)/d$, and the original root and last-date action objectives are strongly convex in action with modulus two. They have unique continuous constrained optimal actors. The last-date V_1 and the root action objective are continuously differentiable on the relative state interior, with the corresponding one-sided boundary interpretation.

For either date where Q_j is continuously differentiable, write $a_j^*(x)$ for its constrained minimizer and

$$\lambda_j(x) = \mathbf{1}_{\{a_j^* = A(x)\}}[-\partial_a Q_j(x, a_j^*)]_+.$$

Then the value gradient is

$$\nabla V_j(x) = \nabla_x Q_j(x, a_j^*(x)) - \lambda_j(x)\nabla A(x). \tag{30}$$

This formula is not a derivative of a rounded deployed actor. Interior actor derivatives require the additional local second derivatives and strict-interiority conditions stated in the retained compiler theorem.

PROOF: For the state part of the jth transition coordinate put $p = x_j$, $q = x_{j+1}$. The function $p/2 + q/8 + p(1-q)/16$ is nondecreasing in both arguments on $[0,1]^2$, because its two partial derivatives are $1/2 + (1-q)/16$ and $1/8 - p/16$. Its range lies in $[0, 5/8]$. Since $0 \leq b_j a \leq 1/8$ and $|z| \leq 1/32$, adding the constant $1/16$ proves (29). The affine capacity gives the stated correspondence properties. The inequalities $|x_j - 5/8| \leq 5/8$, $|x_j - x_{j+1}| \leq 1$, and $(1/2 - 2\bar{x})_+ \leq 1/2$ give the

stated state-cost bounds. Moreover $h(a) \le h(1/4) = 5/64$, which gives the full stage bound. Polynomial transitions, squared positive parts, and the fixed compact innovation interval have the required continuity. The previous measurability proof applies to the risk primitive.

For completeness, the inherited joint curvature bound follows directly from the original primitives. On each smooth region, $\nabla^2 S \succeq (4/d) I$, $\nabla^2 g \succeq 0$, and $|\partial_j g| \le 10/d$. The only nonzero second derivative of the transition is the mixed derivative of $-x_j x_{j+1}/16$. In a joint state-action direction (v, w), the Hessian chain rule therefore gives

$$D^2(g \circ F)[(v,w)]^2 \ge -\frac{1}{8} \sum_j \partial_j g(F) v_j v_{j+1} \ge -\frac{5}{4d} \|v\|_2^2.$$

The omitted Jacobian–Hessian–Jacobian term is nonnegative in the full joint direction, including its state-action mixed terms. Together with $h'' \ge 2$, this proves for every fixed shock the lower curvature

$$\frac{4 - 5\beta/4}{d} \|v\|_2^2 + 2w^2.$$

The squared-hinge derivatives match at the junction, so integrating the almost-everywhere second-derivative inequality along a segment extends it across the junction. Integration under any feasible AVaR density preserves the bound. Subtracting its common quadratic term leaves convex functions; their supremum is convex, so the risk supremum preserves the same bound. It follows that the joint objective has the state term $(4 - 5\beta/4)\|x - y\|_2^2/(2d)$ in its strong-convexity inequality. Interpolating two optimal state-action pairs preserves feasibility because the capacity is affine. Taking the value at that feasible interpolation and dropping the nonnegative action-distance term proves strong convexity of V_1 with the announced modulus. This supplies the state convexity used in the active-tail theorem, rather than assuming it for a fitted continuation.

For each fixed root state, the successor is affine in its scalar action. Convexity of V_1, followed by monotonicity and convexity of AVaR, makes $a \mapsto \rho_\alpha(V_1(F(x, a, \cdot)))$ convex. Adding $h'' \ge 2$ proves root-action strong convexity; the same property of the last-date objective follows from the original primitive curvature. The preceding

compact theorem now gives existence, uniqueness, and continuity of both optimal actors.

The retained exact last-date formula is C^1 in state and action: the positive-part square and its explicit tail polynomial have matching values and first derivatives at their junctions. Normalize the action by $a = A(x)v$, with $v \in [0,1]$. The normalized objective has a unique continuous minimizer. Comparing the value increment at the old optimizer and at the new optimizer, and using uniform first-order expansion on this fixed compact action interval, gives

$$\nabla V_1 = \nabla_x Q_1 + v^* \partial_a Q_1 \nabla A.$$

At a lower endpoint $v^* = 0$, at an interior optimum $\partial_a Q_1 = 0$, and at capacity $v^* = 1$, $\partial_a Q_1 \le 0$. This proves (30) and continuity of the value gradient. In particular the multiplier tends to zero at a switch between an interior and a binding-capacity solution.

For the root objective, invariance puts every successor strictly inside the state cube, where ∇V_1 is continuous and bounded on the compact successor range. Along the innovation, $V_1(F(x,a,z))$ is strongly convex with modulus $181/64$. Every one of its level sets has at most two points, so its uniform-law distribution has no atoms. Continuity of the function on the interval makes its distribution function strictly increasing between its minimum and maximum: the preimage of every nonempty open value interval in that range contains a nonempty open innovation interval. Its quantile is therefore unique at the declared risk levels. The differentiability conditions of the retained AVaR envelope argument apply, with bounded action/state derivatives. Equivalently, the active central interval has a unique continuous minimizer and the value envelope differentiates the integral with that interval held fixed. Thus, writing $E_{x,a}$ for the optimal two-tail set,

$$\partial_a Q_0(x,a) = h'(a) + \frac{\beta}{2R(1-\alpha)} \int_{E_{x,a}} b^\mathsf{T} \nabla V_1(F(x,a,z))\, dz.$$

The analogous state derivative has $D_x F^\mathsf{T} \nabla V_1$ and ∇S. Continuity of the integrands and tail endpoints gives continuity of these derivatives. Applying the same normal-

ized action envelope proves (30) for V_0. Endpoint actions are included; no global differentiability of the actor across active-set changes is asserted. *Q.E.D.*

Observation, feasibility, and terminal-policy transfer. The implemented terminal controller observes a downward 40-bit state $y \le x$ with $\|x - y\|_1 \le d2^{-40}$ and returns an action feasible at y. Since $A(y) \le A(x)$, it remains feasible at the original state. Its certified action gap at y is at most 2^{-21}. The primitive fixed-action objective and the optimal value both have state Lipschitz constant at most $15/d$, so its original-state gap is at most

$$2^{-21} + 30\,2^{-40} < 2^{-20}.$$

The fixed-action modulus follows from $\|\nabla S\|_\infty \le 8/d$, the transition column-sum bound $11/16$, $\|\nabla g\|_\infty \le 10/d$, and $8 + \beta(11/16)10 < 15$. The value modulus is the retained capacity-aware bound. The controller's rounding and exact fallback must actually enforce its premise; an optimizer return code alone does not do so. This explains why its allowance remains valid when the new method changes the locations of original-law integration queries while retaining the same deployed controller.

10.3. *Residual validation and the optional infinite-horizon case*

For the finite-horizon one-sided sandwich, suppose $f_j - \mathcal{T}_j f_{j+1} \le u_j$, $\mathcal{T}_j^\pi f_{j+1} - f_j \le v_j$, and $\ell_T \le f_T - g \le u_T$. Monotonicity and cash invariance give separately

$$f_0 - V_0 \le \sum_{j<T} B_j u_j + B_T u_T, \qquad J_0^\pi - f_0 \le \sum_{j<T} B_j v_j - B_T \ell_T.$$

Each follows by backward induction, starting from its corresponding terminal inequality. Adding them proves the retained complete-policy bound, and feasibility supplies $J_0^\pi \ge V_0$. This validates an NBO's own attained continuation without assuming optimal-continuation training labels. Date-specific constant shifts of f_j cancel in the paired residuals and in $u_T - \ell_T$.

For a stationary infinite-horizon model with the same compact and kernel conditions, bounded cost, and $0 \le \beta < 1$, the Bellman map is a β-contraction on the

complete space $C(X)$. The finite-horizon continuity proof shows that it maps this space into itself. Thus it has a unique continuous fixed point V. For any Borel feasible stationary policy, the policy map is a β-contraction on bounded Borel functions by (27); it has a unique bounded Borel fixed point J^π. A Borel minimizer of the continuous action objective at V satisfies $T_{\pi^*}V = TV = V$, so uniqueness of its policy fixed point gives $J^{\pi^*} = V$. For another policy, $V \leq T_\pi V$; monotone iteration and contraction give $V \leq J^\pi$.

The infinite-history criterion is well-defined as the limit of the same finite nested recursions: the omitted cost after n dates is bounded uniformly by $\|c\|_\infty \beta^n/(1-\beta)$. Passing the finite-horizon comparison to this limit proves optimality against the same admissible history-policy class. This argument does not assert that infimizing arbitrary Borel action objectives preserves Borel measurability. The more general policy-iteration theorem in the retained supplement explicitly assumes that additional preservation and selection property when it applies a Bellman infimum to possibly discontinuous policy values.

10.4. *The critic and actor computational graph*

For a smooth parameterized critic and differentiable feasible actor, the derivative must correspond to the objective actually optimized. For example, at a fixed original state write

$$r_\theta = c(x, a_\theta) + \beta\rho(f_{+,\theta}(F(x, a_\theta, Z))) - f_\theta(x).$$

Suppose differentiation is justified by integrable local domination, and either ρ is expectation or it is AVaR at an atomless unique quantile, as in the retained tail-gradient theorem. Let q_θ be its maximizing tail density, with $q_\theta = 1$ for expectation. The chain and risk-envelope rules give

$$\nabla_\theta r_\theta = (D_\theta a_\theta)^\mathsf{T} \left[\nabla_a c + \beta \mathbb{E}_{q_\theta}(D_a F)^\mathsf{T} \nabla_x f_{+,\theta}(F)\right] + \beta \mathbb{E}_{q_\theta} \nabla_\theta f_{+,\theta}(F) - \nabla_\theta f_\theta(x).$$

The last expectation is a partial parameter derivative at a fixed successor. The actor contribution is written separately to avoid double counting. A risk-envelope deriva-

tive holds the maximizing density fixed in this formula; its validity comes from the stated regularity assumptions, not from differentiating a sorting permutation. The half-squared-residual objective $\mathcal{L}(\theta) = \frac{1}{2}\mathbb{E}[r_\theta^2]$ has gradient $\mathbb{E}[r_\theta \nabla_\theta r_\theta]$ when its own domination condition holds. Detaching the future target defines a different frozen-target gradient or a semigradient algorithm. It does not establish that this full original coupled gradient was used, and neither formula proves optimizer convergence.

At a projection or rounding discontinuity these differentiability premises may fail. Piecewise training derivatives can be specified as an algorithmic choice, but the global economic certificate still uses the actual feasible action and its original-law inequalities. This is consistent with the own-future constructive theorem, whose validity does not depend on a successful differentiable optimizer.

11. STOPPING BOUNDARIES AND THE CONTINUOUS-TIME RECURSION

The retained Brownian Markov bridge is a separate classical recursive utility verification theorem (Pardoux and Peng, 1990, El Karoui et al., 1997). Its forward coefficients and driver are understood to be jointly Borel in their displayed deterministic arguments, so composition with each admissible progressively measurable control yields progressively measurable coefficients. The forward state-Lipschitz constants and driver (y,z)-Lipschitz constants are uniform in the control. Terminal payoff and the driver at $(y,z) = (0,0)$ are bounded. A feedback must define its own well-posed closed-loop state process. These conditions make the retained existence proof precise; they do not establish existence for a fully coupled FBSDE or an arbitrary discontinuous feedback.

PROPOSITION S.12—Stopped original-BSDE verification: *Let $\tau^a \le T$ be a specified stopping time for each admissible control, such as exit from a declared domain capped at T, with bounded Borel original stopping payoff $G(\tau^a, X_{\tau^a})$. Before stopping use exactly the driver and forward coefficients of the retained BSDE bridge. Let bounded $u \in C^{1,2}$ have the bounded derivatives required for stopped Itô calculus and continuous values on the declared stopping boundary. Suppose*

$$\mathcal{H}^a u \ge -e_-(s) \text{ for every admissible action}, \qquad \mathcal{H}^\pi u \le e_+(s)$$

on the unstopped domain, and at every possible stopping location and at the terminal part of the boundary suppose

$$-\eta_- \le G(s,x) - u(s,x) \le \eta_+.$$

Here $e_\pm$ are nonnegative deterministic integrable functions and $\eta_\pm \ge 0$. The stopped BSDE has a unique solution in $S^2 \times H^2$, and

$$0 \le Y_0^\pi - \inf_a Y_0^a \le e^{L_y T}(\eta_- + \eta_+) + \int_0^T e^{L_y s}(e_-(s) + e_+(s))\, ds.$$

The stopping rule and boundary payoff are parts of the original economic criterion. A smaller interior residual does not remove their errors or permit them to be changed.

PROOF: Embed the stopped equation on $[0,T]$ with terminal variable $\xi^a = G(\tau^a, X_{\tau^a})$ and driver $\mathbf{1}_{\{s<\tau^a\}} f(s, X_s, y, z, a_s)$. It is progressively measurable, has the same uniform Lipschitz constants, and has square-integrable terminal data and bounded driver at zero. The weighted Picard argument in the retained proof therefore gives a unique $S^2 \times H^2$ solution. After τ^a it equals ξ^a with $Z = 0$; before stopping it is exactly the declared stopped equation.

Apply stopped Itô calculus to $u(s \wedge \tau^a, X_{s\wedge\tau^a})$. Subtracting gives the same linearized error BSDE as in the retained proof, with its coefficients set to zero after τ^a, terminal error $\xi^a - u(\tau^a, X_{\tau^a})$, and residual driver $\mathbf{1}_{\{s<\tau^a\}} \mathcal{H}^{a_s} u$. Its measurable coefficients satisfy $|A_s| \le L_y$ and $\|\vartheta_s\| \le L_z$. The bounded Girsanov coefficient has an exponential density with all finite moments. The same Cauchy–Schwarz and H^1 martingale argument used in the retained verification proof therefore justifies expectation of the stopped integration-factor identity,

$$Y_0^a - u(0,x) = \mathbb{E}^\vartheta\left[e^{\int_0^{\tau^a} A_v dv}(G-u)(\tau^a, X_{\tau^a}) + \int_0^{\tau^a} e^{\int_0^s A_v dv}\mathcal{H}^{a_s} u(s, X_s)\, ds\right].$$

The factor is positive and at most $e^{L_y s}$. Applying the lower residual and boundary bounds gives a lower bound for every admissible control. Applying the upper bounds gives an upper bound for the implemented policy. Since $\tau^a \le T$, the deterministic integrals and terminal factors are bounded by those displayed in the

proposition. Taking the infimum of the lower bounds and subtracting proves the result. *Q.E.D.*

Why the boundary and graph are material. For a deterministic state starting at $1/2$ in $(0,1)$, use either of the two constant controls $a = \pm 1$ in $dX_s = a\,ds$, stop on exit before $T = 1$, and assign boundary costs zero at the left endpoint and one at the right. With zero driver and $u = 0$, every interior residual is zero, but the right-exit policy has regret one. The stopping-boundary error exactly supplies the missing term. This is a verification counterexample, not a second numerical application.

The BSDE graph is $z_u = \sigma^\top \nabla u$ for the same control and the original driver. Its $C^{1,2}$ requirement is substantive. A piecewise-affine ReLU critic is generally not C^2 and cannot be inserted into this classical formula by declaring its second derivative zero away from its kinks; a different weak, viscosity, or smoothing argument with the required error account would first be needed. The finite-horizon ReLU Bellman construction uses order and Lipschitz arguments and does not rely on this continuous-time smooth verification theorem.

Finally, a general recursive utility is not automatically cash invariant. Even the globally Lipschitz BSDE driver $f(s,y,z) = \lambda y$ gives $Y_0 = e^{\lambda T}\mathbb{E}\xi$, so adding a constant c to the terminal criterion changes Y_0 by $e^{\lambda T}c$, not generally by c. The continuous-time bridge correctly uses its $e^{L_y T}$ stability factors. It does not inherit the monetary discrete-risk contraction or identify an arbitrary NDU/FBSDE with the original investment AVaR objective.

12. INDEPENDENT FINITE ENCLOSURES FOR THE ORIGINAL POLICY CATALOGUE

This section supplies the mathematical justification for validation D. It is a finite verification procedure for the original investment objective, using classical convexity and the AVaR threshold representation (Rockafellar and Uryasev, 2002, Boyd and Vandenberghe, 2004). It is not a new optimization theorem, an independent population experiment, or a claim of neural exclusivity. The implementation reconstructs the original economic primitives separately from the production terminal compiler. Its mathematical guarantee still uses the original domain, continuation-convexity,

and implemented-terminal-policy bounds; software independence does not remove these assumptions.

12.1. *Original objective and precise validation statement*

Fix $d = 2$, $X = [0,1]^2$, $R = 1/32$, $\beta = 15/16$, and $\alpha \in \{1/2, 3/4\}$. Put $t = 1 - \alpha$ and let P be the uniform law on $[-R, R]$. Indices below are cyclic. The original primitives are

$$A(x) = \frac{1}{8} + \frac{x_1 + x_2}{16}, \qquad h(a) = a^2 + 4a^4,$$

$$F_j(x, a, z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + b_j a + s_j z,$$

$$b = (1/2, 1/4), \quad s = (1, -1),$$

$$S(x) = \sum_{j=1}^{2} (x_j - 5/8)^2 + \frac{1}{8} \sum_{j=1}^{2} (x_j - x_{j+1})^2 + 2(1/2 - x_1 - x_2)_+^2,$$

$$S_T(x) = 2 \sum_{j=1}^{2} (x_j - 5/8)^2 + \frac{1}{8} \sum_{j=1}^{2} (x_j - x_{j+1})^2 + 2(1/2 - x_1 - x_2)_+^2.$$

For bounded w, write

$$\rho_\alpha(w) = \inf_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{t} \mathbb{E}_P (w - \eta)_+ \right\}.$$

The last-date optimal value and the two root fixed-action objectives are

$$G_y(v) = S(y) + h(v) + \beta \rho_\alpha(S_T(F(y, v, \cdot))),$$

$$V_1(y) = \min_{0 \le v \le A(y)} G_y(v),$$

$$Q_x(a) = S(x) + h(a) + \beta \rho_\alpha(V_1(F(x, a, \cdot))),$$

$$J_x(a) = S(x) + h(a) + \beta \rho_\alpha(J_1(F(x, a, \cdot))).$$

Here J_1 is the unchanged implemented terminal policy, not a newly optimized or independently reconstructed controller. The original results in Proposition S.11 establish domain invariance, existence, and strong state convexity of V_1 with modulus $(4-5\beta/4)/2$. The implemented terminal guarantee is

$$V_1(y) \le J_1(y) \le V_1(y) + \varepsilon_1, \qquad y \in X, \qquad \varepsilon_1 = 2^{-20}. \tag{31}$$

The validation below relies on this proved uniform guarantee and makes no convexity assumption about J_1.

PROPOSITION S.13—Finite original-policy validation enclosure: *Fix an original state $x \in X$ and feasible action $a \in [0, A(x)]$. Let $M \ge 1$ be an integer and put*

$$z_k = -R + \frac{Rk}{M}, \qquad y_k = F(x, a, z_k), \qquad k = 0, \dots, 2M.$$

At each y_k, choose any feasible v_k and an exactly maximizing AVaR density q_k for $S_T(F(y_k, v_k, \cdot))$. Define

$$\begin{aligned} g_k &= h'(v_k) + \beta \mathbb{E}_P[q_k \, \partial_v S_T(F(y_k, v_k, Z))], \\ \delta_k &= \mathrm{proj}_{[-v_k, A(y_k)-v_k]}(-g_k/2), \\ \ell_k &= G_{y_k}(v_k) + g_k \delta_k + \delta_k^2, \qquad\qquad u_k = G_{y_k}(v_k). \end{aligned}$$

These are valid bounds $\ell_k \le V_1(y_k) \le u_k$, including boundary actions. For an n-vector w, let $\rho_\alpha^{[n]}(w)$ denote AVaR under equal atom weights $1/n$. Form the vectors

$$\ell^{\mathrm{mid}} = (\ell_1, \ell_3, \dots, \ell_{2M-1}), \qquad u^{\mathrm{end}} = (u_0, u_2, u_2, u_4, \dots, u_{2M-2}, u_{2M}).$$

Thus each cell contributes both endpoints to u^{end}; its interior endpoints occur twice. With $c = S(x) + h(a)$, define

$$L_Q = c + \beta \rho_\alpha^{[M]}(\ell^{\mathrm{mid}}), \qquad U_Q = c + \beta \rho_\alpha^{[2M]}(u^{\mathrm{end}}). \tag{32}$$

Then the original objectives satisfy

$$L_Q \le Q_x(a) \le U_Q, \qquad L_Q \le J_x(a) \le U_Q + \beta\varepsilon_1. \tag{33}$$

Every quantity in the construction is rational when the original state, action, and primitive-node actions are rational. A finite terminal search affects the widths, not the validity, of these inequalities.

12.2. *Proof: exact original primitives and boundary-safe node bounds*

PROOF: We first verify that all evaluations concern the original feasible economy. For $p, q \in [0,1]$, the state part $p/2 + q/8 + p(1-q)/16$ is increasing in both arguments and lies in $[0, 5/8]$. Since $A(x) \in [1/8, 1/4]$, $0 \le b_j a \le 1/8$, and $|s_j z| \le 1/32$, every successor belongs to $[1/32, 27/32]^2 \subset X$. This applies both to a root action and to each feasible terminal action. The continuous terminal objective attains its minimum on the compact feasible interval.

Fix a terminal state y and feasible v. Write $F(y, v, z) = (w_1 + z, w_2 - z)$. The aggregate-state squared hinge in S_T is constant in z because the two shock loadings sum to zero. The first two squared deviations contribute $4z^2$ and the cyclic coupling term contributes z^2. Consequently

$$p(z) := S_T(F(y, v, z)) = p_0 + p_1 z + 5z^2.$$

This is a primitive identity. Its evaluation at $-R, 0, R$ determines the polynomial exactly; three-point interpolation would not identify a general nonlinear function. At rational inputs, all three coefficients are rational.

Here is an explicit exact maximizing density. Set $L = 2R\alpha$ and

$$u = \mathrm{proj}_{[-R, R-L]}\left(-\frac{p_1}{2p_2} - \frac{L}{2}\right), \quad p_2 = 5, \qquad q(z) = t^{-1}\mathbf{1}_{[-R,u]\cup[u+L,R]}(z).$$

The central interval has length L, so $0 \le q \le t^{-1}$ and $\mathbb{E}_P q = 1$. To verify maximality directly, let $\eta = \max\{p(u), p(u+L)\}$. An interior cut is centered on the quadratic's vertex, so its endpoint values coincide and $p \le \eta$ inside the central interval and $p \ge \eta$ on its two tails. At the left boundary the clipped vertex lies weakly to the left of the central midpoint, $p(u+L) \ge p(u)$, and the only nonempty tail is to the right of $u + L$; it again has $p \ge \eta$. The right-boundary case is symmetric. Thus, up

to endpoints of probability zero,

$$\eta + t^{-1}\mathbb{E}_P(p-\eta)_+ = \mathbb{E}_P(qp).$$

For any feasible density $\widetilde{q}$ and any threshold e,

$$\mathbb{E}_P(\widetilde{q}p) = e + \mathbb{E}_P[\widetilde{q}(p-e)] \le e + t^{-1}\mathbb{E}_P(p-e)_+.$$

Taking an infimum in e proves weak duality; the displayed equality proves that q attains the original AVaR. Integrating the quadratic over its two rational tail intervals gives the exact rational risk. This proof includes clipped cuts and does not differentiate a moving cut.

The function S_T is convex and continuously differentiable, including at the squared-hinge junction. For any other feasible terminal action w, its supporting inequality along the affine action direction gives

$$S_T(F(y,w,z)) \ge S_T(F(y,v,z)) + \partial_v S_T(F(y,v,z))(w-v).$$

Use the same fixed feasible density q at every w. Weak duality and its equality at v, followed by integration of this support, show

$$\rho_\alpha(S_T(F(y,w,\cdot))) \ge \rho_\alpha(S_T(F(y,v,\cdot))) + \mathbb{E}_P[q\,\partial_v S_T(F(y,v,Z))](w-v).$$

Since $h'' \ge 2$, adding the original stage term proves the global action minorant

$$G_y(w) \ge G_y(v) + g(w-v) + (w-v)^2, \qquad 0 \le w \le A(y). \tag{34}$$

The action derivative inside the expectation is affine in z: the quadratic parts of S_T have affine gradients, while the squared-hinge gradient is constant in z along the zero-aggregate shock. Evaluations at $0, R$ therefore reconstruct this affine derivative exactly. No derivative of the optimal tail endpoints is used.

The minimum of $g\delta + \delta^2$ over the full feasible displacement interval is attained at $\delta = \mathrm{proj}_{[-v,A(y)-v]}(-g/2)$. Minimizing the minorant gives the stated lower bound; evaluating the feasible action v gives the upper bound. This reasoning applies if $v = 0$, $v = A(y)$, or an imperfect interior search iterate. In particular, a nonnegative support slope at zero certifies that endpoint and a nonpositive support slope at

capacity certifies that endpoint. At an interior point, a negative support slope rules out a minimizer to its left, and a positive support slope rules out one to its right. Thus rational bisection is legitimate, but the support bound remains sound after any finite number of steps. The separately recorded check $u_k - \ell_k \le 2^{-20}$ is a width check; it is not the justification of the lower bound.

The continuous-risk outer enclosure. The original strong state convexity of V_1 implies convexity of $f(z) = V_1(F(x,a,z))$, because the root transition is affine in z. For every real threshold η, the function $\phi_\eta(z) = (f(z) - \eta)_+$ is convex. On the cell $[z_{2j}, z_{2j+2}]$, Jensen's inequality and convex secant domination give, with cell length $\Delta = 2R/M$,

$$(\ell_{2j+1} - \eta)_+ \le \phi_\eta(z_{2j+1}) \le \frac{1}{\Delta}\int_{z_{2j}}^{z_{2j+2}} \phi_\eta(z)\,dz,$$

$$\frac{1}{\Delta}\int_{z_{2j}}^{z_{2j+2}} \phi_\eta(z)\,dz \le \frac{\phi_\eta(z_{2j}) + \phi_\eta(z_{2j+2})}{2} \le \frac{(u_{2j} - \eta)_+ + (u_{2j+2} - \eta)_+}{2}.$$

Summing cells with weights $1/M$ and adding the threshold yields

$$\eta + \frac{1}{Mt}\sum_{j=0}^{M-1} (\ell_{2j+1} - \eta)_+ \le \eta + t^{-1}\mathbb{E}_P(f - \eta)_+,$$

$$\eta + t^{-1}\mathbb{E}_P(f - \eta)_+ \le \eta + \frac{1}{2Mt}\sum_{j=0}^{M-1} \{(u_{2j} - \eta)_+ + (u_{2j+2} - \eta)_+\}.$$

These hold at every common threshold. Taking infima over all real thresholds preserves each inequality and proves

$$\rho_\alpha^{[M]}(\ell^{\mathrm{mid}}) \le \rho_\alpha(f) \le \rho_\alpha^{[2M]}(u^{\mathrm{end}}).$$

Neither the lower nor the upper node array has to be convex. The required convexity is that of the original optimal continuation along the innovation. A bound only on the ordinary integral of f would not prove this risk enclosure. Adding c and multiplying by the positive β gives (32).

Finally, the threshold formula is monotone in its random argument, and replacing its threshold by $\eta - c_0$ proves $\rho_\alpha(w + c_0) = \rho_\alpha(w) + c_0$ for every constant c_0. Apply

these facts to (31) at the original successors to obtain

$$Q_x(a) \le J_x(a) \le Q_x(a) + \beta\varepsilon_1.$$

Combining with the Q bounds proves (33). The node-optimization uncertainty is already contained in ℓ_k, u_k. The distinct implemented-future allowance $\beta\varepsilon_1$ is added exactly once at the root. No convexity or differentiability of the rounded deployed future controller is asserted. *Q.E.D.*

12.3. *Exact atom handling and finite policy comparisons*

For completeness, the equal-mass risk in the proposition can be computed without floating cutoffs. Sort $w_{(1)} \ge \cdots \ge w_{(n)}$, put $r = n(1-\alpha) > 0$, $k = \lfloor r \rfloor$, and $\theta = r - k$. Then

$$\rho_\alpha^{[n]}(w) = \frac{\sum_{i=1}^{k} w_{(i)} + \theta w_{(k+1)}}{r}, \tag{35}$$

where the final term is omitted if $\theta = 0$, including $k = n$. Indeed, assign masses $\lambda_i \in [0,1]$ with $\sum_i \lambda_i = r$. The threshold inequality in the proof gives $r^{-1}\sum_i \lambda_i w_i \le \rho_\alpha^{[n]}(w)$. The assignment of unit mass to the k largest values and mass θ to the next value attains the threshold expression at the cutoff value. This proves the formula, including ties and split atoms. At $\alpha = 0$ it reduces to the ordinary mean; $\alpha = 1$ is outside this definition. At the two frozen risks and the frozen $M = 64$, both atom counts are integral. The implementation nevertheless uses exact rational masses and preserves the repeated endpoints.

Suppose a fixed catalogue contains n_0 original states and each complete policy r has a valid interval $[L_{jr}, U_{jr}]$ at state j. Summing its pointwise inequalities gives the deterministic mean bound

$$\frac{1}{n_0}\sum_j L_{jr} \le \frac{1}{n_0}\sum_j J_{jr} \le \frac{1}{n_0}\sum_j U_{jr}.$$

For two policies r, s, use the interval $[L_{jr} - U_{js}, U_{jr} - L_{js}]$ for their difference at state j. If they have the same exact action, the same original state and risk, and the identical future controller, the two original defining objectives are equal, and

that state's difference is exactly zero. Replacing its subtraction interval by $[0,0]$ is then an identity, not an independence assumption. Summing these statewise difference intervals proves the reported mean-contrast interval. A strictly negative upper endpoint certifies a lower cost for the left policy; a strictly positive lower endpoint certifies a higher cost. An interval containing zero remains unresolved, except that $[0,0]$ certifies equality. No statistical or population inference follows from these deterministic sums.

In D, exact stored binary64 identities are converted to their precise rational values. The input binding checks the common state across every action and service and the unchanged future-controller identity. The frozen catalogue has 144 states per risk, 983 distinct state/action/risk keys, and 13,824 primary policy records. The prescribed output includes all 96 arm/service/risk means, all 56 unordered arm contrasts with the fixed compiled-service labels, and all 240 within-arm service-pair identity cells. These are declared coverage requirements, not assertions that execution succeeded. No favorable sign is a pass condition.

12.4. *What interval compatibility does and does not establish*

Let I be an independently justified interval for an original policy value and P_0 a recorded service or fine-reference interval. If they are disjoint, both cannot enclose the same value: at least one enclosure argument, arithmetic step, input identity, or model premise requires investigation. The case must be retained. If they intersect, their intersection alone does not establish correctness of P_0; for example, $I = [0,2]$ and $P_0 = [1,3]$ overlap although a true value $1/2$ lies outside P_0. The stronger inclusion $I \subseteq P_0$ would transfer coverage from I to P_0, conditional on validity of I. The reverse inclusion does not. D records overlap and both inclusion directions separately and uses overlap only as a compatibility diagnostic. It does not thereby verify every endpoint of the old 1,024-cell reference, which shares the production terminal compiler.

The lower bound L_Q in (32) concerns the specified action. It does not by itself provide a valid lower bound on $\inf_a Q_x(a)$ and therefore creates no new global regret or action-distance certificate. The procedure leaves all original A/B/C policies, ob-

servations, timing records, negative results, and qualification proofs unchanged. Its full actual work is additional author-side validation cost. The subsequent execution receipt, rather than this mathematical proposition, determines whether every fixed case was actually completed and passed.

12.5. *Proof of the original comparison modulus*

PROOF OF THEOREM S.3: Throughout this proof $d = 2$. Let $\boldsymbol{b} = (1/2, 1/4)^\top$ and $\boldsymbol{s} = (1, -1)^\top$. For a terminal state y, normalize its feasible action by $a = A(y)v$, where $v \in [0,1]$ is held fixed when differentiating in y. Put

$$m(y,v) = F(y, A(y)v, 0), \qquad k(y,v) = 5\{m_1(y,v) - m_2(y,v)\}.$$

The original domain argument in Proposition S.11 puts m, and every $m + \boldsymbol{s}z$ used here, in the state cube. The zero-sum shock leaves the aggregate squared hinge unchanged. Direct expansion of the original terminal cost therefore gives

$$g(m + \boldsymbol{s}z) = g(m) + kz + 5z^2.$$

By the exact original tail formula in Theorem A.20, the normalized last-date objective is

$$\widetilde{Q}(y,v) = S(y) + h(A(y)v) + \beta\{g(m(y,v)) + \mathcal{R}_\alpha(k(y,v))\}.$$

The function $\mathcal{R}_\alpha$ is continuously differentiable and

$$\mathcal{R}'_\alpha(k) = \operatorname{clip}\left(\frac{\alpha k}{10t}, -\alpha R, \alpha R\right).$$

Its derivative has Lipschitz constant $\alpha/(10t)$, including the two clipping junctions.

We establish a uniform upper state-curvature bound for $\widetilde{Q}(\cdot, v)$. For $d = 2$, the target, cyclic, and squared-hinge terms in S have Hessian operator norms bounded by 2, 1, and 8, respectively. Thus $\nabla^2 S \preceq 11I$ wherever the second derivative is defined. The same decomposition for g gives $0 \preceq \nabla^2 g \preceq 13I$. Its gradient coordinates satisfy

$$|\partial_j g(y)| \le 4|y_j - 5/8| + \tfrac{1}{2}|y_j - y_{j+1}| + 4(1/2 - y_1 - y_2)_+ \le 5.$$

The state Jacobian of m is nonnegative on the cube. Its row sums are at most $3/4$, and its column sums at most $47/64$. Indeed the unnormalized transition has both sums at most $11/16$; the term $\boldsymbol{b}v\nabla A^{\top}$ adds at most $1/16$ to a row and $3/64$ to a column. Hence

$$\|D_y m\|_2 \le \sqrt{(3/4)(47/64)} \le 3/4.$$

The only second state derivatives of m_j are the mixed derivatives $-1/16$ of $-y_1y_2/16$; the normalized capacity term is affine. In a state direction w,

$$\left|\sum_{j=1}^{2} \partial_j g(m)\, D_y^2 m_j[w,w]\right| \le \frac{10}{8}|w_1w_2| \le \frac{5}{8}\|w\|_2^2.$$

The chain rule consequently gives the uniform upper bound

$$D_y^2(g\circ m)[w,w] \le \left\{13(3/4)^2 + 5/8\right\}\|w\|_2^2 = \frac{127}{16}\|w\|_2^2.$$

Also $h''(a) = 2 + 48a^2 \le 5$ on $[0,1/4]$ and $\|\nabla A\|_2^2 = 1/128$. The stage upper bound is therefore $11 + 5/128 = 1413/128$.

For the risk-motion term, the bilinear transition terms cancel in the difference of the two coordinates:

$$k(y,v) = \frac{35}{16}(y_1 - y_2) + \frac{5}{4}A(y)v.$$

This is affine in y for fixed v, and

$$\nabla_y k = (35/16 + 5v/64,\ -35/16 + 5v/64)^{\top},$$

$$\|\nabla_y k\|_2^2 = 2(35/16)^2 + 2(5v/64)^2 \le \frac{19625}{2048}.$$

It follows that the upper Hessian contribution of $\mathcal{R}_\alpha(k(y,v))$ is at most $19625\alpha/(20480t)$. Combining these estimates gives $D_y^2\widetilde{Q}[w,w] \le M_\alpha\|w\|_2^2$. Squared hinges and the tail formula have matching first derivatives at their junctions. Their restrictions to a line have absolutely continuous first derivatives, so integrating this almost-everywhere second-derivative bound along a segment proves that

$$y \longmapsto \widetilde{Q}(y,v) - \tfrac{1}{2}M_\alpha\|y\|_2^2$$

is concave on the cube, uniformly over $v \in [0,1]$.

An infimum of concave functions is concave: its value at a convex combination is at least the same combination of the two infima. Since the normalized feasible interval is fixed, this proves semiconcavity of $V_1(y) - M_\alpha \|y\|_2^2/2$. The retained original joint curvature and the affine capacity graph give convexity and a unique continuous minimizing $v(y)$. For clarity, continuous differentiability does not require differentiability of that minimizer. Evaluate a value increment at the old minimizer for one inequality and at the new minimizer for the opposite inequality. Uniform first-order expansion of the C^1 objective on the compact normalized interval and continuity of $v(y)$ give $\nabla V_1(y) = \partial_y \widetilde{Q}(y, v(y))$ in the interior. This formula extends continuously to the boundary, and incorporates the capacity term already displayed in (30).

Convexity together with M_α-semiconcavity now implies the claimed gradient modulus. One explicit justification is to mollify V_1 on any smaller interior cube. Convolution preserves convexity and semiconcavity, so the smooth Hessian has every eigenvalue between 0 and M_α. Integrating the Hessian along a segment bounds the difference of the mollified gradients by M_α times its length. The gradients converge locally uniformly because V_1 is C^1. Pass to the limit and then use continuity at the boundary.

Finally the two successor points at a fixed innovation differ by $\boldsymbol{b}(b-a)$. Differentiation in the innovation gives

$$\begin{aligned}\left|\frac{d}{dz}\{f_b(z) - f_a(z)\}\right| &\le \|\boldsymbol{s}\|_2 \|\nabla V_1(F(x,b,z)) - \nabla V_1(F(x,a,z))\|_2 \\ &\le M_\alpha \|\boldsymbol{s}\|_2 \|\boldsymbol{b}\|_2 |b-a| \\ &= \frac{\sqrt{10}}{4} M_\alpha |b-a| \le \frac{4}{5} M_\alpha |b-a|.\end{aligned}$$

All points and joining segments are in the invariant convex cube. Integration in z proves (8). *Q.E.D.*

13. EXACT PRIMITIVE VALIDATION OF ORIGINAL-OPTIMUM REGRET

The R119 independent catalogue enclosure bounded $Q_x(a)$ and $J_x(a)$ at fixed actions. The following additional construction also provides an independent lower bound on $\inf_a Q_x(a)$. It uses the retained original curvature proposition and explicit primitive gradients; it is a validation strengthening, not a new claim that a scalar fixed-action lower value already bounded the all-action optimum.

PROPOSITION S.14—An independent rational all-action witness: *Fix the original $d = 2$ model and a rational original state and root action. At a terminal anchor y, choose any feasible rational v and any feasible original tail density q. Let W be the exact terminal objective under this fixed density, and let (p_y, p_v) be its state and action gradients. For arbitrary rational $\lambda, \nu \geq 0$, define*

$$e = p_v + \lambda - \nu, \qquad c = W + \lambda\{v - A(y)\} - \nu v - e^2/4,$$
$$r = p_y - \frac{\lambda}{8d}\mathbf{1}. \tag{36}$$

Then $V_1(y') \geq c + r^\top(y' - y)$ for every original state y'. At a boundary one may choose $\lambda = \mathbf{1}_{\{v=A(y)\}}(-p_v)_+$ and $\nu = \mathbf{1}_{\{v=0\}}(p_v)_+$; no exact stationarity is required.

Partition the root innovation interval into n equal cells, with midpoints z_i and successors $y_i = F(x, a, z_i)$. Acquire such supports (c_i, r_i), and choose rational masses $q_i \in [0, 1/(nt)]$ summing to one. Put

$$W_0 = S(x) + h(a) + \beta \sum_i q_i c_i,$$
$$p_0 = h'(a) + \beta \sum_i q_i r_i^\top \boldsymbol{b}, \qquad a_* = \mathrm{proj}_{[0,A(x)]}(a - p_0/2),$$
$$B_0 = W_0 + p_0(a_* - a) + (a_* - a)^2. \tag{37}$$

Then $B_0 \leq V_0(x)$. Together with any separately valid original complete-policy upper bound $U_J(a)$, this proves

$$0 \leq J_x(a) - V_0(x) \leq U_J(a) - B_0, \qquad |a - a_0^*(x)|^2 \leq U_J(a) - B_0.$$

The maximum of finitely many such lower bounds acquired at the same original state is also valid. No support or optimum bound is transferred across distinct states without an additional justified state comparison.

PROOF: For a fixed feasible original density, the joint terminal state-action objective W_q has the lower curvature $m_x\|dy\|_2^2 + 2\,dv^2$, where $m_x = (4 - 5\beta/4)/d > 0$. This follows directly from $\nabla^2 S \succeq (4/d)I$, $h'' \geq 2$, $\nabla^2 g \succeq 0$, and $|\partial_j g| \leq 10/d$: the only potentially negative transition-chain term is $-\sum_j \partial_j g(F)\,dy_j dy_{j+1}/8$, bounded below by $-5\|dy\|_2^2/(4d)$. The other chain Hessian is positive semidefinite even in a joint state-action direction. Integration under the fixed nonnegative density preserves this bound. Matching first derivatives extend it across squared-hinge junctions. Dropping the positive state-curvature term gives

$$Q_1(y', v') \geq W_q(y', v') \geq W + p_y^\top (y' - y) + p_v(v' - v) + (v' - v)^2.$$

Replace p_v by $e - \lambda + \nu$. For a feasible $v' \in [0, A(y')]$, $-\lambda v' + \nu v' \geq -\lambda A(y')$. The affine capacity and the inequality $e(v' - v) + (v' - v)^2 \geq -e^2/4$ therefore give $Q_1(y', v') \geq c + r^\top (y' - y)$. Taking the infimum over the original feasible v' proves (36). In particular the intercept must be this valid global-support intercept; a larger scalar lower bound obtained by clipping a quadratic at only the anchor state cannot be attached to r without a further proof.

Here are explicit gradients suitable for an independent rational implementation. Put $m = F(y, v, 0)$. The original terminal polynomial in the innovation is $g(m) + kz + 5z^2$, whose coefficients can be reconstructed by evaluating the primitives at $-R, 0, R$. For its exact clipped quadratic tail with left cut u, let

$$L = 2R\alpha, \quad B = 2Rt, \quad \mu_1 = -L(2u + L)/(2B), \qquad G_j = \partial_j g(m) + (10/d)s_j\mu_1.$$

The squared hinge is constant along the zero-sum shock, and direct differentiation of the remaining quadratics gives $\partial_j g(m + sz) = \partial_j g(m) + (10/d)s_j z$. Thus G_j is the exact fixed-density mean of that derivative. With cyclic indices,

$$p_v = 2v + 16v^3 + \beta \sum_j b_j G_j,$$

$$p_{y,j} = \partial_j S(y) + \beta \left[G_j \left\{ \frac{1}{2} + \frac{1 - y_{j+1}}{16} \right\} + G_{j-1} \left\{ \frac{1}{8} - \frac{y_{j-1}}{16} \right\} \right]. \tag{38}$$

These formulas differentiate the fixed density, not its optimizing cut. The density supplied by the independent quadratic threshold construction is exactly feasible and maximizing at the anchor, but feasibility alone suffices for the global support proof. All quantities are rational at the stated inputs.

For the root, the masses q_i define a piecewise constant feasible AVaR density with value nq_i on cell i. For another feasible root action w and any z in cell i,

$$F(x, w, z) - y_i = \boldsymbol{b}(w - a) + \boldsymbol{s}(z - z_i).$$

Integrating its affine support over the symmetric cell removes the last term exactly. The same fixed feasible density consequently yields

$$Q_x(w) \geq S(x) + h(w) + \beta \sum_i q_i \{c_i + r_i^\top \boldsymbol{b}(w - a)\} \geq W_0 + p_0(w - a) + (w - a)^2.$$

Its quadratic minimum over the full original capacity interval is (37). Taking the infimum of Q_x proves the optimum lower bound. For equal cells with integral nt, selecting the largest nt values of c_i with fixed ties and assigning mass $1/(nt)$ to them is feasible and maximizes this finite linear witness. This sorting changes the chosen original density, not the original continuous innovation law.

Every implemented feasible policy costs at least V_0, so $J_x(a) \leq U_J(a)$ gives the regret inequality. The original Q_x is strongly convex in its action with modulus two. Its constrained minimizer obeys the boundary variational inequality. Hence $Q_x(a) - V_0(x) \geq |a - a_0^*(x)|^2$. Use $Q_x(a) \leq J_x(a) \leq U_J(a)$ and $B_0 \leq V_0(x)$ to prove the action-distance bound. A maximum of valid lower bounds on the same scalar optimum remains a valid lower bound. This completes the proof. *Q.E.D.*

The independent fixed-action upper value is still constructed by the original midpoint/duplicated-endpoint convex-order argument in Proposition S.13, with $\beta\varepsilon_1$ added once for the actual future policy. The new all-action witness does not assert containment of a fine production interval inside an arbitrarily selected coarse one, nor equality of any fine endpoint to the truth. It supplies its own exact, reproducible bounds on the original optimum, regret, and action error.

14. ORIGINAL-RISK MARGINAL ACQUISITION, TRAINING AND POLICY CONTROL

This section proves the statements in Section 3. The economic assumptions are the exact primitives in (13)–(14), with $d = 2$, $\beta = 15/16$, $R = 1/32$, and a separately fixed $\alpha \in \{1/2, 3/4\}$. The original strong convexity, domain invariance and constrained value gradient are proved in Proposition S.11; the original V_1 gradient modulus M_α is proved in Theorem S.3. Neither property is an assumption about a fitted critic. The null-space identity, finite-difference arguments, box dual bound and constrained strong-convexity inequalities use elementary or classical arguments; their proofs are included to specify exactly which training and policy claims they support.

14.1. *Constrained motion of the original adverse tail*

Let

$$\boldsymbol{b} = (1/2, 1/4)^\top, \qquad \boldsymbol{s} = (1, -1)^\top,$$
$$B = 2R(1-\alpha), \qquad L = 2R\alpha, \qquad I = [-R, R-L].$$

For $u = (x, v) \in \Omega$, put

$$m(u) = F(x, A(x)v, 0), \qquad f_u(z) = V_1(m(u) + \boldsymbol{s}z), \qquad H_u^{\mathrm{cen}}(c) = \int_c^{c+L} f_u(z)\, dz.$$

The letter H_u^{cen} denotes the central-window integral only; $H_\alpha(x, v)$ continues to denote the action marginal. The original invariant domain places every successor, and the segments used below, in the convex state cube. The inherited lower curvature of V_1 is $181/128$. Consequently f_u is $m_z = 181/64$-strongly convex in z. It is continuously differentiable, and

$$(H_u^{\mathrm{cen}})'(c) = f_u(c + L) - f_u(c)$$

is strongly monotone with modulus $m_z L > 0$. Thus H_u^{cen} has a unique minimizer $c(u) \in I$.

LEMMA S.15—An envelope and a constrained cut modulus: *For every u,*

$$\rho_\alpha(f_u) = \frac{1}{B}\left\{\int_{-R}^{c(u)} f_u(z)\,dz + \int_{c(u)+L}^{R} f_u(z)\,dz\right\}. \tag{39}$$

If two such functions satisfy $\mathrm{Lip}_z(f_u - f_{\widetilde{u}}) \le E$, their minimizing cuts satisfy

$$|c(u) - c(\widetilde{u})| \le E/m_z.$$

Both assertions include an endpoint minimizing cut and a tail of zero length.

PROOF: For any $c \in I$, the indicator of the two tails divided by $t = 1 - \alpha$ is a feasible density relative to the original uniform law. Its objective is the total integral of f_u minus $H_u^{\mathrm{cen}}(c)$, divided by B. If the minimizing cut is interior, its two endpoint values agree. Convexity then places all values inside the central interval below those on its complement. If the cut equals $-R$, its right derivative is nonnegative, so $f_u(-R+L) \ge f_u(-R)$. Convexity makes f_u nondecreasing to the right of $-R+L$, and every value in the central interval is at most $f_u(-R+L)$. Thus the right tail is again a set of the largest values. The other endpoint is symmetric. The tail density therefore maximizes the risk envelope. One can verify maximality directly: if q_* has cap $1/t$ on these tails and zero on the center, choose a threshold ζ separating their values. For any other feasible q,

$$\mathbb{E}(q - q_*)f_u = \mathbb{E}(q - q_*)(f_u - \zeta) \le 0,$$

using equal total mass and the signs on the two sets. Strong convexity makes every level set contain at most two points, which have zero uniform-law mass, so no atom or fractional threshold mass changes this argument.

Write $c = c(u)$, $\widetilde{c} = c(\widetilde{u})$. Their constrained optimality inequalities are

$$(H_u^{\mathrm{cen}})'(c)(\widetilde{c} - c) \ge 0, \qquad (H_{\widetilde{u}}^{\mathrm{cen}})'(\widetilde{c})(c - \widetilde{c}) \ge 0.$$

Strong monotonicity of $(H_u^{\mathrm{cen}})'$, combined with these two inequalities, gives

$$\begin{aligned} m_z L|c - \widetilde{c}|^2 &\le \left\{(H_u^{\mathrm{cen}})'(\widetilde{c}) - (H_{\widetilde{u}}^{\mathrm{cen}})'(\widetilde{c})\right\}(\widetilde{c} - c) \\ &\le LE\,|\widetilde{c} - c|. \end{aligned}$$

The last bound follows from the Lipschitz constant of the difference across an interval of length L. Divide when the cuts differ; the coincident-cut case already satisfies the claim. The variational inequalities are valid at the endpoints, so no equation $(H_u^{\mathrm{cen}})'(c) = 0$ is imposed there. *Q.E.D.*

PROOF OF THEOREM S.4: The original V_1 is continuously differentiable with M_α-Lipschitz gradient on the cube, including its continuous boundary extension. Since the normalized feasible set for c is compact and the minimizer unique, $c(u)$ is continuous. This also follows quantitatively from Lemma S.15 and the estimates below. The objective under any fixed feasible cut is continuously differentiable in its parameters, uniformly over this compact cut set. Comparing a value increment at the old and at the new maximizing cut and using uniform first-order expansion therefore gives the envelope derivative with the cut held fixed:

$$H_\alpha(u) = \frac{\beta}{B}\left\{\int_{-R}^{c(u)} m_u^a(z)\,dz + \int_{c(u)+L}^{R} m_u^a(z)\,dz\right\}, \qquad m_u^a(z) = \boldsymbol{b}^\top \nabla V_1(m(u) + \boldsymbol{s}z). \tag{40}$$

The derivative of the normalized transition in v is $A(x)\boldsymbol{b}$; this factor cancels on dividing the envelope derivative by $A(x)$. The argument does not differentiate $c(u)$, an optimizing last-date action, or the rounded deployed controller. Continuity of the integrands and of the cuts proves the continuous marginal extension at all boundaries.

First fix x and use the original action coordinate a. The original gradient bound gives

$$|m_a^a(z) - m_b^a(z)| \le M_\alpha \|\boldsymbol{b}\|_2^2 |a-b| = \frac{5M_\alpha}{16}|a-b|,$$

$$\mathrm{Lip}_z\, m_a^a \le M_\alpha \|\boldsymbol{b}\|_2 \|\boldsymbol{s}\|_2 = \frac{\sqrt{10}}{4} M_\alpha \le C_\alpha,$$

$$\mathrm{Lip}_z(f_a - f_b) \le C_\alpha |a-b|.$$

Hence the cuts differ by at most $C_\alpha |a-b|/m_z$. For a fixed integrand m_a^a, changing a tail cut from c to c' changes its normalized tail integral by at most

$$\frac{1}{B}\left|\int_{c'}^{c} \{m_a^a(z) - m_a^a(z+L)\}\,dz\right| \le C_\alpha \frac{\alpha}{t}|c-c'|.$$

Changing both the integrand and the cut in (40) therefore proves

$$|\partial_a R_\alpha(x, a/A(x)) - \partial_a R_\alpha(x, b/A(x))| \le \beta D_\alpha |a - b|.$$

Here ∂_a means differentiation of the displayed composition as a function of the original action. Since $h''(a) = 2 + 48a^2$ lies between 2 and 5 on the original action interval, the claimed Lipschitz constant of Q'_x is $L_\alpha = 5 + \beta D_\alpha$. Convexity of V_1, the affine action transition and the monotone convex risk envelope make the continuation term convex in a. The stage cost gives 2-strong convexity. Multiplying action increments by $A(x)$ proves the normalized curvature statement.

It remains to bound variation with both state and normalized action. The original calculation in Theorem S.3 gives, for fixed v, $\|D_x m\|_2 \le 3/4$. Also $D_v m = A(x)\boldsymbol{b}$, so $\|D_v m\|_2 \le \sqrt{5}/16$. For the full three-dimensional parameter,

$$\|D_u m\|_2 \le \sqrt{(3/4)^2 + (\sqrt{5}/16)^2} = \frac{\sqrt{149}}{16} \le \frac{49}{64}.$$

The last inequality follows on squaring from $2384 \le 2401$. Integrating this Jacobian on a segment in the convex parameter cube gives the same global Lipschitz constant for m. Since $\|\boldsymbol{b}\|_2 = \sqrt{5}/4 \le 9/16$,

$$|m^a_u(z) - m^a_{\widetilde{u}}(z)| \le \frac{441 M_\alpha}{1024} \|u - \widetilde{u}\|_2.$$

The innovation derivative of $f_u - f_{\widetilde{u}}$ is bounded by

$$M_\alpha \|\boldsymbol{s}\|_2 \|m(u) - m(\widetilde{u})\|_2 \le \frac{147 M_\alpha}{128} \|u - \widetilde{u}\|_2,$$

using $\sqrt{2} \le 3/2$. Lemma S.15 bounds the change of cut by this quantity divided by m_z. Applying the fixed-integrand tail bound above and $C_\alpha = 4M_\alpha/5$ proves

$$|H_\alpha(u) - H_\alpha(\widetilde{u})| \le \beta \left\{ \frac{441 M_\alpha}{1024} + \frac{147 M^2_\alpha \alpha}{160 m_z t} \right\} \|u - \widetilde{u}\|_2.$$

All denominators are positive for the declared risk levels. The ordinary-expectation case $\alpha = 0$ can omit the cut argument, but no $\alpha = 1$ normalized-tail statement is asserted here. *Q.E.D.*

14.2. *What the finite-risk oracle has to enclose*

The label theorem assumes an enclosure of the original optimal-future objective $Q_x(a)$. A deployed terminal policy value is a different oracle unless its error is explicitly removed or enclosed. For clarity, a sufficient original-law construction is the following. Partition $[-R, R]$ into cells $B_i = [z_i^-, z_i^+]$, with lengths h_i, midpoint z_i, and probability $p_i = h_i/(2R)$. Suppose actual paid last-date calculations supply

$$l_i \le V_1(F(x,a,z_i)), \qquad V_1(F(x,a,z_i^{\pm})) \le u_i^{\pm}.$$

Let Z_- be the discrete law taking values l_i with masses p_i, and let Z_+ take values u_i^-, u_i^+, each with mass $p_i/2$. Then

$$S(x) + h(a) + \beta\rho_\alpha(Z_-) \ \le Q_x(a) \le \ S(x) + h(a) + \beta\rho_\alpha(Z_+). \tag{41}$$

Here the discrete laws are computational bounds for the original continuous law, not substituted economic risk distributions.

PROOF: For each real threshold ζ, the function $(V_1(F(x,a,z)) - \zeta)_+$ is convex in z. Its midpoint value minorizes its cell average and its endpoint secant majorizes that average. Monotonicity in the enclosed node values consequently gives

$$\mathbb{E}(Z_- - \zeta)_+ \le \mathbb{E}(V_1(F(x,a,Z)) - \zeta)_+ \le \mathbb{E}(Z_+ - \zeta)_+.$$

For completeness, for a bounded random variable W,

$$\rho_\alpha(W) = \inf_{\zeta \in \mathbb{R}} \{\zeta + t^{-1}\mathbb{E}(W - \zeta)_+\}.$$

To verify this identity, any feasible risk density satisfies $\mathbb{E}qW \le \zeta + t^{-1}\mathbb{E}(W - \zeta)_+$. Choose an α-quantile as the threshold, put density $1/t$ above it and zero below it, and, if needed, assign the remaining feasible mass on its atom. This attains the bound. Boundedness ensures a quantile exists. Apply the identity to the three hinge expectations while retaining the same threshold in their pointwise inequalities, then take infima. Addition of the exact stage term proves (41). No differentiability of a finite quadrature, sort operation or last-date compiler is used. *Q.E.D.*

Every approximate last-date optimization, endpoint value, finite-risk calculation and arithmetic operation must be enclosed in (41). A valid analytic terminal compiler can supply these intervals; its supporting state slope is not thereby an exact derivative of V_1. The current finite-label construction needs only its valid optimal-value intervals.

14.3. *Derivative probes, boundaries and their precision cost*

PROOF OF PROPOSITION S.5: Fix x, write $q = Q_x$, $g = q'(a)$, and $L_q = L_\alpha$. For $0 \le s \le d$, strong convexity and derivative Lipschitzness imply

$$2s \le q'(a+s) - g \le L_q s.$$

Integrating on the forward interval gives

$$g + d \le \frac{q(a+d) - q(a)}{d} \le g + L_q d/2.$$

Thus the first interval in (13) contains g, including at the lower action boundary. On the backward interval, $-L_q s \le q'(a-s) - g \le -2s$, which gives the second interval, including at the upper boundary. If both probes are feasible, then

$$-(L_q - 2)s \le q'(a+s) + q'(a-s) - 2g \le (L_q - 2)s.$$

Integrate from 0 to d and divide by $2d$ to prove the central radius $(L_q - 2)d/4$. The outward value differences contain the exact secants, so the conclusions survive any declared valid oracle error. Their intersection contains g; subtraction of the known stage derivative leaves the original continuation marginal. If a floating implementation subtracts $h'(a)$, it must use a valid outward enclosure or its exactly represented rational value. The same requirement applies to dividing a normalized derivative by the positive capacity. *Q.E.D.*

These inequalities quantify the precision allocation even before an experiment. If each value interval has width at most $2e$, the central derivative interval has half-width at most

$$e/d + (L_\alpha - 2)d/4.$$

A smaller probe magnifies the paid value uncertainty. With $L_\alpha > 2$, the unconstrained balancing choice for this upper bound is $d = 2\sqrt{e/(L_\alpha - 2)}$; feasibility and a prospectively fixed probe rule still have to be respected. This formula does not prescribe changing the current experiment's probe after seeing its labels.

PROPOSITION S.16—A limit of the specified three-probe interface: *Fix an interior point with feasible symmetric step $d > 0$, and a scalar class with derivative curvature between $\mu > 0$ and $L_q \geq \mu$. An algorithm supplied only values at the three locations $-d, 0, d$, relative to that point, cannot in general return a derivative interval of width less than $(L_q - \mu)d/6$, even with exact values. If each of the three supplied value intervals has width $2e$, there are compatible functions whose derivatives differ by $2e/d$. The central-interval radius $e/d + (L_q - \mu)d/4$ therefore has the correct order for this particular information interface.*

PROOF: Put $\kappa = (L_q + \mu)/2$ and, on $[-d, d]$, define

$$p(s) = \frac{s(s^2 - d^2)}{d^2}, \qquad q_\pm(s) = \frac{\kappa s^2}{2} \pm \frac{(L_q - \mu)d}{12} p(s).$$

Their three values agree, because $p(-d) = p(0) = p(d) = 0$. Also $p'(0) = -1$, so their derivatives at zero differ by $(L_q - \mu)d/6$. Since $p''(s) = 6s/d^2$, both second derivatives lie in $[\mu, L_q]$. Extending p linearly outside each endpoint with its endpoint derivative makes p continuously differentiable; the second derivative then vanishes outside. The same curvature bounds hold on any larger feasible interval. Any valid output interval for the common three exact observations must contain both derivatives.

For the uncertainty assertion use

$$\widetilde{q}_\pm(s) = \frac{\kappa s^2}{2} \pm (e/d)s.$$

At either endpoint their two values differ by $2e$; at zero they agree. The common value intervals with midpoint $\kappa s^2/2$ and radius e contain both functions at all three queried sites. Their derivatives at zero differ by $2e/d$, and both curvatures are κ. The displayed upper radius follows by the secant proof with μ in place of 2. *Q.E.D.*

This is a local, three-observation oracle statement. Other paid values, gradients, global shape restrictions or access to an original primitive formula can provide more information. It is not a lower bound for all algorithms in the fixed investment model, for the complete acquisition bundle, or for a neural representation class.

14.4. *Finite-box training and exact optimization evidence*

Let each row index i include a site and one feasible one-sided derivative branch. The bounds $\ell_i \le H_\alpha(\xi_i) \le u_i$ are replicated at that site's branches. At an endpoint, only the inward branch is meaningful. The finite-dimensional feasible set is the same declared box Θ for the value-loss and marginal-loss fits within a family and budget.

PROOF OF PROPOSITION S.6: For a scalar predicted derivative z,

$$\max_{h\in[\ell_i,u_i]} |z-h| = \max\{z-\ell_i, u_i-z\}.$$

Thus $T(\theta)$ is exactly the robust maximum node error. It is continuous as a finite maximum of affine functions. The coefficient box is nonempty and compact, so its minimum is attained. In particular, (15) is a linear program in (θ, T), even when the represented critic is not convex in its action.

For any feasible θ, each of the affine errors is at most $T(\theta)$. Multiply the positive and negative rows by y_i^+, y_i^-, respectively, and add. The normalization of their combined mass gives

$$T(\theta) \ge -\sum_i \ell_i y_i^+ + \sum_i u_i y_i^- + r^\top \theta.$$

The minimum of the last linear term over the coordinate box is the sum of the smaller endpoint products. This proves $B_y \le T_*$. Each robust row error is nonnegative, so also $0 \le T_*$. Evaluating the finite maximum at the exactly checked stored $\widehat{\theta}$ gives $T_* \le P$ and the stated gap. No assertion about a floating solver's dual feasibility has entered the argument. *Q.E.D.*

All quantities in this certificate can be checked using rational arithmetic for the declared features at the dyadic acquisition sites. A stored binary coefficient

is converted to the rational number represented by its bytes, not to an intended decimal value. Nonnegative approximate dual weights may be rounded to rationals and renormalized; the resulting weights, rather than their unrounded counterparts, must satisfy the conditions of the proposition. If all proposed weights are zero, another specified nonnegative normalized choice gives a valid, potentially weak bound. A coefficient outside its box is not made feasible by reporting the solver's constraint tolerance.

The certificate also enables an exact comparison of the actual two learned readouts. Evaluate $P_{\mathrm{V}} = T(\widehat{\theta}_{\mathrm{V}})$ and $P_{\mathrm{M}} = T(\widehat{\theta}_{\mathrm{M}})$ with the same rows and labels. If their checked values satisfy $P_{\mathrm{M}} < P_{\mathrm{V}}$, the marginal fit has strictly reduced this particular robust node error. Its certified distance from the best coefficient vector in the box is at most $P_{\mathrm{M}} - \max\{0, B_y\}$. Neither inequality by itself orders off-design regret, rejection probability or complete computational work. The value-loss control is an actual bounded least-squares fit. The following certificate is evaluated on both stored readouts, with the same exact design and midpoint targets. It quantifies the value-loss optimization gap without assuming that a floating solver attained the global minimum.

PROPOSITION S.17—An exact certificate for the value-loss control: *Let the finite value design rows be Φ_i, let the stored midpoint targets be y_i, and set*

$$F(\theta) = \sum_i (\Phi_i \theta - y_i)^2, \qquad F_* = \min_{\theta \in \Theta} F(\theta).$$

For an exactly checked $\widehat{\theta} \in \Theta$, define

$$g = 2 \sum_i \Phi_i^\top (\Phi_i \widehat{\theta} - y_i),$$

$$L_F = \max \left\{ 0, F(\widehat{\theta}) - g \cdot \widehat{\theta} + \sum_j \min\{\underline{\theta}_j g_j, \overline{\theta}_j g_j\} \right\}.$$

Then

$$L_F \le F_* \le F(\widehat{\theta}), \qquad 0 \le F(\widehat{\theta}) - F_* \le F(\widehat{\theta}) - L_F.$$

These are finite-design optimization bounds. They do not remove the error in the numerical value labels or imply uniform approximation or policy optimality.

PROOF: Continuity of F and compactness of the nonempty box give existence of its minimum. Expanding each square, for every $\theta \in \Theta$ one obtains

$$F(\theta) = F(\widehat{\theta}) + g \cdot (\theta - \widehat{\theta}) + \sum_i \{\Phi_i(\theta - \widehat{\theta})\}^2.$$

Drop the nonnegative last sum and minimize the linear term separately over each box coordinate. This gives the expression inside the maximum in L_F. The additional lower bound zero follows from the sum of squares. Feasibility of $\widehat{\theta}$ gives the upper bound. Subtraction proves the gap bound. All identities hold for the actual rational numbers represented by the checked design, targets and coefficients; no differentiability of the Bellman oracle or fitted-feature knot is involved in differentiating this coefficient loss. *Q.E.D.*

14.5. *Constructive ranges for all declared feature families*

The stored level is the fitted or interpolated $Q_x(0)$. Accordingly the continuation anchor itself is that level minus the known $S(x)$; the production full action critic adds $h(A(x)v)$ to the stored level and to the learned continuation difference. This is the full-value identity used by the recursive anchor theorem, not an assumption that the stored anchor is exact off the acquisition grid.

For any fixed state, anchoring $\phi(x, v)$ by $\phi(x, 0)$ leaves its action derivative unchanged. The denominator $A(x) = (2 + x_1 + x_2)/16$ is affine and at least $1/8$. On a rational box $C \subset \Omega$, its range $[A_-, A_+]$ is computed at state endpoints. If $D_v\phi \in [p_-, p_+]$, take the minimum and maximum of the four rational products

$$p_-/A_-, \quad p_-/A_+, \quad p_+/A_-, \quad p_+/A_+$$

to enclose $D_v\phi/A(x)$. This does not require independence of numerator and denominator. It is an outward rectangle bound. Unions of boxes or polynomial pieces use the extrema of their already valid bounds.

Quadratic and polynomial features. The anchored degree-two pool is $v, x_1 v, x_2 v, v^2$. Its first three coefficients are signed. The quadratic family restricts the last coefficient to $[0, 128]$; the polynomial family permits $[-128, 128]$. The degree-three pool contains exactly

$$x_1^p x_2^q v^r, \qquad r \geq 1, \quad p, q \geq 0, \quad p + q + r \leq 3.$$

It has ten elements. Each action derivative numerator is $r x_1^p x_2^q v^{r-1}$. All coordinates lie in $[0, 1]$, so endpoint powers and products give exact range bounds for each nonnegative monomial before division by A. The coefficients, not the basis monomials, can have either sign; signed interval summation accounts for that fact.

ReLU and exactly matched hinges. In addition to $v, x_1 v, x_2 v$, the pool has 36 features

$$\left(\frac{w_1(x_1 - 1/2) + w_2(x_2 - 1/2) + v - q}{4} \right)_+ - \left(\frac{w_1(x_1 - 1/2) + w_2(x_2 - 1/2) - q}{4} \right)_+,$$

where $w_1, w_2 \in \{-1, 0, 1\}$ and $q \in \{1/8, 3/8, 5/8, 7/8\}$. For the first affine argument z, the derivative numerator is 0 when $z < 0$ and $1/4$ when $z > 0$. It has these two one-sided values at $z = 0$. Compute the minimum and maximum of the affine z on the box: a strictly negative box needs only 0, a strictly positive box only $1/4$, and a crossing or touching box retains both whenever the corresponding side is present. Using $[0, 1/4]$ at every such boundary is conservative and valid. The 36 coefficients are nonnegative; the three base coefficients remain signed. A conventional coordinate-loop hinge with these same arrays has identical mathematical derivatives and values. A different arithmetic encoding can incur different rounding or runtime, which must be recorded separately.

Clamped tensor cubic splines. The coarse univariate knot vector is clamped at 0 and 1 with no interior knots; the fine vector additionally has one simple interior knot at $1/2$. There are respectively four and five cubic basis functions per coordinate. The tensor pool omits the first action basis, leaving $4 \cdot 4 \cdot 3 = 48$ and $5 \cdot 5 \cdot 4 = 100$ coefficients. The omitted basis is the only one nonzero at $v = 0$; thus these features

already vanish at the anchor. The standard degree-k basis is built by the finite recursion

$$B_{i,k}(s) = \frac{s - t_i}{t_{i+k} - t_i} B_{i,k-1}(s) + \frac{t_{i+k+1} - s}{t_{i+k+1} - t_{i+1}} B_{i+1,k-1}(s),$$

with a zero term for a zero denominator and the half-open indicator bases at degree zero. The continuous endpoint limits define clamped values. On each knot interval this is a polynomial with rational coefficients, and differentiating that polynomial gives the required action derivative, including endpoint limits. Tensor products preserve this finite piecewise-polynomial form.

Tensor linear hats. Use the declared five- or nine-point state mesh in each state coordinate and the five-point action mesh. The tensor hats that use the first action hat are omitted, leaving $5 \cdot 5 \cdot 4 = 100$ or $9 \cdot 9 \cdot 4 = 324$ coefficients. Within a mesh box the state hats are nonnegative affine functions and the action-hat derivative is its constant one-sided slope, zero, or the opposite slope. Their products give numerator bounds. At an action knot both incident slopes are retained. Crossing a state knot does not justify dropping an incident polynomial piece.

For completeness, polynomial ranges in the spline or tensor pieces can be constructed without a sampled derivative test. On a closed coordinate interval map the variable to $[0,1]$ and write a degree-n polynomial in its Bernstein basis

$$P(s) = \sum_{k=0}^{n} b_k \binom{n}{k} s^k (1-s)^{n-k}.$$

The coefficients are obtained by finite rational basis conversion. The basis functions are nonnegative and sum to one, so $P(s)$ lies between the smallest and largest b_k. Tensor products have the same nonnegative partition-of-unity property and bound multivariate polynomial pieces by their smallest and largest tensor coefficients. Intersecting a cell with the finitely many knot boxes, or merely enclosing each intersection in a knot box, gives valid ranges. Affine hinge cuts can also be used to subdivide cells; retaining both sides avoids that extra subdivision cost. Every option has an explicit finite construction, and its actual arithmetic work belongs in acquisition and certificate costs if executed.

These rules provide bounds for all original states, including feature-knot states, and all feasible one-sided action limits. They do not assume that a ReLU or tensor-grid derivative has a global Lipschitz constant.

14.6. *Finite-cover and path-integral certificates*

PROOF OF THEOREM S.7: Fix a nonempty cell C. For every $u \in C$, Theorem S.4 and the label at any retained node ξ_i imply

$$\ell_i - \Lambda_\alpha r_i(C) \le H_\alpha(u) \le u_i + \Lambda_\alpha r_i(C).$$

Taking the maximum of lower endpoints and the minimum of upper endpoints yields the target range. The independent feature range and interval linear combination contain the fitted action marginal. The difference of two real numbers in these two intervals is bounded in absolute value by the two opposite endpoint differences. This proves the cell error bound $\epsilon(C)$. Limits from every incident feature piece are included by construction, and the target marginal is continuous, so the same bound holds for the required one-sided action derivatives.

For clarity, a version that links this certificate directly to the training loss is also available. Choose a reference derivative row A_i at a label node, and define

$$\Delta_{ij}(C) \ge \sup_{u \in C} |A(x)^{-1} D_v \phi_j(u) - A_{ij}|, \qquad T_i(\theta) = \max\{A_i\theta - \ell_i, u_i - A_i\theta\}.$$

The feature intervals just described give such a Δ_{ij} by two endpoint differences. The triangle inequality proves

$$\sup_{u \in C} |\widehat{H}_\theta(u) - H_\alpha(u)| \le T_i(\theta) + \Lambda_\alpha r_i(C) + \sum_j |\theta_j| \Delta_{ij}(C). \tag{42}$$

Take the smaller of all such bounds and (17) if desired. Even an exactly minimized T_i leaves the target coverage and feature-variation terms. This explicitly identifies the additional information required to move from fit nodes to a service state.

For fixed x, the fitted continuation is continuous and piecewise polynomial in v; hence it is absolutely continuous. The true continuation is continuously differen-

tiable. For all v except the finitely many feature knots,

$$\left|\partial_v\{\widehat{R}_\theta(x,v) - R_\alpha(x,v)\}\right| \le A(x)\epsilon_x(v).$$

The choice of cell may be made by a fixed ordering of the finite cells, or by taking the smallest valid bound. It is measurable for the piecewise-algebraic covers used here. Integrating this inequality between $a/A(x)$ and $b/A(x)$ gives (19). The exact stage cost and the state-only anchor cancel from the error increment.

Equation (10) gives $-\delta \le \eta_x(b) - \eta_x(a) \le \delta$. Adding that difference proves the implemented-policy contrast. For every feasible b,

$$Q_x(a) - Q_x(b) \le \widehat{q}_x(a) - \widehat{q}_x(b) + \mathcal{D}_x(a,b).$$

Take the supremum in b and add $0 \le \eta_x(a) \le \delta$ to obtain the original optimum bound. The supremum is nonnegative because the term at $b = a$ is zero. Replacing it by a larger certified upper value preserves the bound. None of these steps uses convexity of the fitted critic. *Q.E.D.*

The scalar all-action calculation in (20) is constructive. For a rectangular or polyhedral finite cover and a fixed represented state, split the feasible action interval at the cover crossings and the finitely many feature knots. A piecewise constant ϵ_x makes $\mathcal{D}_x(a,b)$ piecewise affine in b. The fitted action objective, including h, is a polynomial of degree at most four on each piece for every declared family. The required supremum is attained at a piece endpoint or at a zero of its derivative polynomial. For rational coefficients, signs and isolating intervals can be checked by polynomial remainder sequences and rational interval refinement; outward evaluation of all these intervals gives a certified upper bound. An alternative is interval subdivision with valid polynomial range bounds. Approximate floating roots without an enclosing calculation are not, by themselves, this certificate.

The associated action seminorm can be written as

$$\|e\|_{\mathrm{act}} = \sup_{x\in[0,1]^2} \operatorname*{ess\,sup}_{0<v<1} |A(x)^{-1}\partial_v e(x,v)|.$$

The supremum over states is literal; replacing it by an essential supremum over state space could lose a particular queried state. For the present absolutely continuous

action sections, this seminorm is zero exactly when $e(x, v) = b(x)$ is independent of v at each state. Indeed a zero derivative integrates to a constant, and an action constant has zero derivative. This is the inherited action-null quotient identity. The new finite-cover construction bounds an approximate, learned member of that quotient under the original integrated AVaR objective.

14.7. *Constrained actor error and the deployed future*

PROOF OF PROPOSITION S.8: For every feasible displacement d, original strong convexity gives

$$Q_x(a + d) \geq Q_x(a) + Q'_x(a)d + d^2.$$

Taking the infimum over the original feasible interval proves $Q_x(a) - V_0(x) \leq \mathcal{K}_x(a; Q'_x(a))$. The map $\mathcal{K}_x(a; g)$ is a supremum of affine functions of g, and hence convex. Its maximum over a closed scalar interval is at an endpoint, which proves the bound $K_x(a)$. Completion of the square gives the projected minimizer in the statement. At the lower boundary $d \geq 0$; if the full derivative interval is nonnegative, $gd + d^2$ is minimized at zero. At the upper boundary the signs reverse. This proves both exact boundary claims.

Let a_0^* be the unique original optimizer. Its constrained first-order condition gives $Q'_x(a_0^*)(a - a_0^*) \geq 0$. Strong convexity at this point implies

$$|a - a_0^*|^2 \leq Q_x(a) - V_0(x) \leq K_x(a).$$

Adding the single nonnegative allowance in (10) proves the statement for the implemented controller.

Here is a precise interpretation of the final stationarity assertion. At an interior smooth action let $\widehat{g}$ be the derivative of the fitted action objective. At an interior kink it may be any supplied convex combination of the two one-sided derivatives. At either endpoint use the inward derivative. Every such $\widehat{g}$ differs from the continuous true derivative by at most ϵ, provided the required one-sided errors have that bound. Let

$$N_{[0,A(x)]}(a) = \{n : n(b - a) \leq 0 \text{ for every feasible } b\}.$$

A checked constrained stationarity residual means that a supplied n in this cone satisfies $|\widehat{g}+n| \le r$. Use the interval $[\widehat{g}-\epsilon, \widehat{g}+\epsilon]$, or its intersection with any other valid derivative interval. For any g in this interval and any feasible d,

$$gd + d^2 = \{g - \widehat{g} + \widehat{g} + n\}d + d^2 - nd \ge -(\epsilon + r)|d| + d^2.$$

The last expression is at least $-(\epsilon+r)^2/4$. This proves $K_x(a) \le (\epsilon+r)^2/4$ for the interval supplied by this error bound. It also explains why an unrelated wider derivative interval need not attain the same numerical bound. *Q.E.D.*

A global exact minimizer of the continuous piecewise-polynomial fitted objective supplies zero constrained residual: at an interior minimum its left derivative is nonpositive and its right derivative nonnegative; at an endpoint the inward derivative has the appropriate sign. The same sign fact applies to a checked local minimum. An implemented polynomial-root enumeration may have rounding error and should report its actual residual or invoke the retained original all-action certificate. No theorem here identifies a floating root with an exact stationary action.

For a different useful form, suppose an implemented actor is known only to have fitted objective suboptimality at most $\zeta \ge 0$, and the original-action marginal error is uniformly at most ϵ along all feasible actions at this state. Write $\Delta = Q_x(a) - V_0(x)$. The path inequality and the strong-convexity distance bound give

$$\Delta \le \zeta + \epsilon|a - a_0^*| \le \zeta + \epsilon\sqrt{\Delta}.$$

Solving this scalar inequality yields

$$\Delta \le \left\{ \frac{\epsilon + \sqrt{\epsilon^2 + 4\zeta}}{2} \right\}^2. \tag{43}$$

The once-only allowance δ gives the implemented-policy bound. This objective-gap form is weaker than the checked-stationarity bound when $\zeta = 0$, but remains valid without a fitted derivative optimality certificate.

14.8. *Complete-value anchors in a finite recursive composition*

The current-state seminorm must not be used to discard a state-dependent value when that value becomes a future continuation. This issue can be checked within the original investment primitives, without another experimental model. For a fixed $\gamma \in \mathbb{R}$, put $W_1(y) = V_1(y) + \gamma(y_1 + y_2)$. Since $s_1 + s_2 = 0$,

$$W_1(F(x,a,Z)) = V_1(F(x,a,Z)) + \gamma\{F_1(x,a,0) + F_2(x,a,0)\}.$$

The second summand is constant in the innovation. Cash invariance of the same original risk map proves

$$Q_x^{W_1}(a) - Q_x(a) = \beta\gamma\{F_1(x,a,0) + F_2(x,a,0)\}.$$

The sum of the two action coefficients is $3/4$. Its derivative contribution is therefore $3\beta\gamma/4$. An additive state-only level in a last-date action critic leaves that last-date minimizer unchanged, but exporting this level as a continuation changes the root marginal as displayed.

THEOREM S.18—Anchor, quotient and policy errors in the same recursion: *Consider a finite horizon $t = 0, \dots, T$ with the same bounded primitives, original feasible actions and original transition at every comparison. Assume the Bellman and policy recursions are well-defined, for example under the compact, continuous, measurable-selector conditions of Proposition S.10. For $t < T$, let*

$$Q_t^W(x,a) = c_t(x,a) + \beta_t \rho_{t,x,a}\{W(F_t(x,a,\cdot))\}, \qquad V_t(x) = \inf_a Q_t^{V_{t+1}}(x,a),$$

where $0 \le \beta_t < \infty$ and each original risk map is monotone and cash invariant. All continuations under consideration are bounded and belong to a class on which these recursions are justified.

At each date choose an original feasible reference action $a_t^0(x)$ and write an acquired action critic as $\widehat{Q}_t(x,a) = \widehat{A}_t(x) + \widehat{D}_t(x,a)$, with $\widehat{D}_t(x,a_t^0(x)) = 0$. For its supplied continuation W_{t+1}, suppose

$$a_t^-(x) \le \widehat{A}_t(x) - Q_t^{W_{t+1}}(x,a_t^0(x)) \le a_t^+(x),$$
$$b_t^-(x,a) \le \widehat{D}_t(x,a) - \{Q_t^{W_{t+1}}(x,a) - Q_t^{W_{t+1}}(x,a_t^0(x))\} \le b_t^+(x,a).$$

Let finite uniform bounds d_t^-, d_t^+ enclose the sums of these two errors over all original feasible pairs. Suppose the exported value satisfies

$$|W_t(x) - \inf_a \widehat{Q}_t(x,a)| \le \zeta_t$$

uniformly, where $\zeta_t \ge 0$ is checked. Given $E_T^- \le W_T - V_T \le E_T^+$, define $E_t^\pm$ by (22). Then $E_t^- \le W_t - V_t \le E_t^+$ at every date and state.

Let the implemented actor satisfy the local original-action bound

$$Q_t^{W_{t+1}}(x, \pi_t(x)) - \inf_a Q_t^{W_{t+1}}(x,a) \le g_t$$

uniformly, and let its continuation policy error be bounded by $0 \le J_{t+1}^\pi - V_{t+1} \le P_{t+1}$. Then

$$0 \le J_t^\pi - V_t \le g_t + \beta_t(E_{t+1}^+ - E_{t+1}^-) + \beta_t P_{t+1}. \tag{44}$$

Starting from the original terminal payoff, $P_T = 0$, these bounds compose for the full finite-horizon policy.

PROOF: Monotonicity and cash invariance give, for every state and feasible action,

$$\beta_t E_{t+1}^- \le Q_t^{W_{t+1}} - Q_t^{V_{t+1}} \le \beta_t E_{t+1}^+.$$

This uses uniform scalar bounds on the entire relevant continuation, not the removal of a state-dependent term from an expectation. The anchor and contrast inequalities add to

$$d_t^- \le \widehat{Q}_t - Q_t^{W_{t+1}} \le d_t^+.$$

Combining these two inequalities, taking the infimum over the identical original feasible set, and including the two signs of ζ_t proves the value recurrence. Backward induction from T gives its claim at all dates.

For the actor, the upper comparison at $\pi_t(x)$ and the lower comparison at the infimum give

$$Q_t^{V_{t+1}}(x, \pi_t(x)) - \inf_a Q_t^{V_{t+1}}(x,a) \le Q_t^{W_{t+1}}(x, \pi_t(x)) - \inf_a Q_t^{W_{t+1}}(x,a)$$

$$+ \beta_t (E^+_{t+1} - E^-_{t+1}).$$

The actual policy recursion at the same action replaces V_{t+1} by J^π_{t+1}. Monotonicity and cash invariance bound that increment between 0 and $\beta_t P_{t+1}$. This proves the upper inequality in (44). The lower inequality follows by induction: the terminal policy uses the original payoff; a feasible action cannot beat the Bellman infimum; and the monotone recursive risk cannot lower a nonnegative future policy error. The same proof applies if a nonnegative terminal policy error has separately been specified and bounded. *Q.E.D.*

The local quantities g_t can be supplied by the path-integral or constrained marginal certificate when its target is the indicated $Q_t^{W_{t+1}}$. They cannot be transferred to a different future continuation without the displayed span term or a sharper justified comparison. If $E^-_{t+1} = E^+_{t+1}$, the continuation error is a single global constant, and its monetary action effect does cancel. A state-dependent anchor typically has a nonzero span and does not meet that condition.

For a noncash-invariant recursive utility, state a different assumption explicitly. Suppose the same original one-step aggregation map, including its stage arguments, is monotone and satisfies

$$|Q_t^W(x,a) - Q_t^{\widetilde{W}}(x,a)| \le L_t \|W - \widetilde{W}\|_\infty$$

uniformly on the relevant bounded class, with a supplied finite L_t. Let $d_t = \max\{|d_t^-|, |d_t^+|\}$. Then the same addition and infimum argument gives

$$\|W_t - V_t\|_\infty \le d_t + \zeta_t + L_t \|W_{t+1} - V_{t+1}\|_\infty.$$

Comparing the actor and the infimum pays two continuation errors, so

$$0 \le J_t^\pi - V_t \le g_t + 2L_t \|W_{t+1} - V_{t+1}\|_\infty + L_t P_{t+1}.$$

To verify these formulas, apply the assumed absolute Lipschitz inequality once at every action for the value comparison, once at the actor and once at the infimum for its local regret, and once to the actual policy continuation. Monotonicity supplies the nonnegative-policy-error induction. No cash-translation identity is used. An NDU

or recursive-utility specification with singular domains or unbounded continuation derivative must establish this assumption on an invariant domain before using the result.

In the present two-date study, the acquisition target is the original $V_1 = T_1 g$, evaluated by paid primitive calculations. Thus its mathematical continuation approximation in the root comparison is zero, while the actual last-date policy error is at most $\varepsilon_1 = 2^{-20}$. Equation (44) reduces to the root local action gap plus $\beta\varepsilon_1$. Numerical uncertainty in evaluating $T_1 g$ is already included in the finite label and policy intervals; it is not permission to treat that operator evaluation as a free oracle.

14.9. *A total original-policy service and its uniform range*

The following argument verifies the constants of the retained exact finite-cover fallback `rational96.py`. It does not require a favorable first proposal or convergence of a floating polynomial-root solver. The declared service retains a recoverable numerical failure in the proposal or certification stages and then invokes this fallback at the same state. Source-identity, domain, corrupt-model and infrastructure failures remain failed executions; they cannot be removed from or replaced within the evaluation sample.

PROPOSITION S.19—Finite all-state qualification and primitive range: *At every represented original state $x \in [0,1]^2$ and either declared risk level, the exact fallback with 1025 equally spaced feasible actions, 256 innovation cells and exact last-date value gap at most 2^{-30} returns a feasible action whose original complete-policy regret is at most $\tau = 1/256$. Its policy-cost interval has width at most $1/8192$. Every exact last-date bisection needs at most 31 iterations, within its stated cap of 80. These are finite arithmetic guarantees; all their actual cost is charged when entered.*

The original primitive ranges satisfy

$$0 \le S \le 49/32, \qquad 0 \le g \le 37/16, \qquad 0 \le h \le 5/64, \tag{45}$$

and hence

$$0 \le V_1 \le 967/256, \qquad 0 \le V_0 \le K := 21097/4096.$$

Any two total qualified services on this same initial-state domain have an original-policy cost contrast in $[-\tau, \tau]$.

PROOF: For $d = 2$, the target part of S is at most $2(5/8)^2 = 25/32$, its cyclic term at most $1/4$, and its squared hinge at most $1/2$. The extra target term in g is at most $25/32$. The action interval lies in $[0, 1/4]$, so $h(a) \le 1/16 + 1/64 = 5/64$. These prove (45). Monotonicity and constant preservation of the original AVaR give

$$V_1 \le 49/32 + 5/64 + (15/16)(37/16) = 967/256,$$

and a second application gives

$$V_0 \le 49/32 + 5/64 + (15/16)(967/256) = 21097/4096.$$

Nonnegativity follows from the same property and the nonnegative primitives.

We next verify the finite terminal solve. For a fixed terminal-decision state y, the original one-date objective has derivative bounded in absolute value by

$$|Q_1'(y, a)| \le 3/4 + \beta(5)(3/4) = 273/64 < 7,$$

because each coordinate of ∇g has absolute value at most 5 and $b_1 + b_2 = 3/4$. The derivative is Lipschitz with constant less than 10. To see the latter directly, put $m = F(y, a, 0)$, $k = 5(m_1 - m_2)$. The exact original risk representation is

$$Q_1(y, a) = S(y) + h(a) + \beta\{g(m) + \mathcal{R}_\alpha(k)\}.$$

Here $\nabla^2 g \preceq 13I$ almost everywhere, $m_a = \boldsymbol{b}$, $k_a = 5/4$, and $\mathcal{R}_\alpha'$ is $\alpha/(10t)$-Lipschitz. Consequently a derivative Lipschitz constant is

$$5 + \beta\{65/16 + 5\alpha/(32t)\} < 10$$

for both declared risks. The first derivatives match at the squared-hinge and tail-clipping junctions, so this almost-everywhere bound integrates to the global one.

The exact bisection starts with $[0, A(y)]$ and evaluates its midpoint a_k after k halvings. The constrained optimum remains in the bracket: a negative derivative

discards the left half and a nonnegative derivative the right half. Its bracket length is $A(y)2^{-k}$. At each step the strong-convexity lower support has gap

$$\Gamma_k = -\min_{d\in[-a_k,A(y)-a_k]}\{Q_1'(y,a_k)d + d^2\}.$$

If the optimum is interior, its derivative is zero, so

$$|Q_1'(y,a_k)| \le 10A(y)2^{-k-1}, \qquad \Gamma_k \le \frac{25}{64}4^{-k}.$$

This is below 2^{-30} already at $k = 15$. If the optimum is the lower boundary, strong convexity implies $Q_1'(y,a_k) \ge 2a_k$; the support minimizer is then the lower boundary and $\Gamma_k \le 7a_k \le 7\,2^{-k-3}$. The upper-boundary argument is symmetric. This is below 2^{-30} at $k = 30$. Thus at most 31 midpoint evaluations suffice, including endpoints with zero limiting derivative. The exact clipped-quadratic risk, its derivative and every arithmetic denominator in this calculation are finite for the declared rational inputs.

For the root, the inherited state derivative bound $|\partial_j V_1| \le 15/2$ gives

$$|Q_x'(a)| \le 3/4 + \beta(15/2)(3/4) = 771/128 < 7.$$

Let $m = 1024$ and use actions $a_j = A(x)j/m$, $j = 0,\dots,m$. Every feasible action is within $A(x)/(2m) \le 1/(8m)$ of this grid. If $[L_j, U_j]$ is the complete-policy interval at a_j, its lower endpoint also minorizes $Q_x(a_j)$. The retained convex-order width bound, with $K_z = 15$, $R = 1/32$, $t \ge 1/4$, terminal point error $e = 2^{-30}$, and terminal-policy allowance $\varepsilon_1 = 2^{-20}$, gives

$$U_j - L_j \le W,$$
$$W = \beta\left\{\frac{15}{8\cdot 256^2} + 2e + \varepsilon_1\right\} = \frac{238095}{2^{33}} < 2^{-15}. \tag{46}$$

The convex-order inequality acts on the original V_1. Adding $\beta\varepsilon_1$ to the upper endpoint controls the actual future policy without assuming that policy's value is convex.

The action-cover lower optimum

$$B_0 = \min_j L_j - \frac{7}{8m}$$

is valid. Indeed each arbitrary feasible action is within the stated distance of a grid point, and its Q_x value is at least that grid value minus $7/(8m)$. Choose j_* with the smallest U_j, with the prescribed deterministic tie rule. Since $\min_j U_j - \min_j L_j \le \max_j (U_j - L_j)$,

$$U_{j_*} - B_0 \le W + \frac{7}{8m}.$$

The exact service rounds a_{j_*} down to 40 binary places. The action remains feasible and its displacement is less than 2^{-40}. The original Q_x value changes by less than 72^{-40}. Apply this perturbation to the lower and upper endpoints while retaining the single future allowance already in U_{j_*}. This step uses Lipschitzness of Q_x, not of the possibly discontinuous deployed policy value. The exact returned width and gap are therefore at most

$$W + 142^{-40}, \qquad W + \frac{7}{8192} + 72^{-40},$$

respectively.

The final floating conversion in the retained service rounds each rational endpoint outward by an adjacent binary64 number. Every such endpoint has absolute value below 8, by the primitive range just proved and the displayed allowances. A nearest binary64 conversion followed by one outward adjacent value has absolute error less than 2^{-48} on this range. The two endpoint conversions and the outward final subtraction enlarge a gap or width by less than 2^{-46}. Direct rational inequalities now give

$$W + 142^{-40} + 2^{-46} < 2^{-14} < 1/8192,$$

$$W + 7/8192 + 72^{-40} + 2^{-46} < 2^{-10} < 1/256.$$

Thus the strict margins survive the declared conversion. All 1025 actions use finitely many primitive evaluations: 513 terminal contexts per action, each with the finite

bisection bound above. The earlier numerical stages have finite caps and can terminate in this fallback, so they cannot remove a valid state from the mathematical service's domain.

Finally each qualified service has $0 \le J_x - V_0(x) \le \tau$. Subtracting two such inequalities with the same original optimum gives $[-\tau, \tau]$ for their contrast at every state. *Q.E.D.*

15. PROOFS FOR THE PAID-SUPPORT POLICY SAFEGUARD

All claims below are deterministic statements about the original economic operator. Conditional sampling and population inference, if used in an experiment, require a separately frozen protocol. None of the proofs changes the current information filtration or the terminal controller.

15.1. *Original-gradient cap and the producer's support semantics*

The original state Lipschitz bound (16), specialized to $d = 2$, and the action exposure $\boldsymbol{b} = (1/2, 1/4)$ give

$$|V_1(F(x,a,z)) - V_1(F(x,b,z))| \le \frac{15}{2}\frac{3}{4}|a-b| = \frac{45}{8}|a-b|.$$

Monotonicity and cash invariance imply the sup-norm Lipschitz bound $|\rho(U) - \rho(V)| \le \|U - V\|_\infty$ for the original AVaR. Since $0 \le h'(a) = 2a + 16a^3 \le 3/4$ on $[0, 1/4]$, it follows that

$$|Q_x(a) - Q_x(b)| \le \left(\frac{3}{4} + \frac{15}{16}\frac{45}{8}\right)|a-b| = \frac{771}{128}|a-b| < 7|a-b|.$$

The established differentiability of the original Q_x therefore gives $|Q_x'| \le 771/128 < 7$, including inward endpoint derivatives. This argument holds for both declared risk levels and every original state; it does not estimate a derivative cap from sample gradients.

Here is why the existing original-risk payload supplies (20). Freeze any feasible risk density constant on the original innovation cells B_i, with exact masses w_i and midpoints z_i. At $y_i = F(x,a,z_i)$, retain a global affine terminal minorant $V_1(y) \ge$

$\ell_i + p_i^{\mathsf{T}}(y - y_i)$. Within a cell, the original transition satisfies

$$F(x,b,z) - F(x,a,z_i) = \boldsymbol{b}(b-a) + \boldsymbol{s}(z - z_i).$$

The integral of $z - z_i$ under its constant density is zero. Thus the same fixed feasible risk witness, evaluated at an arbitrary feasible action b, gives

$$Q_x(b) \geq S(x) + h(b) + \beta \sum_i w_i \{\ell_i + p_i^{\mathsf{T}} \boldsymbol{b}(b-a)\}.$$

Using $h(b) \geq h(a) + h'(a)(b-a) + (b-a)^2$ proves the global quadratic support with

$$W = S(x) + h(a) + \beta \sum_i w_i \ell_i, \qquad p = h'(a) + \beta \sum_i w_i \boldsymbol{b}^{\mathsf{T}} p_i.$$

An outward lower W and an interval enclosing the actual p preserve it. The witness may have been selected by sorting lower midpoint values at a; its weights are held fixed in the displayed comparison. It is not implicitly differentiated as an optimizing density at another action. The same proof covers a feasible active-tail density.

The original upper U comes from the convex-order endpoint-mixture calculation for V_1 under the original continuous law. The implementation exposes its separately recomputed upper `(base + BETA * upper_risk).hi` before the terminal-controller allowance. It preserves the actual lower witness separately from the reported policy-interval lower endpoint; they need not have identical rounding. This avoids treating a generic policy upper bound as an exact optimal-future value or subtracting an unidentified error term.

PROOF OF THEOREM 6: For $0 < h \leq A - a$, derivative Lipschitzness and the lower support give

$$W + ph + \frac{\mu}{2}h^2 \leq q(a+h) \leq q(a) + q'(a)h + \frac{L}{2}h^2 \leq U + q'(a)h + \frac{L}{2}h^2.$$

Rearranging gives $q'(a) \geq p - e/h - ch$ and therefore the lower inequality after $p \geq p_-$ and minimization in h. For $0 < h \leq a$ use $a - h$ in the same calculation. Dividing by the negative displacement gives $q'(a) \leq p + e/h + ch \leq p_+ + e/h + ch$. This proves the upper inequality. Integrating the continuous derivative proves the

smooth upper Taylor inequality used here; strong convexity supplies the lower quadratic term. One-sided boundary derivatives have the indicated inward interpretation.

For $e, c > 0$, $e/h + ch$ decreases until $h = \sqrt{e/c}$ and increases thereafter, which proves (21). At $e = 0$ its infimum on $(0, r]$ is zero. If $c = 0$, the infimum is e/r, including zero. When $r = 0$ there is no admissible displacement in that derivation and no finite inference on that side.

We next show sharpness, which concerns the stated information rather than the particular investment formula. Keep an exact slope p, the support level W, the upper level U, and the action interval fixed. Consider

$$q_g(a + d) = U + gd + \frac{L}{2}d^2.$$

This function has L-Lipschitz derivative and is μ-strongly convex because $L \geq \mu$. Its difference from the supplied support is

$$q_g(a + d) - \{W + pd + \tfrac{\mu}{2}d^2\} = e + (g - p)d + cd^2.$$

For all positive feasible d, nonnegativity of this expression is exactly $g - p \geq -\inf_{0<d\leq A-a}(e/d + cd)$. For all negative feasible d it is exactly $g - p \leq \inf_{0<-d\leq a}(e/(-d) + c(-d))$. At $d = 0$ it is $e \geq 0$. Thus every derivative in

$$[p - \Psi(e, c, A - a),\ p + \Psi(e, c, a)]$$

is realized by this quadratic with the same supplied information, and neither side can be tightened. At $a = 0$, arbitrarily large positive g remain feasible, and at $a = A$, arbitrarily negative g do so. If an extra original-gradient bound is intersected, validity persists; sharpness after additional model-specific restrictions is not asserted.

For a concrete boundary warning, set $A = 1/4$, $a = 0$, $\mu = 2$, $L = 10$, $W = U = 0$, and $p = -1$. Both $q_-(b) = -b + 5b^2$ and $q_+(b) = b + 5b^2$ lie above the common support $-b + b^2$ on $[0, 1/4]$. Their inward derivatives at zero have opposite signs. Treating the touching boundary support slope as the actual derivative would falsely certify descent for q_+. *Q.E.D.*

PROOF OF THEOREM 7: Integration of the derivative inequalities, or the defining strong convexity and smoothness inequalities, gives for every feasible d

$$q'(a)d + \tfrac{\mu}{2}d^2 \le q(a+d) - q(a) \le q'(a)d + \tfrac{L}{2}d^2.$$

Taking the extrema over the valid derivative interval proves $\underline{m}(d) \le q(a+d) - q(a) \le \overline{m}(d)$. The future-policy difference belongs to $[-\delta, \delta]$, so addition proves (25). This step uses no convexity, continuity, or differentiability of the implemented future policy's error.

The saved upper value gives $J_x(a+d) \le U + \overline{m}(d) + \delta = U_b$. The quadratic lower support and $J_x \ge q$ give the lower endpoint L_b. For every feasible action, the saved support minorizes q. Its infimum for the actual unknown slope is at least the smaller of the infima for p_- and p_+, since for each displacement the affine function of p is bounded below by its endpoint minimum. Thus $B \le \min_{b\in[0,A]} q(b) = V_0(x)$. Taking the maximum with another valid original-optimum lower bound preserves this inequality. Subtracting proves the regret bound. A feasible implemented policy cannot beat V_0, by the monotone original recursion and its feasible last-date controller, which gives the nonnegative side.

On $d \ge 0$, the robust upper model equals $g_+ d + Ld^2/2$; on $d \le 0$ it equals $g_- d + Ld^2/2$. Each is convex. If the interval straddles zero, both pieces are nonnegative and $d = 0$ minimizes them. If $g_- > 0$, the nonpositive part has its unconstrained minimizer $-g_-/L$, clipped to $[-a, 0]$; the positive part cannot improve on zero. The negative-gradient case is symmetric. This proves the displayed proposal formula. The three actual acceptance checks concern only the realized feasible action, so a rounded action is valid when its newly computed inequalities pass. No bound for a different action is reused.

A strict negative upper contrast proves actual improvement. The two absolute tests prove qualification. At an accepted sequence, sum the statewise inequalities $J(a_{j-1}) - J(a_j) \ge -\overline{m}_j(d_j) - \delta$; the actual costs telescope. Retention steps add zero. Every accepted action is feasible and qualified by its own three tests, completing the induction. A finite measurable rule with fixed tie and failure conventions supplies

an economic policy. The test is valid even when the proposal was produced by a nonconvex critic, since the premises concern the original q.

For the claimed information sharpness only, take an interior anchor, an exact supplied support slope, and the exact untruncated interval (22), before imposing any further original-model restriction. Use the quadratic q_g in the preceding proof with the endpoint of that exact feasible gradient interval maximizing gd. It gives $q_g(a+d) - q_g(a) = \overline{m}(d)$. There is no representation-dependent slack to remove under those data. The additional future-policy floor can also be attained within the interface $0 \le \eta \le \delta$ at two distinct actions by setting $\eta(a) = 0, \eta(a+d) = \delta$. This reproduces the retained R120 future-information limitation and is not a new claim about what the particular terminal controller must realize. *Q.E.D.*

PROPOSITION S.20—A checkable interior precision ladder: *Let $a \in (0, A)$, $L > \mu$, and suppose level k supplies the information in (20), with $e_k \le e_0 4^{-k}$ and $p_{k,+} - p_{k,-} \le w_0 2^{-k}$. At every level satisfying $e_0 4^{-k} \le (L-\mu)\min\{a, A-a\}^2/2$, the exact derivative interval has width at most $D2^{-k}$, where $D = w_0 + 2\sqrt{2(L-\mu)e_0}$. Let $\Gamma_k = -\min_d \overline{m}_k(d)$ be the unrounded robust smooth-model gain. Then*

$$0 \le G_L(a, q'(a)) - \Gamma_k \le AD2^{-k}.$$

With a realized rounding displacement error at most ρ, an additional conservative loss allowance $(G + LA)\rho$ is valid whenever the gradient interval used in the robust model lies in $[-G, G]$.

PROOF: The interior condition permits the minimizing positive h on both sides. The uncertainty on each side is $\sqrt{2(L-\mu)e_k}$, which yields the width bound. Let $I = [l, u]$ be that interval and let $g = q'(a) \in I$. Here I is the exact-infimum interval; a larger finite-radius numerical interval requires its actual width in this argument. The scalar g_0 in I closest to zero is l if $l > 0$, u if $u < 0$, and zero otherwise. The formula for the robust proposal shows $\Gamma_k = G_L(a, g_0)$. One may also see this by separating the positive and negative displacement domains. If $g_0 = 0$, the robust gain is zero; otherwise it uses the same sign as every $g \in I$ and the smallest magnitude. Therefore $G_L(a, g_0) \le G_L(a, g)$. The maximum of the functions $-gd - Ld^2/2$ is A-Lipschitz in

g, since every feasible d satisfies $|d| \leq A$. Also $|g - g_0| \leq u - l \leq D2^{-k}$. These prove the assertion.

For rounding, on the feasible displacement interval each piece of $\overline{m}$ has derivative of magnitude at most $G + LA$. The pieces meet continuously at zero, so the same bound is global. A displacement perturbation of at most ρ therefore increases the robust upper model by at most $(G + LA)\rho$. In actual code it is preferable to reevaluate the outward model at the realized action, rather than to spend this extra allowance. *Q.E.D.*

This proposition gives a finite gain stopping bound under explicit oracle guarantees and a strict margin. Its scope must be kept separate from the qualification-width test: as e and slope uncertainty tend to zero, (24) can still have width $(L - \mu)d^2/2 + \delta$. Additional candidate information is necessary when that curvature width exceeds the prescribed requirement. The $L = \mu$ case has $\Psi(e, 0, r) = e/r$ and admits the corresponding faster algebraic bound, but no nonzero-capacity or interior room can be inferred at a boundary.

PROPOSITION S.21—A priced stopping region: *Suppose an implemented incumbent has certified original regret at most $K \geq 0$. Suppose every admissible continuation of a single-decision solver incurs at least c_* extra computation units before it can change the decision, these units cost $\lambda \geq 0$ per unit in original-objective units, and no benefits on other decisions or later reuse are included. If $\lambda c_* \geq K$, stopping weakly dominates every such continuation in original objective plus computation cost.*

PROOF: Let J be the incumbent's original cost and J' the final cost of any feasible continuation. Feasibility gives $J' \geq V_0$, so $J - J' \leq J - V_0 \leq K$. The continuation's incremental computation cost is at least λc_*, which is no smaller than that maximum possible saving. Its total cost cannot be lower than stopping. Sunk computation cancels from the comparison. The conclusion fails to cover reuse benefits by construction; those require their own economic account and cannot be added to a single-decision calculation afterwards. *Q.E.D.*

16. MATCHING GUARANTEES AT A FIXED PAID-INFORMATION INTERFACE

The sharpness witnesses in Theorem 6 imply a conditional information bound for the gain component of a decision. We state the deduction explicitly because comparing representations requires the same information and the same feasible actions. The conclusion concerns a particular scalar support interface. It does not assert a general new minimax optimization principle, a lower bound for access to the full investment formulas, or a neural-specific advantage.

Fix numbers

$$\mathcal{I} = (A, a, \mu, L, W, U, p), \qquad A > 0, \quad 0 < a < A, \quad 0 < \mu \le L, \quad U \ge W.$$

Here the support slope p is supplied exactly. Let $\mathcal{D} = [-a, A - a]$, $e = U - W$, $c = (L - \mu)/2$, and

$$I = [g_-, g_+] = [p - \Psi(e, c, A - a),\ p + \Psi(e, c, a)].$$

Both endpoints are finite. This is the exact, untruncated interval: there is no finite-probe enlargement and no additional gradient bound. Let $\mathcal{Q}(\mathcal{I})$ consist of all continuously differentiable functions $q : [0, A] \to \mathbb{R}$ that are μ-strongly convex, have L-Lipschitz derivative, and satisfy

$$q(a) \le U, \qquad q(a + d) \ge W + pd + \frac{\mu}{2} d^2 \quad (d \in \mathcal{D}).$$

No other restrictions on q are imposed. In particular, membership does not assert that q is generated by the investment model. Write

$$\sigma_I(d) = \max_{g \in I} gd, \qquad \overline{m}(d) = \sigma_I(d) + \frac{L}{2} d^2, \qquad \Gamma = -\min_{d \in \mathcal{D}} \overline{m}(d) \ge 0.$$

The robust proposal $d^\star$ in Section 6 attains this minimum.

An admissible method receives only the fixed information $\mathcal{I}$ and chooses a Borel probability law ν on $\mathcal{D}$, without another objective, derivative, or supporting-value query. A deterministic method is a point mass. Write $Z \sim \nu$. The adverse function is selected knowing the announced law, before the displacement is drawn. Thus the adversary does not observe or respond to a realized random draw. Arbitrary internal calculations and representation labels do not enlarge the supplied information.

PROPOSITION S.22—A matching bound for the fixed support interface: *For every admissible law ν,*

$$\sup_{q\in\mathcal{Q}(\mathcal{I})} \mathbb{E}_\nu[q(a+Z)-q(a)] = \sigma_I(\mathbb{E}_\nu Z) + \frac{L}{2}\mathbb{E}_\nu[Z^2] \tag{47}$$
$$= \overline{m}(\mathbb{E}_\nu Z) + \frac{L}{2}\operatorname{Var}_\nu(Z).$$

The supremum is attained. Consequently

$$\inf_\nu \sup_{q\in\mathcal{Q}(\mathcal{I})} \mathbb{E}_\nu[q(a+Z)-q(a)] = \min_{d\in\mathcal{D}} \overline{m}(d) = -\Gamma. \tag{48}$$

The deterministic robust proposal attains the value. Randomization cannot increase the guaranteed gain.

For the implemented-policy interface, let $\mathcal{E}_\delta$ be the set of all Borel measurable functions $\eta : [0,A] \to [0,\delta]$, where $\delta \geq 0$, and let $J = q + \eta$. Impose no continuity, convexity, or other information on η. Then for every admissible ν,

$$\sup_{\substack{q\in\mathcal{Q}(\mathcal{I})\\ \eta\in\mathcal{E}_\delta}} \mathbb{E}_\nu[J(a+Z)-J(a)] = \sigma_I(\mathbb{E}_\nu Z) + \frac{L}{2}\mathbb{E}_\nu[Z^2] + \delta\,\mathbb{P}_\nu\{Z\neq 0\}. \tag{49}$$

This supremum also is attained, and

$$\inf_\nu \sup_{\substack{q\in\mathcal{Q}(\mathcal{I})\\ \eta\in\mathcal{E}_\delta}} \mathbb{E}_\nu[J(a+Z)-J(a)] = \min\{0, -\Gamma+\delta\}. \tag{50}$$

The largest uniformly guaranteed implemented-policy gain over this interface is therefore $\max\{0,\Gamma-\delta\}$. *It is attained by retaining the incumbent when $\Gamma \leq \delta$ and choosing $d^\star$ when $\Gamma > \delta$. At equality either action is minimax, although retention incurs no policy change.*

PROOF: Theorem 6 gives $q'(a) \in I$ for every $q \in \mathcal{Q}(\mathcal{I})$. Conversely, for each $g \in I$ define

$$q_g(a+d) = U + gd + \frac{L}{2}d^2, \qquad d \in \mathcal{D}.$$

This function has constant curvature L, is μ-strongly convex, and has the required derivative Lipschitz constant. The support difference is

$$q_g(a+d) - \left(W + pd + \frac{\mu}{2}d^2\right) = e + (g-p)d + cd^2.$$

For $d > 0$, its nonnegativity is equivalent to $g \geq p - e/d - cd$; requiring this for all positive feasible d is exactly $g \geq g_-$. For $d < 0$, the equivalent condition for all negative feasible d is $g \leq g_+$. At $d = 0$ the difference is $e \geq 0$. Thus every q_g belongs to the information class and has $q_g(a) = U$. This verifies the witnesses against all the supplied information, including the support away from the anchor.

For an arbitrary q in the class, the smooth upper Taylor inequality with its single derivative $g = q'(a)$ holds simultaneously at every feasible displacement:

$$q(a+d) - q(a) \leq gd + \frac{L}{2}d^2.$$

The compact displacement interval makes these random differences integrable. Taking expectations and then the supremum gives

$$\sup_q \mathbb{E}_\nu[q(a+Z) - q(a)] \leq \max_{g \in I} \left\{ g\mathbb{E}_\nu Z + \frac{L}{2}\mathbb{E}_\nu[Z^2] \right\}.$$

Choose an endpoint g maximizing $g\mathbb{E}_\nu Z$, or any $g \in I$ when the mean is zero. The single function q_g constructed above gives equality for every displacement, hence for the announced law. This proves the first line of (47); the variance identity proves its second line. Since $\mathbb{E}_\nu Z \in \mathcal{D}$, the right side is at least $\min_{\mathcal{D}} \overline{m}$. The point mass at $d^\star$ attains that lower bound. This proves (48). In particular the argument does not exchange a supremum and an expectation of pointwise adversarial choices.

For the implemented future, the difference $\eta(a+d) - \eta(a)$ is at most δ if $d \neq 0$, and is exactly zero if $d = 0$. Consequently its expectation is bounded above by $\delta \mathbb{P}_\nu\{Z \neq 0\}$. The single Borel function

$$\eta_*(b) = \begin{cases} 0, & b = a, \\ \delta, & b \neq a \end{cases}$$

attains this bound for every displacement at once. It can be combined with the same q_g that attained the preceding expectation. This proves the exact formula (49). In particular a penalty δ cannot be charged to retaining the identical action.

It remains to show that mixing retention and replacement cannot improve the worst-case guarantee. Put $r = \mathbb{P}_\nu\{Z \neq 0\}$. If $r = 0$, (49) equals zero. If $r > 0$, write $v = \mathbb{E}_\nu[Z \mid Z \neq 0] \in \mathcal{D}$ and $s^2 = \mathrm{Var}_\nu(Z \mid Z \neq 0) \geq 0$. Positive homogeneity of the interval support function gives

$$\sigma_I(\mathbb{E}_\nu Z) + \frac{L}{2}\mathbb{E}_\nu[Z^2] + \delta r = r\left\{\overline{m}(v) + \frac{L}{2}s^2 + \delta\right\} \geq r(-\Gamma + \delta) \geq \min\{0, -\Gamma + \delta\}.$$

For the last inequality, if $-\Gamma + \delta$ is nonnegative the left bound is nonnegative; if it is negative, multiplication by $r \leq 1$ increases it. Retention attains zero. When $\Gamma > \delta$, $\Gamma > 0$ implies $d^\star \neq 0$, and the point mass at $d^\star$ attains $-\Gamma + \delta$. If $\Gamma = \delta > 0$, that same nonzero point mass and retention both attain zero. If $\Gamma = \delta = 0$, the robust minimizer is retention, also attaining zero. These cases establish (50) and the stated guarantee. *Q.E.D.*

The upper and lower guarantees match for every deterministic or randomized method at this same interface, whether its proposal is represented by a neural network, a hinge basis, a spline, a polynomial, or no fitted representation. The conclusion concerns achievable certified improvement after one fixed payload; it is not a lower bound on the number of original risk evaluations needed to acquire that payload, on global learning, or on multistage Bellman error propagation. Nor is it a dominance result after charging distinct methods' computational costs.

The investment problem supplies additional structure, including its exact stage cost, risk formulas, derivative cap, and fixed terminal controller. Its implemented finite-radius, outward-rounded intervals, six-candidate menu, candidate precision requirement, and complete-policy qualification checks also differ from this exact interface. Accordingly Proposition S.22 does not assert minimax optimality of the executed service or impossibility of improvement using those additional data. The functions q_g and η_* prove a lower bound only over the stated abstract information class; they are not asserted to be attainable investment-model instances or error func-

tions of the fixed terminal controller. The original-policy acceptance requirements remain those of Theorem 7.

This analytic corollary was derived after the R122 scientific freeze. It changes neither the frozen proposal rule nor the experiment catalogue, samples, selected parameters, numerical results, or inference procedure.

17. FROM ACTION SAFEGUARDS TO GAINS IN THE SAME RECURSION

The paid-support safeguard compares two actions while retaining a specified future policy. Extending the economic gain to several changed dates requires an additional recursion. The physical transition law alone cannot propagate that gain under a nonlinear risk preference. We give the appropriate lower recursion, retaining the original state, feasible policy, conditional risk primitive and information set. This is an action-gain application of the reference-centered representation in Section 4 and of risk-averse dynamic programming (Ruszczyński, 2010, 2014). It is not a new general policy-iteration principle, or a numerical result for a longer horizon than the executed investment experiment.

17.1. *Original-policy advantages and centered lower evaluations*

At dates $t = 0, \dots, T$, let $\mathcal{X}_t$ be measurable state spaces. For each complete feasible measurable policy ζ, define its original cost by

$$V_T^\zeta = g, \qquad V_t^\zeta(x) = c_t(x, \zeta_t(x)) + \beta_t \rho_{t,x,\zeta_t(x)}(V_{t+1}^\zeta), \qquad \beta_t \geq 0. \tag{51}$$

The terminal cost and the selected stage costs are bounded and measurable. For every selected action, the original conditional evaluation is finite on bounded measurable functions, is monotone, cash invariant and normalized at zero, and maps bounded measurable continuations to bounded measurable state functions. These hypotheses ensure that the finite recursions below are defined. They also specify the economic preference; replacing a static risk objective by a nested one is not allowed by these hypotheses.

Suppose the selected original risk primitives have the dual representation (8) on this same function domain. The admissible q are probability measures, their penalties $\psi(q)$ may take the value $+\infty$, and the set of finite-penalty measures is

nonempty. All integrals below are with respect to these original successor-state measures. Attainment of a risk supremum is not assumed.

Fix an incumbent π and a proposed complete feasible policy π'. The original incumbent continuation, not a newly fitted or altered objective, is used in the one-step advantage

$$b_t^*(x) = V_t^\pi(x) - c_t(x, \pi_t'(x)) - \beta_t \rho_{t,x,\pi_t'(x)}(V_{t+1}^\pi). \tag{52}$$

Assume a bounded measurable certificate supplies

$$0 \le b_t(x) \le b_t^*(x) \quad \text{at every state and date.} \tag{53}$$

It suffices to impose this on an explicitly specified invariant reachable state domain supporting every kernel admitted by either policy's original risk evaluation and by the exact or conservative lower operator. This includes a kernel whose originally infinite penalty is relaxed to a finite penalty, including zero. A finite sample check alone is not this all-state hypothesis. Retaining the incumbent at a state gives $b_t^*(x) = b_t(x) = 0$ exactly.

For $v = V_{t+1}^\pi$, center at the *updated* action $a = \pi_t'(x)$:

$$k_t^\pi(x,q) = \rho_{t,x,a}(v) - \int v\, dq + \psi_{t,x,a}(q), \tag{54}$$

$$\mathcal{L}_t^\pi h(x) = \rho_{t,x,a}(v) - \rho_{t,x,a}(v-h) = \inf_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int h\, dq + k_t^\pi(x,q) \right\}. \tag{55}$$

The equality is algebraic: subtract the risk supremum at $v - h$ from the finite risk value at v. In particular, this is an infimum with a *positive* penalty. Using the supremum operator for an upper error account here would reverse the economic guarantee.

LEMMA S.23—The lower centered operator: *The penalties in* (54) *are nonnegative and have infimum zero. The operator* $\mathcal{L}_t^\pi$ *is monotone, cash invariant, concave, normalized at zero and nonexpansive in the supremum norm. If verified penalties satisfy*

$$0 \le \underline{k}_t(x,q) \le k_t^\pi(x,q) \quad (q \in \mathcal{Q}_{t,x,\pi_t'(x)}),$$

then the operator

$$\underline{\mathcal{L}}_t h(x) = \inf_{q \in \mathcal{Q}_{t,x,\pi'_t(x)}} \left\{ \int h \, dq + \underline{k}_t(x,q) \right\}$$

has these same properties and satisfies $\underline{\mathcal{L}}_t h \le \mathcal{L}_t^{\pi} h$. For this second operator, measurability of its statewise outputs is also assumed. Zero penalties are an allowed conservative choice.

PROOF: Each feasible dual objective at v is bounded above by its supremum, so $k_t^{\pi} \ge 0$. Approximation of that supremum by feasible measures gives $\inf_q k_t^{\pi} = 0$, including when no maximizing measure exists. Since $0 \le \underline{k}_t \le k_t^{\pi}$, the lower penalties also have infimum zero. An infimum of affine functions of h is concave. Probability measures preserve order and integrate a constant to that constant; taking infima proves monotonicity and cash invariance. The zero infimum of the penalties proves normalization. If $\|h - \widetilde{h}\|_{\infty} \le \epsilon$, monotonicity and cash invariance give $\mathcal{L}_t^{\pi}\widetilde{h} - \epsilon \le \mathcal{L}_t^{\pi} h \le \mathcal{L}_t^{\pi}\widetilde{h} + \epsilon$, and the same proof applies to $\underline{\mathcal{L}}_t$. Taking infima after the pointwise penalty inequality gives the claimed operator order. These properties also show that each output lies between $-\|h\|_{\infty}$ and $\|h\|_{\infty}$; with the stated measurability hypothesis it is a bounded measurable state function. *Q.E.D.*

THEOREM S.24—Propagation of certified action gains: *Under the preceding assumptions, let $D_t = V_t^{\pi} - V_t^{\pi'}$. Define the exact and conservative lower-gain recursions by*

$$\begin{aligned} H_T = \underline{H}_T &= 0, \\ H_t &= b_t + \beta_t \mathcal{L}_t^{\pi} H_{t+1}, \\ \underline{H}_t &= b_t + \beta_t \underline{\mathcal{L}}_t \underline{H}_{t+1}. \end{aligned} \tag{56}$$

Then, at every state and date,

$$0 \le \underline{H}_t \le H_t \le D_t, \qquad V_t^{\pi'} \le V_t^{\pi}. \tag{57}$$

Moreover, any bounded measurable nonnegative functions h_t with $h_T = 0$ and

$$h_t \le b_t + \beta_t \underline{\mathcal{L}}_t h_{t+1}$$

are valid lower certificates: $h_t \le \underline{H}_t \le D_t$. Thus a verified subsolution can certify the gain without solving the lower recursion exactly.

PROOF: Subtract the two original-policy recursions, add and subtract $c_t(x, \pi'_t(x)) + \beta_t \rho_{t,x,\pi'_t(x)}(V^{\pi}_{t+1})$, and use (55). This gives the exact identity

$$D_T = 0, \qquad D_t = b^*_t + \beta_t \mathcal{L}^{\pi}_t D_{t+1}. \tag{58}$$

The terminal equality uses the same original terminal cost for both policies. If $D_{t+1} \ge H_{t+1} \ge \underline{H}_{t+1} \ge 0$, monotonicity, the nonnegative discount and $b^*_t \ge b_t \ge 0$ give $D_t \ge H_t \ge \underline{H}_t \ge 0$. Backward induction from the terminal equality proves (57). The same induction using the subsolution inequality gives $h_t \le \underline{H}_t$. No physical expectation, fixed adverse kernel, change of risk preference, or equality of successive policies has been inserted in this calculation. *Q.E.D.*

The theorem supplies the missing link between a safe action update and its economic effect through a common recursion. If only a single date changes, its advantage is propagated through the unchanged earlier risk operators. If several dates change, each advantage is still computed against the same incumbent continuation sequence; evaluating datewise advantages under unrelated fitted futures would not prove the result. Repeated complete-policy updates can be certified by applying the theorem anew to each incumbent. For finitely many such updates, the sum of their statewise lower-gain certificates is a lower bound on the total gain, because the actual original values telescope. This is not a convergence rate to the globally optimal policy.

17.2. *A finite occupation calculation with the correct orientation*

There is an implementable finite-dimensional form when the state sets are finite and each infimum in the selected lower operator has an exact representation by a nonempty finite complete menu with finite real penalties. Any piece with penalty $+\infty$ is omitted: bounded continuations and normalization make the infimum finite, and such a piece cannot affect it. Write the resulting entire menu at (t, x) as q_{txj}, $j \in \mathcal{J}_{tx}$, and its verified nonnegative finite penalties as $\underline{k}_{txj}$. The word complete

means that minimizing over this menu gives the specified lower operator for every continuation used below; it is not a list of favorable witnesses sampled from a larger risk set. Let ν_0 be an initial-state probability vector and put $B_0 = 1$, $B_{t+1} = B_t \beta_t$.

PROPOSITION S.25—Lower-gain occupations: *Under this finite complete-menu assumption,* $\nu_0^\top \underline{H}_0$ *equals the minimum of*

$$\sum_{t=0}^{T-1} B_t \sum_x m_t(x) b_t(x) + \sum_{t=0}^{T-1} B_{t+1} \sum_{x,j} y_t(x,j) \underline{k}_{txj} \tag{59}$$

over nonnegative flows satisfying

$$m_0 = \nu_0, \qquad \sum_j y_t(x,j) = m_t(x),$$

$$m_{t+1}(x') = \sum_{x,j} y_t(x,j) q_{txj}(x'). \tag{60}$$

Every verified subsolution in Theorem S.24 gives a lower bound $\nu_0^\top h_0$ *on this minimum. A feasible flow alone gives an* upper *bound on the minimum and therefore does not certify the original policy gain.*

PROOF: The lower recursion is a finite-horizon minimizing control problem with risk index j, stage cost $b_t(x) + \beta_t \underline{k}_{txj}$, transition q_{txj} and zero terminal cost. Finite minima attain a minimizing Markov choice at each date and state. Backward induction therefore gives its value $\underline{H}_t$.

Any possibly history-dependent risk-index rule induces the nonnegative occupation masses in (60). Conversely, given a feasible flow, use the conditional probabilities $y_t(x,j)/m_t(x)$ when $m_t(x) > 0$, and any available index when the mass is zero. Forward induction recovers exactly the given masses. Its expected discounted cost is (59), proving the equality of minima.

Alternatively, multiply $h_t(x) \le b_t(x) + \beta_t \{\sum_{x'} q_{txj}(x') h_{t+1}(x') + \underline{k}_{txj}\}$ by $B_t y_t(x,j)$ and sum over all t, x, j. The flow equalities cancel the intermediate terms; the zero terminal value leaves $\nu_0^\top h_0$ no greater than the flow objective. This proof does not divide by a discount and also covers zero discounts. Minimizing preserves the

lower bound, whereas inserting one feasible flow can only bound the minimum from above. *Q.E.D.*

These are occupations under original risk-envelope kernels, not the physical law of the investment state. For a continuous risk set, replacing the infimum by a subset increases it and can destroy the lower certificate. A numerical implementation must instead bound the complete infimum from below, possibly by a verified relaxation or an explicitly bounded integration error. The positive penalty and the minimization direction are both essential. Discarding nonnegative penalties is conservative here; freezing a single reference-maximizing risk kernel need not be.

17.3. *The paid payload and the original incumbent*

To apply the action safeguard at several dates, its scalar objective must represent the original one-step cost under the incumbent future:

$$Q_t^\pi(x,a) = c_t(x,a) + \beta_t \rho_{t,x,a}(V_{t+1}^\pi).$$

Suppose, at a given state and date, one has proved a decomposition $Q_t^\pi(x,a) = q_t(x,a) + \eta_t(x,a)$, with the strong convexity, smoothness and paid-support assumptions of Theorem 7 for q_t, and an all-action bound $0 \le \eta_t(x,a) \le \delta_t(x)$. Then the realized feasible candidate with checked displacement d has the advantage lower bound

$$b_t(x) = -\overline{m}_{t,x}(d) - \delta_t(x) > 0$$

when its gain test passes. Otherwise retention gives $b_t(x) = 0$. A finite candidate menu and fixed tie conventions give a measurable policy when the evaluated inputs are measurable. The all-state hypotheses and any additional policy-qualification tests must also hold before Theorem S.24 is invoked.

In the executed two-date economy, the unchanged last-date controller provides its already proved allowance. That particular numerical allowance cannot be reused at every date of a longer learned recursion without a new proof. If q_t uses the original optimal future while the actor uses V_{t+1}^π, a valid nonnegative allowance follows, for example, from $0 \le V_{t+1}^\pi - V_{t+1}^* \le \epsilon_{t+1}$ uniformly and the original monotone cash-

invariant risk evaluation: $0 \leq \eta_t \leq \beta_t \epsilon_{t+1}$. The continuation level and its error must therefore be retained and propagated even when the action choice uses marginal information. No cost or sample claim for such a longer implementation is made here.

17.4. *A stationary extension with a complete domain*

Let $\mathcal{X}$ be a measurable state space and let $B_b(\mathcal{X})$ be the Banach space of all bounded measurable real functions with the supremum norm. Fix feasible measurable stationary policies π, π' and a discount $0 \leq \beta < 1$. Assume their selected stage costs are bounded, and their original normalized, monotone, cash-invariant risk maps send *every* member of $B_b(\mathcal{X})$ to a member of $B_b(\mathcal{X})$. Assume the stated dual representations on this entire space. These complete-domain conditions are needed for the contraction argument; a collection containing only the eventual solutions is not a replacement for them.

Define $T_\zeta v(x) = c(x, \zeta(x)) + \beta \rho_{x,\zeta(x)}(v)$ for $\zeta \in \{\pi, \pi'\}$.

PROPOSITION S.26—Stationary certified policy improvement: *The maps $T_\pi, T_{\pi'}$ have unique fixed points $V^\pi, V^{\pi'} \in B_b(\mathcal{X})$. Form the advantage b^*, centered operator $\mathcal{L}^\pi$, and a bounded measurable certificate $0 \leq b \leq b^*$ by their stationary versions above. Let $\underline{\mathcal{L}}$ be a verified lower-penalty operator that maps all of $B_b(\mathcal{X})$ to $B_b(\mathcal{X})$. Then the maps*

$$h \longmapsto b + \beta \mathcal{L}^\pi h, \qquad h \longmapsto b + \beta \underline{\mathcal{L}} h$$

have unique fixed points $H, \underline{H}$, and

$$0 \leq \underline{H} \leq H \leq V^\pi - V^{\pi'}.$$

Iteration of either lower-gain map from zero increases monotonically and converges uniformly to its fixed point. Every bounded nonnegative subsolution of the conservative map is a valid original-gain lower certificate.

PROOF: The monetary risk assumptions imply nonexpansiveness on the supremum norm by the order and cash-invariance argument in Lemma S.23. Thus T_ζ is a β-contraction of the stated Banach space into itself. The contraction theorem

gives its unique fixed point. Lemma S.23 makes each gain map a contraction on the same complete space, proving existence, uniqueness and uniform convergence of its iterates.

Normalization and $b \geq 0$ give a nonnegative first iterate from zero. Monotonicity makes all subsequent iterates nondecreasing. The exact identity

$$D := V^{\pi} - V^{\pi'} = b^* + \beta \mathcal{L}^{\pi} D$$

holds by subtracting the two original fixed-point equations. The map on the right is also a contraction. Its iterates from zero are nonnegative because $b^* \geq b \geq 0$, and converge to D. Comparing the three iterations step by step, using the operator order and $b \leq b^*$, yields $0 \leq \underline{H} \leq H \leq D$ in the limit. If $h \geq 0$ is a conservative subsolution, iteration from h is nondecreasing and, by contraction, converges uniformly to $\underline{H}$. Hence $h \leq \underline{H} \leq D$. *Q.E.D.*

The stationary statement proves improvement under its all-state original- objective premises. It does not prove convergence of an arbitrary learned policy-iteration sequence to the optimum, a complexity bound for computing the risk infima, a neural representation advantage, or a complete-cost adoption result. Its economic use is precise: a verified action advantage can be propagated through the same recursive preference while preserving the continuation information and adverse-kernel effects that a pointwise fit or a physical expectation would miss.

18. JOINT INFERENCE FOR THE FIXED ORIGINAL POLICIES

This section gives the probability argument implemented by `code122/stats122.py`. It concerns costs of the original investment problem and the exact fixed catalogue. It does not provide a population adoption theorem for a neural representation. All statements condition on the information available before the new state acquisition, including the inherited actors, the frozen source, and its deterministic rules. They do not condition on the realized values of the subsequently acquired states.

Population, numerical premises, and the economic random variables. Let

$$\mathcal{X}_{40} = \left\{0, 2^{-40}, \ldots, (2^{40}-1)2^{-40}\right\}^2, \qquad P_{40} = \mathrm{Unif}(\mathcal{X}_{40}).$$

Assume the prescribed entropy acquisition supplies independent uniform 40-bit coordinates, independently of the pre-acquisition information. The sample is then $X_1, \ldots, X_n$ independently distributed according to P_{40}, with $n = 32$. Sampling is with replacement; no state is excluded or replaced because of its economic outcome. Both risk levels and all methods use these same state units. The original economic innovations remain continuously uniform on their declared interval. Discretization of the initial-state sampling law does not replace that continuous innovation law.

For a fixed risk index and state x, let $V(x)$ denote the optimal value of the original problem and let $J_m(x)$ be the original complete policy cost delivered by method m, including its prescribed future policy. The statistical premises are that these policies are well-defined measurable state functions and, uniformly on $\mathcal{X}_{40}$,

$$V(x) \le J_m(x) \le V(x) + \tau, \qquad \tau = 2^{-8}. \tag{61}$$

Every policy uses the same original objective, information set, feasible decisions, and interpretation of its future-policy cost. In particular, agreement of first actions can be used as a cost identity only because the compared methods also use the same prescribed future policy.

For each of the 58 actor/risk cells, denote its separately paid control by $m0$, and its recycled and adaptive services by me. The support-to-action safeguards prove, state by state,

$$0 \le c_{me}(x) \le G_{me}(x) := J_{m0}(x) - J_{me}(x) \le \tau. \tag{62}$$

Here c_{me} is the reported accumulated certified gain; on retention of the incumbent it is zero. The upper bound follows from (61), not from an actor fitting loss. For any two qualified methods m, b at the same risk index,

$$H_{mb}(x) := J_b(x) - J_m(x) \in [-\tau, \tau]. \tag{63}$$

Consequently the support for signed comparisons has twice the width of the support for guaranteed own-control improvement.

The independent evaluator supplies valid outward enclosures of the same original complete-policy costs and their paired differences. Writing its raw safe-gain interval as $[\underline{g}^I_{mj}, \overline{g}^I_{mj}]$, the interval used by the frozen statistics program is

$$\ell_{mj} = \max\{0, c_m(X_j), \underline{g}^I_{mj}\}, \qquad u_{mj} = \min\{\tau, \overline{g}^I_{mj}\}. \tag{64}$$

Under the stated numerical premises, $\ell_{mj} \le G_m(X_j) \le u_{mj}$. The raw independent interval and the proof-supported intersection are distinct objects and are both reported. Taking their intersection does not independently validate the proof-derived endpoint. A conflicting or missing enclosure is a failure to satisfy the numerical premises; it cannot be repaired by deleting a sample or by relabelling an unresolved independent comparison as a confirmation.

For a signed comparison, subtract the independent policy enclosures in the cost-difference direction and intersect with $[-\tau, \tau]$. If the saved exact action identity and the common future-policy rule prove equality on a sampled state, its paired interval is $[0, 0]$. This equality for finitely many sampled states does not make the population difference of two distinct methods identically zero. The frozen program uses the exact population interval $[0, 0]$ only when the full method key equals the comparator key.

Computational domain of the uniform premise. The all-state premises here concern the dyadic initial-state support $\mathcal{X}_{40}$. The finite exact-rational service argument in Section 8, together with the original feasible fallback and policy qualification, is a mathematical totalization on that domain. It is not an observed uniform wall-time or memory bound. Infrastructure failures and incomplete attempts remain recorded; no operational failure is silently converted to an extra accepted observation. A transfer from P_{40} to a continuously uniform initial-state population would require an independently justified modulus for the actual implemented policy cost, or a separate acquisition and rounding argument. Branching, acceptance, and stopping decisions can make the implemented policy discontinuous, so continuity of the original primitive alone does not supply this transfer. These qualifications do not change the continuous economic innovation distribution.

LEMMA S.27—A bounded-mean inequality with interval observations: *Let $Y_1, \dots, Y_n$ be independent copies of $Y \in [a,b]$, $a < b$, and put $\mu = \mathbb{E}Y$. For $r > 0$,*

$$\mathbb{P}\{|\overline{Y} - \mu| > r\} \le 2\exp\{-2nr^2/(b-a)^2\}.$$

If data-dependent outward endpoints obey $\ell_j \le Y_j \le u_j$ for every sample, then, with at least the same coverage, μ belongs to

$$[\max\{a, \overline{\ell} - r\}, \min\{b, \overline{u} + r\}].$$

No independence of the endpoints from their own observations is needed.

PROOF: For any bounded random variable Z on $[a,b]$, under any probability law on that interval,

$$\mathrm{Var}(Z) \le \mathbb{E}\big(Z - (a+b)/2\big)^2 \le (b-a)^2/4.$$

For real t, set $\psi(t) = \log \mathbb{E}\exp(t(Y-\mu))$. Boundedness permits differentiation under the expectation. The second derivative is the variance of Y under the exponentially tilted probability law, hence $0 \le \psi''(t) \le (b-a)^2/4$. Since $\psi(0) = \psi'(0) = 0$, the integral remainder formula gives, also for negative t,

$$\psi(t) = t^2 \int_0^1 (1-s)\psi''(st)\, ds \le t^2(b-a)^2/8.$$

Independence and Markov's inequality therefore imply, for $\lambda > 0$,

$$\mathbb{P}\{\overline{Y} - \mu \ge r\} \le \exp\{-\lambda n r + n\lambda^2(b-a)^2/8\}.$$

Choosing $\lambda = 4r/(b-a)^2$ gives the upper-tail bound; replacing Y by $-Y$ gives the lower-tail bound. Their union gives the displayed two-sided inequality (and also controls strict failure). Finally $\overline{\ell} \le \overline{Y} \le \overline{u}$. On the event $|\overline{Y} - \mu| \le r$, the mean lies between $\overline{\ell} - r$ and $\overline{u} + r$; the known support justifies clipping to $[a,b]$. *Q.E.D.*

LEMMA S.28—Exact one-sided event inversion: *Let $K \sim \mathrm{Bin}(n,p)$, fix $0 < \delta < 1$, and define $L_0 = 0$. For $1 \le k \le n$, let L_k be the unique root in $(0,1)$ of*

$$B_k(v) := \sum_{j=k}^{n} \binom{n}{j} v^j (1-v)^{n-j} = \delta.$$

Then $\mathbb{P}_p\{L_K > p\} \le \delta$. *Replacing every* L_k *by a smaller nonnegative number preserves the bound.*

PROOF: For $1 \le k \le n$, $B_k(0) = 0$, $B_k(1) = 1$, and differentiation and cancellation give

$$B_k'(v) = n\binom{n-1}{k-1} v^{k-1}(1-v)^{n-k} > 0 \quad (0 < v < 1).$$

Thus the root exists and is unique. Since $B_{k+1}(v) \le B_k(v)$, the sequence L_k is nondecreasing. For $p \in (0,1)$, if there is any k with $L_k > p$, let k_0 be the smallest such index. Then $\{L_K > p\} = \{K \ge k_0\}$, whose probability is $B_{k_0}(p) < \delta$. If no such k exists the probability is zero. At $p = 0$, $K = 0$; at $p = 1$, no L_k exceeds p. Decreasing the endpoints only decreases the failure event. *Q.E.D.*

Rational implementation of the event endpoints. The frozen program uses $\delta = 1/23200$ and 48 bisections. For $k \ge 1$, initialize $l = 0, h = 1$. At a rational midpoint $v = c/d$, test the exact integer inequality

$$23200 \sum_{j=k}^{32} \binom{32}{j} c^j (d-c)^{32-j} \le d^{32}. \tag{65}$$

If it holds, replace l by v; otherwise replace h by v. The invariant $B_k(l) \le \delta < B_k(h)$ holds initially and after every step. At completion $h - l = 2^{-48}$ and $l \le L_k < h$. The reported endpoint is l, an outward lower bound, rather than a floating-point approximation to the root. For $k = 0$ the reported lower endpoint is zero.

For each safe-improvement method and each of the two fixed thresholds $t \in \{2^{-24}, 2^{-20}\}$, set

$$I_{mj}(t) = \mathbf{1}\{c_m(X_j) \ge t\}, \qquad K_m(t) = \sum_{j=1}^{32} I_{mj}(t), \qquad p_m(t) = P_{40}\{c_m(X) \ge t\}.$$

These are independent Bernoulli observations across state units, although the 232 event series may be mutually dependent. Equation (62) gives

$$\mathbb{E}_{40} G_m(X) \ge t\, p_m(t).$$

Lemma S.28 and the outward bisection therefore justify $t\, l_{K_m(t)}$ as a one-sided lower bound for the original mean policy gain. An adaptive numerical refinement within one state does not create another Bernoulli observation.

PROPOSITION S.29—Joint 95 percent conditional coverage: *Under the population and uniform numerical premises above, define*

$$r_G = \frac{13\tau}{32}, \qquad r_H = \frac{13\tau}{16}.$$

For each of the 116 safe-improvement methods, use

$$\underline{\mu}_m = \max\left\{0, \overline{\ell}_m - r_G, 2^{-24} l_{K_m(2^{-24})}, 2^{-20} l_{K_m(2^{-20})}\right\}, \tag{66}$$

$$\overline{\mu}_m = \min\{\tau, \overline{u}_m + r_G\}. \tag{67}$$

For each of the 348 prespecified signed comparisons, with paired sample endpoints ℓ^H_{mj}, u^H_{mj}, use

$$[\max\{-\tau, \overline{\ell}^H_m - r_H\}, \min\{\tau, \overline{u}^H_m + r_H\}], \tag{68}$$

with $[0,0]$ for an exactly identical method key. With probability at least $19/20$, simultaneously all 116 intervals in (67), all 232 event-probability lower bounds, and all 348 signed intervals cover their stated population quantities. The probability is conditional on the fixed pre-acquisition information and is over the 32 independent initial states, with no additional state units supplied by repeated clocks.

PROOF: The normalized exponents in Lemma S.27 agree:

$$2(32)\left(\frac{r_G}{\tau}\right)^2 = 2(32)\left(\frac{r_H}{2\tau}\right)^2 = \frac{169}{16} =: z.$$

No numerical logarithm is needed to certify the family allocations. The positive Taylor terms give

$$e^z > \sum_{j=0}^{30} \frac{z^j}{j!} =: T_{30}.$$

For an independently checkable rational witness, the integer floors of the first twelve terms, in increasing order of j, are

$$1,\ 10,\ 55,\ 196,\ 518,\ 1095,\ 1928,\ 2910,\ 3842,\ 4509,\ 4763,\ 4573.$$

Their sum is $24400 > 23200$. Hence $T_{30} > 23200$, with all comparisons obtainable by integer arithmetic. The union bound for the 116 two-sided safe mean bounds is at most

$$116 \cdot 2e^{-z} < 232/T_{30} < 1/100.$$

For the 348 two-sided signed bounds it is at most

$$348 \cdot 2e^{-z} < 696/T_{30} < 3/100.$$

The event inversions have total failure probability at most $232/23200 = 1/100$. The sum of these three family bounds is $1/20$. No independence between methods, comparisons, or event thresholds is used.

On the joint event, each safe mean exceeds both its bounded-mean lower endpoint and both of its threshold-derived lower endpoints, and is below its bounded-mean upper endpoint. Their intersection therefore retains coverage. The signed intervals follow directly from (63) and Lemma S.27; an identical method has zero difference at every state. This proves the proposition. *Q.E.D.*

Meaning and limits of the statement. The family covers fixed original-policy costs, with numerical endpoint uncertainty retained. It neither proves a stronger claim than the uniform qualification premises nor promotes empirical compatibility to independent confirmation. The inherited actors are conditioned on; no result here concerns their retraining on an unseen sample or their optimization error under a different Bellman objective.

The complete-menu price domains use saved paired original-policy intervals and recorded resource costs. They are descriptive conditional scenarios, not additional confidence statements supplied by this proposition. In particular, the timed scenarios use the fixed first 16 states for their economic means, whereas the simultaneous inference above uses all 32 states. The three clocks, any restrictive

additional complete-clock sensitivity, reuse-volume scenarios, and repeated administrative or statistical readers add zero independent population observations. An own-incumbent improvement also does not identify a neural-specific advantage over a traditional representation with the same information, features, feasible policies, and resources.

19. INHERITED ACTION-MEMORY AND JOINT INFORMATION METHODS

The following complete results retain their original source identities. The R123 midpoint contraction and R124 reduced joint interpolation are premises for the current extensions, not newly claimed theorems of this revision.

20. USING THE ACTION INFORMATION IN EVERY PAID BELLMAN SUPPORT

An original Bellman calculation can purchase several valid supports before returning a qualified action. A refusal is an economic decision about a candidate; it need not invalidate the original information acquired at that candidate. The single-payload bound of Theorem 6 uses one such support. We now retain the complete same-state collection, intersect its implications along the feasible action path, and integrate the result. This changes the supplied information interface, so it does not contradict the fixed-payload lower bound in Proposition S.22. It applies equally to neural, polynomial, spline, matched-hinge and original-model proposals.

The economic objective is the same original optimal-future function $q = Q_x$ on $[0, A]$, with $0 < \mu \le L$, and the implemented cost remains $J(b) = q(b) + \eta(b)$, $0 \le \eta(b) \le \delta$. All anchors below concern the same state, original innovation law, risk preference and future target. Supports from another actor's separately charged service are not silently pooled. If an experiment purchases a common pool, every comparator must receive that pool and its full acquisition bill.

At each of $m \ge 1$ paid anchors $a_i \in [0, A]$, suppose the valid original information includes

$$W_i \le q(a_i) \le U_i, \qquad \ell_i \le q'(a_i) \le u_i. \tag{69}$$

The derivative interval can be obtained by the boundary-aware conversion of an original quadratic support, intersected with any separately proved original-gradient

enclosure. A fitted marginal alone is not such an interval. Repeated measurements at one anchor are retained and intersected; an empty intersection is an error, not a favorable certificate.

20.1. *Intersect before integrating*

For $t \in [0, A]$ define

$$\begin{aligned} \lambda_i(t) &= \ell_i + \mu(t - a_i) + (L - \mu)\min\{t - a_i, 0\}, \\ v_i(t) &= u_i + \mu(t - a_i) + (L - \mu)\max\{t - a_i, 0\}, \\ \lambda(t) &= \max_i \lambda_i(t), \qquad v(t) = \min_i v_i(t). \end{aligned} \tag{70}$$

These are bounds on the original derivative along the entire action interval. With the usual signed-integral convention, put

$$C^+(a,b) = \begin{cases} \int_a^b v(t)\,dt, & b \geq a, \\ -\int_b^a \lambda(t)\,dt, & b < a, \end{cases} \tag{71}$$

$$C^-(a,b) = \begin{cases} \int_a^b \lambda(t)\,dt, & b \geq a, \\ -\int_b^a v(t)\,dt, & b < a. \end{cases}$$

THEOREM S.30—Action contrast from the complete paid derivative collection: *Suppose q is continuously differentiable, μ-strongly convex and has L-Lipschitz derivative, and (69) holds. Then*

$$\lambda(t) \leq q'(t) \leq v(t), \qquad C^-(a,b) \leq q(b) - q(a) \leq C^+(a,b). \tag{72}$$

For each individual anchor, define C_i^+ by using λ_i, v_i in (71), and similarly define C_i^-. At every fixed pair of feasible actions,

$$C^+(a,b) \leq \min_i C_i^+(a,b), \qquad C^-(a,b) \geq \max_i C_i^-(a,b). \tag{73}$$

Thus the comparison is with the best individual anchor in the entire paid collection. The inequalities can be strict because the informative anchor can change along the action path. Adding sound information can only narrow these contrast intervals.

For fixed a, the upper contrast $b \mapsto C^+(a,b)$ is continuous and μ-strongly convex. Its unique minimizer on $[0,A]$ is obtained from the following exact rule. If $\lambda(a) \le 0 \le v(a)$, retain a. If $\lambda(a) > 0$, minimize to the left: choose 0 when $\lambda(0) \ge 0$, and otherwise the unique zero of λ in $(0,a)$. If $v(a) < 0$, choose A when $v(A) \le 0$, and otherwise the unique zero of v in (a,A). This is an optimal guaranteed-contrast proposal for the derived upper model, not a claim that the actual economic optimum has been identified.

The global envelopes are continuous piecewise affine. With rational inputs, their breakpoints, integrals and zeros can be computed by rational arithmetic. Section 21 gives an explicit finite partition. The calculation pays for every retained record, intersection, integral, proposal and rejected qualification; zero new original-risk queries does not mean zero computation cost.

20.2. *Absolute qualification and the original implemented future*

Retain the raw quadratic supports as well. If the actual original support slope at a_i is known only to belong to $[p_i^-, p_i^+]$, define

$$s_i(b) = W_i + \min\{p_i^-(b-a_i), p_i^+(b-a_i)\} + \frac{\mu}{2}(b-a_i)^2. \tag{74}$$

Here W_i denotes the lower level of that very support. When a separate value lower bound is available, it can also be used in (69). Each s_i is valid everywhere, but a nonzero slope-interval width can give it a downward derivative jump at its anchor. Its curvature term alone does not make this robust lower function convex.

Define the pointwise value bounds and the original-optimum bound

$$\begin{aligned} \underline{q}(b) &= \max_i \{W_i + C^-(a_i,b),\ s_i(b)\}, \\ \overline{q}(b) &= \min_i \{U_i + C^+(a_i,b)\}, \\ B &= \max\left\{B_{\text{old}},\ \min_{t\in[0,A]} \max_i s_i(t)\right\}. \end{aligned} \tag{75}$$

The inherited B_{old} is any previously verified lower bound on the optimum of this same q; it may be omitted. The minimum in the last line is a global piecewise

calculation, not the output of an unchecked convex optimizer. Then $\underline{q} \le q \le \overline{q}$ and $B \le \min q$.

THEOREM S.31—A qualified policy update from retained information: *For distinct realized feasible actions a, b,*

$$\begin{aligned} J(b) - J(a) &\in [C^-(a,b) - \delta,\ C^+(a,b) + \delta], \\ \underline{q}(b) &\le J(b) \le \overline{q}(b) + \delta, \\ 0 \le J(b) - \min q &\le \overline{q}(b) + \delta - B. \end{aligned} \tag{76}$$

The signed contrast interval may be intersected with $[\underline{q}(b) - \overline{q}(a) - \delta, \overline{q}(b) + \delta - \underline{q}(a)]$. For $b = a$, the same implemented policy has contrast exactly zero.

A candidate is adopted only if its checked contrast upper endpoint is negative and

$$\overline{q}(b) + \delta - B \le \tau, \qquad \overline{q}(b) + \delta - \underline{q}(b) \le \omega. \tag{77}$$

These conditions preserve the original regret and objective-width requirements. The bounds are recomputed at the actually rounded feasible action. A finite sequence of accepted changes has actual total gain at least the sum of its certified gains, by telescoping original costs. No differentiability, continuity or convexity of η is required.

The robust-model minimizer is a proposal. Qualification may reject it. A fixed finite menu can additionally contain prescribed fractions, feasible endpoints, envelope switchpoints and previously completed qualified actions. The implemented rule and its ties must be frozen before an independent experiment. Uniformly improved lower guarantees do not imply that one selected candidate has lower actual cost than another candidate: that comparison requires its own original-policy contrast. All economic and complete-work comparisons therefore remain part of the numerical study.

The declared implementation gives the individual-anchor and fused modes the same candidate menu: the six fractions $1, 1/2, 1/4, 1/8, 1/16, 1/32$ of each individual robust-model displacement and of the joint displacement, all realized on the same downward-rounded 2^{-30} action lattice. Both modes use the same fused

absolute lower bound and raw-support optimum bound. The individual mode uses $\max_i C_i^-$ and $\min_i C_i^+$ for each signed contrast and uses the latter in every candidate upper value. It therefore chooses the best available individual-anchor contrast separately for every queried pair. The fused mode changes precisely the operation of intersecting before integrating. Equal feasible candidates and common lower bounds prevent attributing an advantage to a weaker comparator's missing information or action menu.

20.3. *What can be certified from this information*

There is a sharp interpretation of the derived action model. Discard the absolute levels and raw supports for this paragraph, and let the information consist only of $[0, A]$, μ, L, and the derivative intervals in (69). Suppose this class is nonempty. Both λ and v are themselves feasible derivatives: their increments lie between $\mu(t-s)$ and $L(t-s)$ and their values at every anchor satisfy the supplied intervals. Integrating either one therefore constructs an admissible objective. It follows that $C^+(a,b)$ is the exact worst original contrast in this reduced information class at every fixed b.

Let $\Gamma = -\min_b C^+(a,b) \geq 0$. With no additional query and only an arbitrary Borel error $0 \leq \eta \leq \delta$, the largest uniformly guaranteed implemented-policy improvement is

$$\max\{0, \Gamma - \delta\}. \tag{78}$$

Retention or the deterministic robust-model minimizer attains it; randomization cannot improve the guarantee. The complete proof includes random laws straddling the incumbent. This is an information bound for the stated derivative interface, not an optimality claim after the additional value levels, original investment formulas, qualification width and finite-precision implementation are supplied.

The strict rational example in Section 21.4 has two derivative anchors. Its joint guaranteed original gain is $103/96$, whereas the better individual-anchor guarantee is $33/32$; the difference is exactly $1/24$. It identifies the consequence of intersecting action information before integrating. It is an abstract smooth-convex witness, not a replacement investment application or an observed neural advantage.

20.4. *Purchasing a new anchor only when the paid information is unresolved*

If all free-information candidates fail, a new original calculation should be directed at the unresolved action region. For a continuous strictly increasing function g, define its constrained zero

$$z(g) = \begin{cases} 0, & g(0) \geq 0, \\ A, & g(A) \leq 0, \\ \text{the unique zero of } g, & \text{otherwise.} \end{cases}$$

The optimizer a_q^* of the original optimal-future objective satisfies

$$a_q^* \in [\underline{a}, \overline{a}] := [z(v), z(\lambda)]. \tag{79}$$

Every additional sound payload shrinks this interval. The same information also gives the remaining original-gain ceiling

$$q(a) - \min q \leq K_q(a) := \max\{0, \max_{b \in [\underline{a}, \overline{a}]} [-C^-(a, b)]\}. \tag{80}$$

The ceiling is a global piecewise calculation over the bracket. It cannot be evaluated by assuming that C^- is convex.

PROPOSITION S.32—A charged midpoint probe with an explicit precision condition: *Let the current bracket have width $w = \overline{a} - \underline{a} > 0$. Purchase a valid original derivative interval of full width at most ϵ at a feasible action c with $|c - (\underline{a} + \overline{a})/2| \leq r$. After fusing this new interval, the new optimizer bracket has width*

$$w_{\text{new}} \leq \min\left\{w, \max\left\{\frac{w}{2} + r, \frac{\epsilon}{\mu}\right\}\right\}. \tag{81}$$

Consequently, if a declared sequence of original-risk calculations actually achieves $r_k \leq \xi w_k$ and $\epsilon_k \leq \theta \mu w_k$, with $\max\{1/2 + \xi, \theta\} = \gamma < 1$, then $w_k \leq \gamma^k w_0$. Every queried support, failed accuracy target, refused policy and arithmetic operation remains charged.

The proposition is a precision-qualified bracketing argument, not a new general bisection theorem or a free derivative oracle. A fixed 128-bin original-risk query may

return a derivative interval too wide to reduce the bracket; that is an unresolved result. At an interior probe with room $s = \min\{c, A - c\}$ and support gap $e \le (L - \mu)s^2/2$, the exact R122 conversion supplies the sufficient full-width bound

$$(p^+ - p^-) + 2\sqrt{2(L - \mu)e}.$$

Thus the desired action resolution determines a real required support precision and risk-query bill. A finite-radius implementation must use its actual wider bound. At a capacity boundary the appropriate one-sided formula replaces this interior display.

The bracket concerns a_q^*, not the minimizer of an arbitrary $q + \eta$ with discontinuous η. A bracketed action is still subject to the original-policy gain, regret and width predicates. For a single decision without reuse, a qualified incumbent with regret upper bound K can also be retained whenever the compulsory price of any new probe exceeds K, as in Proposition S.21. A tighter valid ceiling for the possible gain in J is $K_q(a) + \delta$, intersected with K. This provides a stated economic stopping condition; it does not assert optimal sampling from shared-host timings.

21. PROOFS FOR THE COMPLETE PAID-SUPPORT CALCULATION

21.1. *Derivative transport, integration and the optimal upper-model proposal*

PROOF OF THEOREM S.30: For $s < t$, differentiable strong convexity implies $q'(t) - q'(s) \ge \mu(t - s)$: apply its first-order inequality in both directions and add. Derivative Lipschitzness gives $q'(t) - q'(s) \le L(t - s)$. At an anchor a_i, transporting the lower endpoint to the right therefore uses μ, whereas transporting it to the left uses L. The upper endpoint uses L to the right and μ to the left. This gives $\lambda_i(t) \le q'(t) \le v_i(t)$ and hence the first inequality in (72). Integrating it over the oriented interval from a to b proves its second inequality, including the reversal for a negative displacement.

For $b \ge a$, pointwise $v \le v_i$ and $\lambda \ge \lambda_i$ give $C^+ \le C_i^+$ and $C^- \ge C_i^-$. For $b < a$, the integrals acquire a minus sign and the two envelope roles exchange; the same two conclusions follow. Taking the extrema over all i proves (73). Adding a sound anchor increases λ and decreases v pointwise, which proves monotonicity of the

contrast bounds. It is not necessary to assume that the new anchor was an accepted policy.

Every function λ_i and v_i is continuous. For $s < t$ its increment lies in $[\mu(t-s), L(t-s)]$, also when its kink lies between s and t. Maxima and minima preserve these increment inequalities. For example, an index maximizing $\lambda_i(s)$ gives the lower increment inequality for λ, while an index maximizing $\lambda_i(t)$ gives its upper inequality. For a minimum choose the minimizing index at t for the lower inequality and at s for the upper inequality. Thus both global envelopes are continuous, strictly increasing, and L-Lipschitz.

For $b < a$, the derivative of $C^+(a,b)$ in b is $\lambda(b)$; for $b > a$ it is $v(b)$. The one-sided derivatives at a are $\lambda(a)$ and $v(a)$, in that order, and are ordered because they enclose $q'(a)$. On each side subtracting μb from these derivatives produces a nondecreasing function, and their jump at a is nonnegative. Integration proves that $C^+(a,b) - \mu b^2/2$ is convex. Therefore C^+ is μ-strongly convex and has a unique minimizer on the compact interval. The subgradient condition at a, the strict increase of both envelopes, and the constrained endpoint conditions give exactly the stated three cases. If an endpoint derivative equals zero, that endpoint is the unique minimizer. The proof also covers $a = 0$, $a = A$, repeated anchors and $L = \mu$. *Q.E.D.*

The theorem does not infer a derivative bound on the missing side of a boundary support. It starts only after every anchor interval has been validly obtained, using the boundary convention of Theorem 6 or other proved original information. If a missing-side bound is infinite, it may be retained in the relevant envelope algebra, but the finite rational algorithm below presumes finite supplied intervals. The original investment gradient cap supplies finite intervals where applicable.

21.2. *Exact finite arithmetic and robust lower supports*

Suppose the anchors, all interval endpoints, A, μ and L are rational. Split $[0, A]$ first at every anchor. On each resulting interval, each local derivative bound is affine. Include every intersection of every pair of these affine functions which lies in that interval. There are finitely many, and all are rational; an identical pair contributes no new point. Between consecutive partition points, a rational midpoint determines

an active maximum and minimum. Any fixed index order resolves an exact tie without changing its value. Integrating the active affine expression gives the exact envelope integral over that piece. A zero in an active piece is rational because its slope is at least $\mu > 0$.

Equivalently, retain the at most two affine pieces from each anchor, each with its applicable interval, and generate only pairwise intersections whose piece domains overlap. This gives at most a quadratic number of candidate breakpoints in m; sorting and testing all candidates is one finite algorithm. No particular optimized operation count is claimed for an implementation that instead loops over all pieces on all intervals. It must record its actual arithmetic and time costs.

At $L = \mu$, every derivative piece has the same slope. The envelopes are the largest and smallest corresponding intercepts plus μt. There is no division by $L - \mu$. At $A = 0$ the only feasible action is retention and every contrast is exactly zero; the nontrivial statements above are used only for $A > 0$.

For the robust support calculation, fix a piece on one side of every anchor. Subtract $\mu b^2/2$ from (74). The result is affine on that piece:

$$s_i(b) - \frac{\mu}{2} b^2 = (p_i^{\diamond} - \mu a_i) b + W_i - p_i^{\diamond} a_i + \frac{\mu}{2} a_i^2,$$

where $p_i^{\diamond} = p_i^{+}$ on $b < a_i$ and $p_i^{\diamond} = p_i^{-}$ on $b > a_i$. Add the pairwise intersections of these affine functions within their applicable intervals. On each resulting piece, $\max_i s_i(b)$ is a single quadratic with curvature μ. Evaluate it at both endpoints and at its stationary point if that point lies inside the piece. The least of these finitely many rational values is the exact global minimum. The endpoint evaluations include every downward derivative jump. This procedure proves existence and computes the required minimum without incorrectly calling the whole robust envelope convex.

With exact slopes, $p_i^{-} = p_i^{+}$, the robust supports really are convex quadratics with common curvature and their maximum is μ-strongly convex. That useful special case does not justify the same assertion for interval slopes.

PROOF OF THEOREM S.31: The support hypothesis says that there is an actual $p_i \in [p_i^-, p_i^+]$ with

$$q(b) \geq W_i + p_i(b - a_i) + \mu(b - a_i)^2/2.$$

Minimizing the linear term over its interval gives $q(b) \geq s_i(b)$ for every feasible b. Equation (72), together with the value bounds at a_i, also gives $W_i + C^-(a_i, b) \leq q(b) \leq U_i + C^+(a_i, b)$. Taking the indicated maximum and minimum proves $\underline{q} \leq q \leq \overline{q}$. Minimizing the global support inequality gives $\min_b \max_i s_i(b) \leq \min_b q(b)$, and taking a maximum with another valid optimum lower bound preserves that order.

For distinct actions, $\eta(b) - \eta(a) \in [-\delta, \delta]$. Adding this interval to the original q contrast proves the first line of (76). Since $0 \leq \eta(b) \leq \delta$, the absolute bounds on $q(b)$ imply the stated bounds on $J(b)$. The original implemented future is feasible and has nonnegative error, so $J(b) \geq q(b) \geq \min q$. Subtracting the lower optimum bound proves the regret upper bound. Subtracting the absolute policy bounds at a, b gives the additional signed interval; both contain the same true contrast, so their intersection is valid.

For $b = a$, the actually implemented action and future policy coincide and subtraction is identically zero. An allowance δ is not charged to this identity. For an actual change, the three acceptance inequalities respectively certify strict original-policy improvement, original regret qualification, and original objective width qualification. Rounding an ideal proposal changes the argument of every bound; the proof applies to its actual rounded feasible value because the bounds hold for every such value. For a finite accepted sequence, summing the original-policy contrast inequalities telescopes. This proves all statements. *Q.E.D.*

If a prospective implementation adds anchors after observing earlier certificate outcomes, these deterministic inequalities remain valid for every sound realized collection. They do not establish a probabilistic claim for adaptively acquired labels; a stochastic oracle needs simultaneous coverage for every retained adaptive query. Nor does a retrospective reanalysis become a prospectively frozen experiment by applying deterministic bounds to it.

21.3. *Sharpness at the explicitly reduced derivative interface*

Fix $A > 0$, $0 < \mu \le L$, and finitely many pairs $(a_i, [\ell_i, u_i])$. Let $\mathcal{Q}_D$ be the set of all continuously differentiable, μ-strongly convex functions on $[0, A]$ with L-Lipschitz derivative satisfying all these derivative intervals. No value, support, gradient-cap or investment-formula information beyond these intervals is imposed. Assume $\mathcal{Q}_D \neq \varnothing$. Fix a feasible incumbent a.

PROPOSITION S.33—Sharp deterministic and randomized gain guarantees: *For each feasible b,*

$$\sup_{q \in \mathcal{Q}_D} \{q(b) - q(a)\} = C^+(a, b), \qquad \inf_{q \in \mathcal{Q}_D} \{q(b) - q(a)\} = C^-(a, b).$$

Both extrema are attained. Put $\Gamma = -\min_{b \in [0,A]} C^+(a, b) \ge 0$. An admissible method receives only this fixed information and chooses any Borel probability law ν for $B \in [0, A]$ without new queries. The adverse function knows ν before B is drawn. Then

$$\inf_{\nu} \sup_{q \in \mathcal{Q}_D} \mathbb{E}_\nu[q(B) - q(a)] = -\Gamma.$$

If $J = q + \eta$ and η ranges over all Borel functions from $[0, A]$ to $[0, \delta]$, then

$$\inf_{\nu} \sup_{q, \eta} \mathbb{E}_\nu[J(B) - J(a)] = \min\{0, -\Gamma + \delta\}.$$

Retention or the deterministic minimizer of $C^+(a, \cdot)$ attains the latter value. This is the exact guarantee (78) at the stated interface.

PROOF: Choose any $q_0 \in \mathcal{Q}_D$. The derivative-envelope proof gives $\lambda \le q_0' \le v$ pointwise. At every a_i, $\lambda(a_i) \ge \ell_i$ and $v(a_i) \le u_i$ by the terms carrying that very anchor. Together these facts imply

$$\ell_i \le \lambda(a_i) \le u_i, \qquad \ell_i \le v(a_i) \le u_i.$$

The preceding increment proof shows that each envelope is continuous and has all secants in $[\mu, L]$. Its indefinite integral is therefore continuously differentiable, has L-Lipschitz derivative, and is μ-strongly convex. It also satisfies every anchor

interval. Thus, for any constant c,

$$q_\lambda(t) = c + \int_a^t \lambda(s)\,ds, \qquad q_\upsilon(t) = c + \int_a^t \upsilon(s)\,ds$$

belong to $\mathcal{Q}_D$. If $b \geq a$, the second function attains C^+ and the first C^-; if $b < a$, their roles exchange. The general bounds were already proved, so this establishes both sharp equalities without asserting independent choices of a worst derivative at each point.

For any law ν, convexity of each admissible q and Jensen's inequality give

$$\mathbb{E}_\nu[q(B) - q(a)] \geq q(\mathbb{E}_\nu B) - q(a).$$

All expectations are finite because q is continuous on a compact interval. Taking the supremum and using the just-proved sharp fixed-action bound gives a value at least $C^+(a, \mathbb{E}_\nu B) \geq -\Gamma$. The deterministic minimizer attains $-\Gamma$, proving the first minimax equality.

For the implemented error put $r = \Pr_\nu(B \neq a)$. If $r = 0$, the contrast is exactly zero. If $r > 0$, set $b_* = \mathbb{E}_\nu[B \mid B \neq a] \in [0, A]$. Conditional Jensen gives, for every admissible q,

$$\mathbb{E}_\nu[q(B) - q(a)] \geq r\{q(b_*) - q(a)\}.$$

The single Borel function which is zero at a and δ elsewhere gives expected error difference δr. It may be chosen together with a function attaining the fixed-action supremum at b_*. Consequently

$$\sup_{q,\eta} \mathbb{E}_\nu[J(B) - J(a)] \geq r\{C^+(a, b_*) + \delta\} \geq r(-\Gamma + \delta) \geq \min\{0, -\Gamma + \delta\}.$$

This also covers $b_* = a$ when the conditional law straddles the incumbent. Retention attains zero. If $\Gamma > \delta$, its positive value implies that the minimizing action differs from a, and the sharp fixed-action contrast plus an error difference δ attains $-\Gamma + \delta$. At equality both choices are optimal when distinct, and the zero case is covered by retention. No adversary observes the realized random action. This proves the second equality. *Q.E.D.*

This proposition is deliberately stated for $\mathcal{Q}_D$. Integrating the extremal derivative may fail additional absolute value constraints or the algebraic restrictions of the investment model. Soundness and dominance of the envelope calculation persist after those restrictions, but its sharpness and minimax claims are not transferred to that narrower class. No numerical adoption or representation-specific conclusion follows from the proposition.

21.4. *An exact rational strict-information witness*

Let $A = 3/2$, $\mu = 1$, $L = 4$, incumbent $a = 1/4$, and derivative anchors

$$a_1 = 1/4, \quad \ell_1 = u_1 = -2, \qquad a_2 = 5/4, \quad \ell_2 = u_2 = -1/2.$$

Write $x = t - 1/4$. Between the two anchors the upper envelope is

$$v(t) = \min\{-2 + 4x,\ x - 3/2\}.$$

The first term is active for $0 \le x \le 1/6$ and the second for $1/6 \le x \le 1$. To the right of the second anchor, $v(t) = 4x - 9/2$. Its zero is $x = 9/8$, hence the joint robust proposal is $b = 11/8$. Its exact upper contrast is

$$\begin{aligned} C^+(a, 11/8) &= \int_0^{1/6} (-2 + 4x)\, dx + \int_{1/6}^{1} (x - 3/2)\, dx + \int_1^{9/8} (4x - 9/2)\, dx \\ &= -\frac{5}{18} - \frac{55}{72} - \frac{1}{32} = -\frac{103}{96}. \end{aligned}$$

Using only the first anchor gives minimizer $x = 1/2$ and guaranteed gain $1/2$. Using only the second gives minimizer $x = 9/8$ and upper contrast

$$\int_0^1 (x - 3/2)\, dx + \int_1^{9/8} (4x - 9/2)\, dx = -\frac{33}{32}.$$

The joint guaranteed gain therefore exceeds the better individual guarantee by $103/96 - 33/32 = 1/24$. The admissible function $q(t) = \int_{1/4}^{t} v(s)\, ds$ attains the joint upper contrast. Its derivative is continuous with slopes either one or four, and it satisfies both exact anchor derivatives. Thus the example verifies a realizable smooth-convex information improvement, not an intersection of mutually incompatible derivatives.

At the two anchors this witness has levels $q(a_1) = 0$ and $q(a_2) = -25/24$. Exact touching μ-quadratic supports at these interior anchors are valid for this function. If those exact levels are additionally disclosed, an algorithm can exploit the known cross-anchor value difference as well; the reduced-interface lower bound is not a lower bound for that stronger information set. The example is therefore not advertised as proving a strict advantage over every possible full-payload algorithm.

The example is algebraic validation. It is not an experimental state from the investment application, a new population sample, or evidence of a uniquely neural advantage. An independently implemented rational calculation can reproduce every number without importing an economic or envelope production module.

21.5. *Optimizer brackets and paid action-directed precision*

PROOF OF (79) AND PROPOSITION S.32: The continuously differentiable strongly convex function q has a unique minimum on the compact interval. Its constrained first-order conditions give $a_q^* = z(q')$, with the endpoint conventions stated in the main text. If two continuous strictly increasing functions satisfy $g \le h$, then $z(h) \le z(g)$. To verify this directly, consider an interior zero of either function and use their sign on either side; if either constrained zero is an endpoint the same sign argument gives the weak inequality. Applying this order property to $\lambda \le q' \le v$ proves (79).

An additional anchor increases λ and decreases v, so $z(\lambda)$ can only decrease and $z(v)$ can only increase. Thus the new bracket is a subset of the old one. This gives the first bound $w_{\text{new}} \le w$.

Write the newly purchased derivative interval as $[d_-, d_+]$, with $d_+ - d_- \le \epsilon$. If $d_- \ge 0$, the new lower envelope is nonnegative at c, so its constrained zero is at most c. The new bracket is contained in $[\underline{a}, \min\{\bar{a}, c\}]$ and has width at most $w/2 + r$. If $d_+ \le 0$, its upper envelope is nonpositive at c; the same argument gives containment in $[\max\{\underline{a}, c\}, \bar{a}]$ and the same width bound. These assertions also hold when the new interval has a zero endpoint, because $\mu > 0$ makes both envelopes strictly increasing.

In the remaining case $d_- < 0 < d_+$, the new upper anchor cone on the left of c is $d_+ + \mu(t - c)$. Its constrained zero cannot be less than $c - d_+/\mu$, clipped at zero. The new lower anchor cone on the right is $d_- + \mu(t - c)$, whose constrained zero cannot exceed $c - d_-/\mu$, clipped at A. Since the fused envelope is at least as informative as this single new interval, its bracket is contained in

$$[c - d_+/\mu,\ c - d_-/\mu] \cap [0, A].$$

This has width at most $(d_+ - d_-)/\mu \le \epsilon/\mu$. Combining the three cases and the nested-bracket bound proves (81). It makes no assumption about which sign the purchased interval will have. Applying the stated relative precision and rounding conditions recursively proves $w_k \le \gamma^k w_0$.

For the potential ceiling, the true optimizer belongs to the bracket and the lower contrast bound gives $q(a) - q(a_q^*) \le -C^-(a, a_q^*)$. Maximizing over the entire bracket and taking the maximum with zero proves (80). For every other feasible b, $J(a) - J(b) \le q(a) - \min q + \delta \le K_q(a) + \delta$. The original incumbent's regret bound is another valid ceiling; their minimum is therefore valid. The same proof as Proposition S.21 establishes the stated single-decision cost dominance when a compulsory incremental query price exceeds that ceiling. *Q.E.D.*

The midpoint does not have to be an unrounded real number. The geometric loss caused by its actual feasible representation is the explicit r term. If the prescribed midpoint equals a paid anchor and the protocol declines to buy it again, no contraction is asserted. If the prescribed query returns a broad interval, the exact new bracket may be unchanged. Both cases have their original status and cost rather than being counted as successful action resolution.

22. JOINT ACTION INFORMATION FROM PAID CONTINUATION VALUES

A continuation calculation returns both a value enclosure and action information. Keeping the value enclosures only as separate endpoint bounds can discard restrictions on the derivative between endpoints. We use those restrictions to bound the original action contrast. The economic object remains the original optimal-future objective $q = Q_x$ on $[0, A]$; the implemented controller has cost $J = q + \eta$

with $0 \le \eta \le \delta$. All data below concern this one state, preference, innovation law and future target. A different representation receives the same data and the same feasible actions, and is charged for the same acquisition. The new calculation does not identify a neural-specific approximation capability.

Smooth convex interpolation and its use in performance estimation are established ideas (Taylor et al., 2017). The interpolation lemma below is a scalar constructive specialization. Its role here is to use the joint information in an original Bellman service and to separate an optimizer's numerical proposal from an independently checkable economic certificate. Neither a generic interpolation principle nor numerical conic optimization is claimed as a new mathematical discovery.

Assume $A > 0$, $0 < \mu \le L$, and that q is continuously differentiable, μ-strongly convex, and has L-Lipschitz derivative. At finitely many paid anchors a_i retain

$$W_i \le q(a_i) \le U_i, \qquad \ell_i \le q'(a_i) \le u_i. \tag{82}$$

Repeated anchors impose all their constraints at the same node. At least one finite value interval and one finite derivative interval are available. Every derivative interval must be justified for the original q, rather than inferred from a fitted derivative. A verified global derivative cap $|q'| \le M$ can be added. If no sharper cap is known, a finite one follows from any derivative anchor and the bound L: one may take $M = \max\{|\ell_i|, |u_i|\} + LA$.

Let $\mathcal{Q}_I$ denote exactly this class of functions satisfying the supplied levels, derivative intervals and declared derivative cap. Assume it is nonempty; a sound original calculation provides one member. Raw quadratic supports, additional formulas of the investment model, or an actor's representation can restrict the class further. Unless explicitly encoded, those restrictions are not part of $\mathcal{Q}_I$ and no sharpness statement below includes them.

22.1. *A finite exact interpolation problem*

For a fixed pair of realized feasible actions a, b, form the sorted distinct nodes $0 = x_0 < x_1 < \cdots < x_n = A$ from both endpoints, every anchor, and a, b. Let f_i, g_i denote unknown value and derivative at node x_i. Put $K = L - \mu$ and, for adjacent

nodes, set

$$d_i = x_{i+1} - x_i, \qquad s_i = g_{i+1} - g_i - \mu d_i,$$
$$w_i = f_{i+1} - f_i - g_i d_i - \frac{\mu}{2} d_i^2. \tag{83}$$

For $K > 0$ impose

$$0 \le s_i \le K d_i, \qquad s_i^2 \le 2K w_i, \qquad s_i^2 \le 2K(s_i d_i - w_i). \tag{84}$$

Add the paid bounds (82) at their nodes and $-M \le g_i \le M$. Finite value boxes can be supplied without additional oracle calls:

$$\max_j \{W_j - M|x_i - a_j|\} \le f_i \le \min_j \{U_j + M|x_i - a_j|\}. \tag{85}$$

When $K = 0$, replace (84) by the linear equalities $s_i = w_i = 0$. Let $\mathcal{P}_I(a,b)$ be the resulting finite feasible set. Every inequality has at most quadratic degree; the two quadratic inequalities are convex and have a rotated-cone representation. Only a linear number of adjacent interpolation constraints is required after sorting the nodes.

THEOREM S.34—Exact scalar interpolation with joint levels and marginals: *A list (f_i, g_i) belongs to $\mathcal{P}_I(a,b)$ if and only if it is the restriction of a function in $\mathcal{Q}_I$ to these nodes. Such a function can be constructed with a continuous piecewise-affine derivative. Consequently*

$$U_I(a,b) := \max_{\mathcal{P}_I(a,b)} (f_b - f_a) = \max_{q \in \mathcal{Q}_I} \{q(b) - q(a)\},$$
$$L_I(a,b) := \min_{\mathcal{P}_I(a,b)} (f_b - f_a) = -U_I(b,a). \tag{86}$$

Here subscripts a, b refer to their nodes. These extrema are attained. The result includes boundary actions and repeated measurements. At $A = 0$ the only action contrast is zero and no interpolation problem is needed.

For clarity, consider the derivative-envelope bound made from the same derivative intervals. At $t \in [0, A]$ define

$$\lambda(t) = \max_i \{\ell_i + \mu(t - a_i) + (L - \mu)\min(t - a_i, 0)\},$$

$$v(t) = \min_i \{u_i + \mu(t - a_i) + (L - \mu)\max(t - a_i, 0)\}.$$

One may intersect these with the global derivative cap. Integrating v for rightward moves and $-\lambda$ for leftward moves gives the upper contrast C_D^+; reversing the endpoints gives the lower contrast C_D^-. Put

$$\overline{q}_D(t) = \min_i \{U_i + C_D^+(a_i, t)\}, \qquad \underline{q}_D(t) = \max_i \{W_i + C_D^-(a_i, t)\}.$$

Every member of $\mathcal{Q}_I$ satisfies these inequalities. Therefore

$$U_I(a, b) \le \min\{C_D^+(a, b), \overline{q}_D(b) - \underline{q}_D(a)\}. \tag{87}$$

The inequality can be strict. It does not assert dominance over a different bound using raw supports absent from $\mathcal{Q}_I$. An implementation can always intersect the new valid bound with every previously verified same-objective bound. Thus adding the new calculation cannot weaken the returned certificate, although its computational cost can make the resulting service less attractive.

22.2. *A rational certificate that does not trust solver convergence*

Computing a floating point optimum of (86) does not give a verified upper contrast. Write its minimization form as

$$\min_{z \in [\underline{z}, \overline{z}]} H(z), \qquad c_j(z) \le 0 \quad (j = 1, \dots, m),$$

where $z = (f, g)$, $H = f_a - f_b$, the c_j are differentiable convex linear or quadratic constraints, and the finite rational box contains the values and derivatives of every original function under consideration. The box may be obtained from (85), derivative transport, and their outward intersections. Equality constraints may be represented by a pair of affine inequalities. Empty intersections or an inconsistent original payload must be reported, rather than repaired by discarding an inconvenient observation.

THEOREM S.35—Exact residual correction for an action certificate: *Choose any rational point z_0 and any nonnegative rational multipliers y_j. No primal feasibility or stationarity of z_0 is required. Define*

$$\begin{aligned} \mathcal{L}(z) &= H(z) + \sum_j y_j c_j(z), \\ r &= \nabla \mathcal{L}(z_0), \\ B &= \mathcal{L}(z_0) + \sum_k \min\{r_k(\underline{z}_k - z_{0k}), r_k(\overline{z}_k - z_{0k})\}. \end{aligned} \tag{88}$$

Then every original compatible function satisfies

$$q(b) - q(a) \le -B. \tag{89}$$

All quantities in (88) are rational when the input positions, bounds and curvature constants are rational. The same construction with $H = f_b - f_a$, $H = f_b$, or $H = -f_b$ gives respectively a lower contrast, a lower absolute value, or an upper absolute value. A solver may propose z_0, y, but an independent implementation must verify their finite values, nonnegative multipliers, exact constraint identity, enclosing box and the final rational expression.

This certificate charges any lack of numerical stationarity through the box residual. It remains sound when the solver fails to converge, although it may then be too wide to permit a policy change. Arbitrary nonnegative multipliers, including zero, are valid; a success flag is neither necessary nor sufficient for a useful certificate. Floating primal residuals are diagnostics, not replacements for (88).

For the original investment model, the derivative cap used here is independently justified by the retained value modulus. Since $\|\nabla V_1\|_\infty \le 15/d$ and $\sum_j b_j = 3d/8$, the original continuation term is action-Lipschitz with constant $45\beta/8$. Also $0 \le h'(a) \le 3/4$ on $[0, 1/4]$. Thus

$$|q_x'(a)| \le \frac{3}{4} + \frac{45\beta}{8} = \frac{771}{128} < 7 \qquad (\beta = 15/16).$$

This supplies the conservative cap $M = 7$ without fitting it to outcomes. It bounds the derivative of the original optimal-future objective, not the derivative of the possibly discontinuous deployed-policy error.

22.3. *The same original-policy gain, regret and value requirements*

Let independently verified bounds satisfy $\widehat{L}(a,b) \le q(b) - q(a) \le \widehat{U}(a,b)$, and $\underline{v}_b \le q(b) \le \overline{v}_b$. These can intersect the new joint certificates with retained original-value and support bounds. Let $B_* \le \min_{[0,A]} q$ be any verified original-optimum lower bound. For distinct realized feasible actions,

$$J(b) - J(a) \in [\widehat{L}(a,b) - \delta, \widehat{U}(a,b) + \delta],$$
$$\underline{v}_b \le J(b) \le \overline{v}_b + \delta, \qquad J(b) - \min q \le \overline{v}_b + \delta - B_*. \tag{90}$$

For $b = a$ the actual contrast is exactly zero. In particular, any strictly negative contrast upper endpoint certifies improvement of the same implemented economic objective. Applying the declared regret tolerance and objective-width tolerance to the same absolute enclosures preserves those requirements. All inequalities are evaluated at the actually rounded feasible action, and every rejected candidate remains in the work account.

There is also a useful consequence when an incumbent already has original regret at most K_a. If u is a valid upper bound on $J(b) - J(a)$, then

$$J(b) - \min q \le K_a + u. \tag{91}$$

Strict improvement therefore preserves an incumbent's regret qualification without a fresh global lower-bound solve. The absolute objective-width requirement is a separate checked condition. One must not infer that width from a gain inequality alone.

Applied at a finite set of candidates, this is a certified candidate selection rule. The exact continuous-action optimization in (86), a finite implemented menu, and a numerical search for a useful dual witness are different operations and are reported separately. Repeated accepted updates telescope their verified original-policy gains. Their extension through several Bellman dates uses the retained centered lower-gain

recursion with the same incumbent future, rather than replacing a nonlinear risk evaluation by a physical expectation.

22.4. *A matching information bound and an economic stopping ceiling*

Let

$$\Gamma_I = -\min_{b\in[0,A]} U_I(a,b) \geq 0.$$

The function minimized here is continuous and μ-strongly convex. Its minimizer is an ideal information benchmark; the finite numerical menu is not assumed to attain it.

PROPOSITION S.36—The exact gain at the joint information interface: *An admissible method receives only I and chooses a probability law ν on the same feasible interval without further original queries. The adverse q is chosen knowing this law before the action is drawn. Then*

$$\inf_{\nu}\sup_{q\in\mathcal{Q}_I} \mathbb{E}_\nu[q(B)-q(a)] = -\Gamma_I,$$

$$\inf_{\nu}\sup_{\substack{q\in\mathcal{Q}_I\\ 0\leq\eta\leq\delta}} \mathbb{E}_\nu[J(B)-J(a)] = \min\{0,-\Gamma_I+\delta\}, \tag{92}$$

where η ranges over Borel functions. The robust deterministic minimizer, or retention when $\Gamma_I \leq \delta$, attains the corresponding bound. It applies equally to all representation labels at this same interface. It is not a lower bound for free access to the full investment formulas, additional raw supports or new acquisitions.

To turn possible remaining gain into a priced stopping test, let a finite action set $\mathcal{B}$ cover $[0,A]$ with radius h and suppose k_b is a verified upper bound on $q(a)-q(b)$ at each $b\in\mathcal{B}$. The cap $|q'|\leq M$ gives

$$q(a)-\min q \leq K_I := \max\{0, \max_{b\in\mathcal{B}} k_b + Mh\}. \tag{93}$$

The complete incumbent's regret bound K_a and the future allowance therefore give the possible implemented-gain ceiling $\min\{K_a, K_I+\delta\}$. If every allowed continua-

tion of a single-decision solver must first pay at least $c_* > 0$ additional work units at economic price λ, and information reuse is excluded, then

$$\lambda c_* \geq \min\{K_a, K_I + \delta\}$$

certifies that stopping weakly dominates every such continuation. The work to build the action cover and its certificates has already been charged. This conditional stopping region does not presume a forecast of future solver gains or convert shared-host elapsed times into a universal economic price.

The rational residual formula also provides exact refinement credits. Hold $\mathcal{L}, z_0, r$ fixed and replace its enclosing box by a valid nested box $[\underline{z}', \overline{z}']$. The bound improves by exactly

$$B' - B = \sum_{r_k \geq 0} r_k(\underline{z}'_k - \underline{z}_k) + \sum_{r_k < 0} (-r_k)(\overline{z}_k - \overline{z}'_k) \geq 0. \tag{94}$$

Likewise a paid inequality $f_i - U_i \leq 0$ with multiplier y_i^+ and $W_i - f_i \leq 0$ with multiplier y_i^- contributes the additional credit $y_i^+(U_i - U_i') + y_i^-(W_i' - W_i)$ when only those endpoints tighten. These are action-certificate sensitivities in original objective units. They can prioritize a declared refinement menu and quantify the margin actually purchased. They are not a prior guarantee that a noisy future query will achieve the proposed endpoint changes; each realized query and its cost must still be recorded and verified.

23. PROOFS FOR JOINT ORIGINAL-ACTION INFORMATION

PROOF OF THEOREM S.34: First suppose $K > 0$ and $q \in \mathcal{Q}_I$. On an adjacent interval write $t = x - x_i \in [0, d_i]$ and

$$r_i(t) = q'(x_i + t) - g_i - \mu t.$$

Strong convexity and derivative Lipschitzness imply for $u < v$ $0 \leq r_i(v) - r_i(u) \leq K(v - u)$. Its endpoint values are 0 and s_i, so $0 \leq s_i \leq Kd_i$ and

$$\max\{0, s_i - K(d_i - t)\} \leq r_i(t) \leq \min\{Kt, s_i\}.$$

Integrating these two ramps gives

$$\frac{s_i^2}{2K} \le \int_0^{d_i} r_i(t)\,dt = w_i \le s_i d_i - \frac{s_i^2}{2K}.$$

This proves all adjacent constraints. The paid bounds hold by definition; the derivative cap and its integrated bound give the boxes.

Conversely suppose all finite constraints hold. On each adjacent interval define the lower and upper ramps

$$r_i^-(t) = \max\{0, s_i - K(d_i - t)\}, \qquad r_i^+(t) = \min\{Kt, s_i\}.$$

Both are continuous, nondecreasing and K-Lipschitz, with endpoint values $0, s_i$. Their integral range is exactly the permitted interval for w_i. Choose a convex combination $r_i = (1-\theta_i)r_i^- + \theta_i r_i^+$ whose integral is w_i. If their integrals coincide use either one; otherwise θ_i is the ratio of the requested excess integral to the positive difference of the two integrals and belongs to $[0,1]$. Set

$$q(x_i + t) = f_i + g_i t + \frac{\mu}{2}t^2 + \int_0^t r_i(s)\,ds.$$

The values at both ends equal f_i, f_{i+1}, and the derivatives at both ends equal g_i, g_{i+1}. Hence the pieces join to a continuously differentiable function on the entire interval. Its derivative increments lie in $[\mu(v-u), L(v-u)]$ within each piece and, by addition, across the joins. Integration proves μ-strong convexity, while the upper increment bound proves derivative Lipschitzness. Its derivative lies between the two endpoint derivatives on every interval, so the cap is preserved. It satisfies all measured data and realizes the required list.

If $K = 0$, the derivative increment must be exactly μd_i; integration gives $s_i = w_i = 0$. Conversely those equalities join pieces of a single quadratic with curvature μ. No division by $L - \mu$ is performed. Duplicate observations are constraints on one node and impose no zero-length interval. Endpoints are already nodes, so no boundary extension is missing.

Finally $\mathcal{P}_I(a,b)$ is a nonempty closed subset of a finite box, hence compact. The linear contrast objective has attained extrema. Necessity and the constructive suf-

ficiency identify these extrema with the original function-class extrema, proving (86). *Q.E.D.*

PROOF OF THEOREM S.35: Nonnegative combinations of the convex constraints, added to the linear objective, make $\mathcal{L}$ convex. Therefore for every z

$$\mathcal{L}(z) \geq \mathcal{L}(z_0) + r^{\mathsf{T}}(z - z_0).$$

For any feasible z, $\mathcal{L}(z) \leq H(z)$ because $y_j \geq 0$ and $c_j(z) \leq 0$. Minimizing the affine expression over the enclosing box gives exactly B in (88). Thus $H(z) \geq B$ at every feasible point, including the original values and derivatives. Since $H = f_a - f_b$, this proves (89). Replacing H gives the other three conclusions by the same proof. All derivatives and arithmetic are rational polynomials in the specified rational inputs. The proof needs neither strong duality nor a feasible or stationary proposed z_0. *Q.E.D.*

POLICY, MINIMAX, STOPPING AND REFINEMENT STATEMENTS: For distinct a, b, the only assumed information about the implemented future error gives $\eta(b) - \eta(a) \in [-\delta, \delta]$. Adding this to the original contrast proves (90); adding $0 \leq \eta(b) \leq \delta$ to the absolute bounds proves its remaining inequalities. At equal actions the same implementation is subtracted from itself. Subtracting the same original optimum from both policies gives (91). Summing actual contrasts telescopes, so finitely many accepted changes preserve their cumulative gain.

For every admissible q, the function $b \mapsto q(b) - q(a) - \mu b^2/2$ is convex. Its supremum over $\mathcal{Q}_I$ is convex, proving strong convexity of U_I. The uniform derivative cap makes U_I M-Lipschitz, hence continuous, so its minimum is attained. Also $U_I(a, a) = 0$, giving $\Gamma_I \geq 0$. For every action law ν, Jensen's inequality gives

$$\sup_q \mathbb{E}_\nu[q(B) - q(a)] \geq \sup_q \{q(\mathbb{E}_\nu B) - q(a)\} = U_I(a, \mathbb{E}_\nu B) \geq -\Gamma_I.$$

The deterministic minimizer attains equality. All expectations are finite, since the common derivative cap bounds the contrast.

For the implemented cost let $r = \Pr(B \neq a)$. When $r = 0$ the contrast is identically zero. Otherwise set $b_* = \mathbb{E}[B \mid B \neq a]$ and choose the Borel error $\eta(t) = \delta \mathbf{1}_{\{t \neq a\}}$.

Conditional Jensen and an attaining function for the fixed action b_* give

$$\sup_{q,\eta} \mathbb{E}[J(B) - J(a)] \geq r\{U_I(a, b_*) + \delta\} \geq r(-\Gamma_I + \delta) \geq \min\{0, -\Gamma_I + \delta\}.$$

Retention attains zero; if $\Gamma_I > \delta$, the deterministic robust minimizer differs from a and attains the other expression. This proves (92), including laws straddling the incumbent and an adversary chosen before the random draw.

For the stopping ceiling, let b^* minimize the true q and choose a covering point $b \in \mathcal{B}$ with $|b - b^*| \leq h$. Then $q(a) - q(b^*) \leq q(a) - q(b) + Mh \leq k_b + Mh$, proving (93). No new policy can improve the current $J(a)$ by more than $q(a) - \min q + \delta$, and the incumbent's regret bound supplies the second ceiling. A compulsory incremental economic cost at least their minimum cannot be recovered even by the best feasible subsequent decision. This proves the stated single-decision stopping dominance.

Finally the minimum of $r_k(z_k - z_{0k})$ over a box coordinate is attained at its lower endpoint if $r_k \geq 0$ and its upper endpoint otherwise. Subtraction for two nested boxes gives (94). Tightening a constraint endpoint changes only the stated constant in $\mathcal{L}$; its gradient is unchanged. This proves the additional multiplier credits. *Q.E.D.*

23.1. *A strict rational witness with exactly the same information*

PROPOSITION S.37—A strict joint-information gain: *For the rational information specified below, the joint minimax original gain is* $5/64$, *while integrating the derivative cones and then intersecting the separate endpoint-value bounds guarantees* $1/16$. *The strict difference is* $1/64$ *under identical supplied information.*

Let $A = 1$, $\mu = 1$, $L = 2$, and supply only

$$q(0) = 0, \quad q(1) = 3/16, \qquad q'(0) = -1/2, \quad q'(1) = 1.$$

The incumbent is $a = 0$. A cap $|q'| \leq 2$ is redundant for this information. Define $h(t) = q(t) - t^2/2 + t/2$. Then $h(0) = 0$, $h(1) = 3/16$, $h'(0) = 0$, $h'(1) = 1/2$, and h' is nondecreasing and 1-Lipschitz. The derivative-cone upper contrast from zero is

$$U_0(b) = \begin{cases} b^2 - b/2, & 0 \leq b \leq 1/2, \\ (b^2 - 1/4)/2, & 1/2 \leq b \leq 1. \end{cases}$$

The upper value derived separately from the level at one and the lower derivative envelope is

$$U_1(b) = \begin{cases} 1/16 - b/2 + b^2/2, & 0 \le b \le 1/2, \\ 3/16 - b + b^2, & 1/2 \le b \le 1. \end{cases}$$

Each function has global minimum $-1/16$, at $b = 1/4$ and $b = 1/2$ respectively. Therefore their pointwise minimum also has minimum $-1/16$. The derivative-envelope calculation with this separate absolute-value intersection guarantees at most $1/16$ gain.

The joint information is stronger. At $b = 3/8$ write $v = h'(b)$ and $z = h(b)$. Applying the adjacent interpolation inequalities on both sides gives

$$z \le \frac{3}{8}v - \frac{1}{2}v^2, \qquad z \le \frac{1}{16} - \frac{1}{8}v - \frac{1}{2}v^2.$$

If $v \le 1/8$, the first right side is at most $5/128$; if $v \ge 1/8$, the second is at most $5/128$. Hence $z \le 5/128$ and

$$q(3/8) \le \frac{5}{128} + \frac{9}{128} - \frac{24}{128} = -\frac{5}{64}.$$

This bound is attained. Set

$$h'(t) = \begin{cases} t, & 0 \le t \le 1/8, \\ 1/8, & 1/8 \le t \le 5/8, \\ t - 1/2, & 5/8 \le t \le 1, \end{cases} \qquad h(t) = \int_0^t h'(s)\, ds.$$

Direct rational integration gives $h(1) = 3/16$ and $h(3/8) = 5/128$. Its derivative has slopes zero or one and is continuous. Consequently $q = h + t^2/2 - t/2$ is admissible, with curvature one or two. On the middle interval $q'(t) = t - 3/8$; on the two outside intervals its sign is respectively negative and positive. Thus its unique minimum is at $3/8$ and is $-5/64$. This one admissible witness rules out a uniformly guaranteed gain larger than $5/64$ at any action. The joint calculation attains that gain at $3/8$, so the ideal joint guarantee is exactly $5/64$, strictly exceeding $1/16$ by $1/64$ with the same data and no new original-value query.

With arbitrary $0 \le \eta \le \delta$, the ideal guaranteed implemented gains are respectively $\max(0, 1/16 - \delta)$ and $\max(0, 5/64 - \delta)$. This is an exact algebraic example in the declared smooth-convex information class. It is not an investment sample, evidence of a trained neural advantage, or a substitute for the independent original-economy experiment. Rational independent arithmetic can verify every endpoint, integral, curvature and gain.

24. THE PROSPECTIVELY FIXED R124 INFERENCE FAMILY

The following inference is conditional on the published scientific source, the actual fixed actors, sound numerical enclosures, and independent uniform entropy bits. The population law is the declared two-coordinate 40-bit dyadic law for initial states. The innovations inside each Bellman operator retain their continuous law. All 16 initial states are independent units; different methods, preferences, candidate attempts and timing repetitions on one state supply no additional independent observations.

PROPOSITION S.38—Simultaneous original-policy inference: *Put* $n = 16$, $\tau = 1/256$, *and* $t = 2^{-24}$. *Consider the fixed 116 policy cells and 290 signed comparisons specified in* `REPORTING_SPEC124.md`. *In every cell let* $G(X)$ *be the actual gain relative to its own qualified incumbent. Assume that the service either retains the incumbent or verifies a strictly positive gain, and that both policies have cost between the original optimum and that optimum plus* τ. *Then* $0 \le G \le \tau$. *If the numerical sample mean is enclosed by* $[\bar{\ell}, \bar{u}]$, *the interval*

$$[\max(0, \bar{\ell} - 5\tau/8), \min(\tau, \bar{u} + 5\tau/8)]$$

contains the population mean except on a joint event of probability at most $1/100$ *over all 116 cells.*

In each cell also count states on which the verified gain is at least t. *Let* k *be this count and let* p_k^- *be a lower one-sided Clopper–Pearson probability bound with failure* $1/11600$. *The lower mean-gain bound* tp_k^- *holds simultaneously for the 116 cells except on an event of probability at most* $1/100$.

For each fixed signed comparison, positive means the first policy has lower original cost. The statewise difference belongs to $[-\tau, \tau]$. *Its numerical sample-mean interval enlarged in*

both directions by $5\tau/4$, then intersected with that support, contains its population mean except on a joint event of probability at most $3/100$ over all 290 comparisons. A method compared with itself has exact population difference zero. Observed equality on the sampled states alone establishes only a sample identity.

The intersection of the two valid lower bounds for each nonnegative gain, and all signed intervals, therefore has simultaneous coverage at least $19/20$.

PROOF: An accepted change has nonnegative actual gain by its exact certificate; retention has zero gain. Since the incumbent is no more than τ above the common original optimum and the candidate cannot fall below that optimum, the gain is at most τ. Two qualified policies are both in that same interval of length τ, proving the signed support.

Hoeffding's two-sided inequality for independent observations in an interval of length R has failure at most $2\exp(-2nr^2/R^2)$ at radius r. For $R=\tau$ and $r=5\tau/8$, and for $R=2\tau$ and $r=5\tau/4$, the exponent is $25/2$. Exact rational arithmetic verifies

$$\sum_{j=0}^{30}\frac{(25/2)^j}{j!}>23200.$$

Consequently $232\exp(-25/2)<1/100$ and $580\exp(-25/2)<3/100$. These are union bounds over the fixed families; dependence across methods on the same state does not affect them. Replacing the exact sample mean by a valid enclosing interval can only enlarge the resulting confidence interval.

The threshold event is a deterministic measurable function of the fixed service and the independent state. Its count is binomial with parameters n and its population event probability p. At $k=0$ set $p_k^-=0$. Otherwise the lower endpoint is the solution of

$$\sum_{j=k}^{n}\binom{n}{j}p^j(1-p)^{n-j}=1/11600.$$

Binomial tail inversion gives the stated one-sided coverage. The implementation takes the lower endpoint after 48 exact rational bisections, so numerical rounding cannot raise this lower bound. On the threshold event $G\ge t$, hence $\mathbb{E}G\ge tp\ge tp_k^-$. A union bound over 116 such events costs at most $1/100$. The sum of the three

family failure bounds is 1/20. All claims are conditional on the stated numerical and sampling premises; none uses timing repeats as population data. *Q.E.D.*

The deliberately small sample has a practical limitation already visible before execution: the signed radius exceeds τ, so these broad distribution-free signed intervals cannot certify a nonzero ordering. Their purpose is a complete prospective account and detection of any incorrect transfer of older population results, not a promised ordering. Positive own-incumbent mean-gain lower bounds can still follow from the threshold-event route. Neither route identifies field computation prices or a population adoption decision.

For completeness, if q is μ-strongly convex on the feasible interval and a^* is its constrained minimizer, its variational inequality and strong convexity give

$$q(a) - q(a^*) \geq \frac{\mu}{2}|a - a^*|^2.$$

Since $q(a) \leq J(a)$, any valid upper bound r on $J(a) - q(a^*)$ implies $|a - a^*| \leq \sqrt{2r/\mu}$. The reported root-action RMS bound uses $\mu = 2$, averages the valid regret bounds first, and rounds the square root upward. It is an upper bound on distance to the optimal-future root action, not an observed exact action error or a statement about the derivative of the deployed future-policy error.

25. THE INFORMATION IN A COMPLETE BELLMAN SUPPORT RECORD

A paid Bellman support asserts one inequality on the entire feasible action domain. Retaining only its value and a recovered derivative interval loses part of that assertion. With an uncertain supporting slope, the support assertion must still hold on the whole domain. We give an exact finite convex representation of this restriction. This advances the joint interpolation calculation in Section 22, whose information class explicitly excluded the raw supports. It does not identify a capability exclusive to neural representations. Smooth convex interpolation has an established literature (Taylor et al., 2017); the result below identifies the additional constraints licensed by the original Bellman payload. Optimal lower bounds based on collected first-order oracle information also predate this construction (Florea and Nesterov, 2024). Our contribution is the completion of the declared NBO payload

on its finite constrained action domain, a constructive sufficiency proof, and its use in original-policy certificates. A general new theory of optimal oracle bounds is not claimed.

Fix one state, original risk law and optimal-future objective $q = Q_x$ on $[0, A]$, where $A > 0$. Assume q is continuously differentiable, μ-strongly convex, and has L-Lipschitz derivative, $0 < \mu \le L$. Retain the paid value and derivative intervals, and the proved derivative cap $|q'| \le M$. For each original support record j, the information also asserts that there is a single number $p_j \in [p_j^-, p_j^+]$ such that

$$q(x) \ge S_j(x; p_j) := W_j + p_j(x - a_j) + \frac{\mu}{2}(x - a_j)^2 \qquad (0 \le x \le A). \tag{95}$$

Its anchor, lower level and slope interval must come from the same original calculation. A fitted derivative is not a substitute. The quantifier in (95) is "there exists one slope valid at every action," not a new slope for each action. Exact slopes are included by setting $p_j^- = p_j^+$. The original finite-risk support record has this quantifier; its outward slope interval encloses the single slope of that supporting objective. For convex h below, enforcing the entire robust pointwise lower envelope is equivalent to existence of such a shared slope. The shared variable is an exact convex representation of that global requirement, not an assertion of additional information beyond the full robust obstacle.

Let $\mathcal{Q}_{\text{full}}$ be exactly the nonempty class satisfying these properties and all retained value and derivative intervals. There is at least one finite value interval. Additional formulas of the investment model are not included unless explicitly encoded. Set $K = L - \mu$ and $h(x) = q(x) - \mu x^2/2$. A support becomes the affine function

$$\ell_j(x; p_j) = W_j - p_j a_j + \mu a_j^2/2 + (p_j - \mu a_j)x. \tag{96}$$

Thus h' is nondecreasing and K-Lipschitz, and $h \ge \ell_j$.

For $\ell \le 0 \le r$ define the bounded-domain conjugate

$$\Phi_K(v; \ell, r) = \max_{\ell \le t \le r} \{vt - Kt^2/2\}. \tag{97}$$

When $K > 0$, put $t_* = \mathrm{proj}_{[\ell,r]}(v/K)$; then $\Phi_K = vt_* - Kt_*^2/2$ and $\partial_v \Phi_K = t_*$. When $K = 0$, $\Phi_0 = \max\{v\ell, vr\}$. These formulas are exact rational calculations for rational inputs.

25.1. *A finite convex program that retains every support*

For trial actions a, b, form the sorted distinct nodes $0 = x_0 < \cdots < x_n = A$ from the endpoints, every paid value and derivative location, every support anchor, and a, b. Write $f_i = h(x_i)$ and $g_i = h'(x_i)$. Transform all paid value and derivative bounds accordingly; impose the original derivative cap $|g_i + \mu x_i| \le M$ and finite value boxes obtained by integrating that cap from any paid value interval. On each adjacent interval put

$$d_i = x_{i+1} - x_i, \qquad s_i = g_{i+1} - g_i, \qquad w_i = f_{i+1} - f_i - g_i d_i.$$

For $K > 0$ retain the joint interpolation inequalities

$$0 \le s_i \le K d_i, \qquad s_i^2 \le 2K w_i, \qquad s_i^2 \le 2K(s_i d_i - w_i). \tag{98}$$

For $K = 0$ use $s_i = w_i = 0$. Include one shared variable p_j in its supplied interval for each support, and add, for every node and support,

$$f_i - \ell_j(x_i; p_j) \;\ge\; \Phi_K(p_j - \mu a_j - g_i; -x_i, A - x_i). \tag{99}$$

Denote the resulting finite feasible set by $\mathcal{P}_{\mathrm{full}}(a, b)$. It has a linear number of adjacent interpolation constraints and $(n+1)m$ support inequalities for m retained supports. Duplicate observations are constraints at one node. Both domain endpoints are present even when no measurement was bought there.

THEOREM S.39—Exact interpolation above complete Bellman supports: *A list (f_i, g_i, p_j) belongs to $\mathcal{P}_{\mathrm{full}}(a, b)$ if and only if there is a function in $\mathcal{Q}_{\mathrm{full}}$ realizing the node values and derivatives, with these same p_j satisfying (95). One such interpolant has a continuous piecewise-affine derivative. The feasible set is nonempty, compact and convex. Consequently the exact worst compatible original contrast is*

$$U_{\mathrm{full}}(a, b) = \max_{\mathcal{P}_{\mathrm{full}}(a,b)} \left\{ f_b - f_a + \frac{\mu}{2}(b^2 - a^2) \right\} = \max_{q \in \mathcal{Q}_{\mathrm{full}}} \{q(b) - q(a)\}. \tag{100}$$

The extrema are attained. Removing the support inequalities enlarges the information class, so its exact upper contrast cannot be smaller than (100). The reversed optimization gives the exact lower contrast. These are information bounds under the stated interface; they are not oracle-complexity bounds for unrestricted access to the original economic formulas.

The proof supplies a constructive sufficiency argument, rather than assuming that nodewise lower bounds imply a global support. On an adjacent interval the construction uses derivative slopes $K, 0, K$ for h, with their lengths fixed by the endpoint jets and the required area. The extra term in (99) is essential. At an interior node a positive supporting-slope discrepancy requires a positive value gap even when $f_i \geq \ell_j(x_i; p_j)$ alone holds.

25.2. *Verified contrasts without an exact optimizer*

Let z collect f, g, p in a valid finite rational box, and write all constraints as $c_j(z) \leq 0$. The adjacent constraints are convex quadratics. Each support constraint is convex because it is a convex function of an affine argument minus an affine function. For $K > 0$ it is continuously differentiable, including at its clipping points. At $K = 0$ choose any valid subgradient of its maximum of two affine functions. Equality constraints may be represented by two inequalities.

For the minimization objective $H = f_a - f_b - \mu(b^2 - a^2)/2$, choose any rational point z_0, nonnegative rational multipliers y_j, and valid subgradients $v_j \in \partial c_j(z_0)$. Put

$$\begin{aligned} \mathcal{L}_0 &= H(z_0) + \sum_j y_j c_j(z_0), \\ R &= \nabla H + \sum_j y_j v_j, \\ B &= \mathcal{L}_0 + \sum_k \min\{R_k(\underline{z}_k - z_{0k}), R_k(\overline{z}_k - z_{0k})\}. \end{aligned} \tag{101}$$

Then $q(b) - q(a) \leq -B$ for every compatible original objective. The formula requires neither primal feasibility nor stationarity of z_0, nor strong duality. All terms are independently checkable by rational arithmetic. A solver status alone is not a certificate. An implementation intersects this upper bound with all earlier valid

same-objective bounds, so an unhelpful numerical dual proposal cannot weaken its returned certificate. The arithmetic, failed proposals and all retained supports remain charged.

For the unchanged implemented future $J = q + \eta$, $0 \le \eta \le \delta$, an upper contrast u at a distinct realized action licenses a policy change only when $u + \delta < 0$. Absolute bounds on the same $q(b)$, and a lower bound on its original optimum, must separately establish the declared regret and objective-width requirements. At identical actions the actual contrast is zero. No regularity of η is assumed, and the optimization of q is not relabeled as optimization of an arbitrary discontinuous $q + \eta$.

An exact decision consequence and its comparator. The supplement gives a rational example in which omitting a paid support leaves the exact upper contrast for the action change $1 \to 0$ at $-1/4$, whereas retaining it gives $-5/8$. The additional guaranteed original gain is $3/8$ under the same disclosed payload; arbitrary future error adds δ to either contrast. A cheap direct evaluation of the support at the incumbent attains the stronger bound in this example. It therefore demonstrates the economic loss from omitting support information, and does not establish a computational advantage over a conventional method that retains that information. A separate exact counterexample shows why checking a support only at interpolation nodes does not ensure its validity between nodes. The new theorem supplies the finite exact program for all coupled records, while the present original-economy experiment executes the separately specified acquisition service. An algebraic information result is not relabeled as an executed conic economic comparison.

26. PAID ACTION RESOLUTION AND POLICY CERTIFICATES

The preceding theorem describes what a complete paid record permits. It does not promise that the record resolves the decision or that the finite program is economical to solve. The original-economy study therefore also tests a declared service which buys new original-risk supports when its inherited qualified policy remains unresolved. It uses the retained rounded-midpoint contraction of Proposition S.32, the same original Bellman oracle, and the existing full-record bounds for each actual

acceptance. Midpoint localization is not introduced as a new root-finding principle. The additions here connect its required support precision, a complete work bound, the actor's initial economic certificate, and the separate precision of the returned original policy. The allocation of inexact-oracle accuracy is an established numerical optimization problem (Devolder et al., 2014, Van Dessel and Glineur, 2024). The work statement below is a closed-form solution of the particular guaranteed-width planning problem specified here; its original-risk and complete-policy premises remain part of the result.

An actor with original regret certificate $J(a_0) - \min q \le K_0$ supplies the initial optimizer interval

$$I_0 = [a_0 - \sqrt{2K_0/\mu}, a_0 + \sqrt{2K_0/\mu}] \cap [0, A]. \tag{102}$$

An outward radius and intersection with previously certified derivative brackets preserve validity. Every conventional actor with the same certificate receives the same reduction. A small training loss without that original-regret certificate does not shrink the interval. The executed warm service uses its retained derivative-envelope bracket; (102) is an additional valid initialization rule, not an unrecorded change to that frozen implementation.

THEOREM S.40—A complete precision and work account for paid localization: *Fix a relative rounding allowance $0 \le \xi < 1/2$ and $\gamma = 1/2 + \xi < 1$. Let w_0 bound the width of I_0, and choose planned widths $w_{k+1} \in [\gamma w_k, w_k]$. Query an actually feasible rounded midpoint of the current certified interval with rounding error at most ξ times its actual width. Suppose an oracle can return an original support with level gap $e = U - W$ and slope width $v = p^+ - p^-$ satisfying, for the requested next width $t = w_{k+1}$,*

$$h = \chi t, \qquad \chi = \begin{cases} \min\{1/2 - \xi, \mu/(4K)\}, & K > 0, \\ 1/2 - \xi, & K = 0, \end{cases} \qquad e \le \frac{\mu\chi t^2}{4}, \qquad v \le \frac{\mu t}{4}. \tag{103}$$

Use a query only while the actual interval width exceeds t; otherwise its planned contraction is already achieved. Then both displacements $\pm h$ are feasible from the midpoint, and the

finite support conversion gives a derivative interval of width at most μt. The new optimizer interval has width at most t.

Assume, as a separately verified oracle premise, that all computations needed for one such support, including its unsuccessful precision levels, cost at most $C_0 + C_1 t^{-p}$, with $C_0, C_1 \geq 0$ and $p > 0$. For a target $0 < w_* < w_0$, put

$$N = \left\lceil \frac{\log(w_0/w_*)}{\log(1/\gamma)} \right\rceil, \qquad w_j = w_* \gamma^{-(N-j)}, \quad 1 \leq j \leq N.$$

After at most N completed precision-qualified probes, the optimizer interval has width at most w_*. Their total work is bounded by

$$NC_0 + C_1 w_*^{-p} \frac{1 - \gamma^{pN}}{1 - \gamma^p}. \tag{104}$$

Among all nonincreasing planned-width schedules with guaranteed contraction floor γ, final width at most w_* and this same declared nonincreasing per-probe cost bound, this schedule minimizes the sum of those bounds. It is an optimum for the stated certificate-planning problem, not for all possible original-model algorithms or actual shared-host elapsed times. Querying every probe at the terminal target instead gives the bound $NC_0 + NC_1 w_*^{-p}$.

To return a policy, let a paid original-value interval at a feasible b be $[W_b, U_b]$, of width e_b, and let $r_b = \max\{|b - \underline{a}|, |b - \overline{a}|\}$ for the final optimizer interval. Then

$$\begin{aligned} |b - a_q^*| &\leq r_b, \\ W_b \leq J(b) &\leq U_b + \delta, \\ 0 \leq J(b) - \min q &\leq \delta + M r_b. \end{aligned} \tag{105}$$

If the optimizer is certified to be stationary, Mr_b can additionally be replaced by $Lr_b^2/2$. This regret bound does not require the value width e_b: a certified original optimizer interval already controls the economic action error. In particular, an interval strictly inside $(0, A)$ certifies stationarity. A returned action must separately satisfy $e_b + \delta \leq \omega$ and the declared regret tolerance. Any fresh value calculation needed for this final gate is charged in addition to (104), as are actor acquisition, the inherited service and the certificate algebra. The theorem does not turn a narrow optimizer interval into an unpaid absolute value certificate.

The proof uses a finite feasible radius in (103); it does not assume that an implemented finite-radius conversion attains a real-valued infimum. The oracle-work premise includes the required slope enclosure and original continuation accuracy. A terminal-solver floor, a maximum risk resolution, an action lattice floor or a resource cap can prevent it from holding. The executed study records such cases as unresolved and retains the last qualified policy. It neither fits a favorable power law to the realized times nor claims (104) as an empirical complexity theorem without verifying that premise. The experiment executes its frozen finite integration ladder and records achieved contractions. It does not claim to execute the ideal cost-minimizing planned-width schedule in the theorem.

Every accepted replacement still requires its own strictly negative upper bound on $J(b) - J(a_0)$ under the same future controller, as well as the original regret and width gates. The study targets a stricter regret tolerance as an additional outcome; a failure to attain that target does not revoke the already qualified incumbent. Failed and rejected queries remain in its complete service bill. These distinctions connect information acquisition to an economic action rather than to an internal fitting error.

27. COMPLETE SUPPORTS AND ORIGINAL-ACTION INFORMATION

PROOF OF THEOREM S.39: For necessity let q and its supporting slopes be compatible. The function $h = q - \mu x^2/2$ is differentiable with a nondecreasing K-Lipschitz derivative. On an adjacent interval the derivative increment $r(t) = h'(x_i + t) - g_i$ starts at zero, ends at s_i, and satisfies

$$\max\{0, s_i - K(d_i - t)\} \le r(t) \le \min\{Kt, s_i\}.$$

Its integral is w_i. Integrating the two ramps gives exactly (98). If $K = 0$, the derivative is constant, and $s_i = w_i = 0$ instead. At every feasible displacement t from x_i, the smooth upper bound and the global support give

$$\ell_j(x_i; p_j) + (p_j - \mu a_j)t \le h(x_i + t) \le f_i + g_i t + Kt^2/2.$$

Rearranging and maximizing over $[-x_i, A - x_i]$ proves (99). This step uses the same p_j at every node and displacement. All paid intervals and the integrated derivative-cap boxes hold by construction.

For sufficiency first take $K > 0$. Fix an adjacent interval and temporarily suppress its index. Thus $d > 0$, $0 \le s \le Kd$, and $s^2/(2K) \le w \le sd - s^2/(2K)$. If $D = d - s/K > 0$, define

$$c = \frac{w - s^2/(2K)}{D} \in [0, s], \qquad r(t) = \min\{Kt, c\} + \max\{0, s - c - K(d - t)\}.$$

The first ramp ends at c/K, and the last starts at $d - (s - c)/K$. Their ordered endpoints follow from $s \le Kd$. The function r is continuous, nondecreasing and K-Lipschitz; its endpoint values are $0, s$. Direct integration yields

$$\int_0^d r(t)\, dt = cd - \frac{c^2}{2K} + \frac{(s - c)^2}{2K} = cD + \frac{s^2}{2K} = w.$$

Set $h(x_i + t) = f_i + g_i t + \int_0^t r(u)\, du$. If $D = 0$, then $s = Kd$ and both allowed area endpoints equal $Kd^2/2$; set $r(t) = Kt$. These definitions realize both endpoint values and derivatives. Joining the pieces gives a continuously differentiable function on $[0, A]$. Its derivative increments are between zero and K times distance within each piece and, by addition, across joins.

It remains to verify every affine support everywhere. On one adjacent interval write an affine support as $\ell(x_i + t) = \ell(x_i) + pt$, and let $G_0 = f_i - \ell(x_i)$ and $G_1 = f_{i+1} - \ell(x_{i+1})$ be the endpoint gaps. If $p \le g_i$, the gap has nonnegative derivative throughout and its minimum is $G_0 \ge 0$. If $p \ge g_{i+1}$ its minimum is $G_1 \ge 0$. In the remaining case put $v = p - g_i \in (0, s)$. For $D > 0$, if $v \le c$, the derivative of the gap first reaches zero at $t = v/K$ on the first ramp; its minimum is

$$G_0 - v^2/(2K).$$

If $v \ge c$, it reaches zero at $t = d - (s - v)/K$ on the last ramp, and its minimum is

$$G_1 - (s - v)^2/(2K).$$

At $v = c$ the gap is constant between the ramps and the two expressions agree, by the area identity above. Because $v/K \le d$ and $(s - v)/K \le d$, both displacements are feasible in the full domain from the corresponding endpoint. The two node inequalities (99) therefore make both displayed minima nonnegative. For $D = 0$ the

derivative is the single K-ramp, and the first expression proves the same conclusion. The endpoint-slope cases $v = 0, s$ are already covered. Thus the constructed function lies above every support on every interval, using the fixed shared slopes in the proposed list. No assertion about an unconstrained optimizer is used in this construction.

For $K = 0$, the adjacent equalities join to one affine function $h = f_0 + g_0 x$. For a support of slope p, its global minimum gap equals the value at the appropriate domain endpoint. Equivalently the exact condition at any node is $f_i - \ell(x_i) \geq \max\{(p - g_i)(-x_i), (p - g_i)(A - x_i)\}$, which is (99). Therefore this case is also necessary and sufficient without division by K.

Adding back $\mu x^2/2$ gives the asserted strongly convex, L-smooth q. Its derivative is nondecreasing and lies between the original q' values at the endpoints of each piece. Thus the node derivative cap implies the cap everywhere. The paid intervals are realized at their nodes, and every original quadratic support holds globally. This proves sufficiency.

Each adjacent inequality defines a convex set: its nonaffine term is the square of an affine function. For fixed feasible displacement t, the expression $(p_j - \mu a_j - g_i)t - Kt^2/2$ is affine in the optimization variables. Its supremum is convex, and the left side of (99) is affine. Consequently all support inequalities are convex even when the support slopes are variables. All constraints are closed. The finite value, derivative and supplied slope boxes make the feasible set compact; compatibility makes it nonempty. Linear contrasts therefore have attained extrema. Necessity and constructive sufficiency identify these extrema with the function-class extrema, including the deterministic curvature adjustment when returning from h to q. Removing constraints enlarges this class and proves the claimed dominance. *Q.E.D.*

THE RESIDUAL CERTIFICATE (101): For every feasible z, $H(z) \geq H(z) + \sum_j y_j c_j(z)$ because $y_j \geq 0$ and $c_j(z) \leq 0$. Convexity and the chosen subgradients give

$$H(z) + \sum_j y_j c_j(z) \geq \mathcal{L}_0 + R^\mathsf{T}(z - z_0).$$

The minimum of this affine expression over the enclosing box is exactly B. Hence $H(z) \geq B$ for every feasible point and $q(b) - q(a) \leq -B$ for every compatible original objective. For $K > 0$ the maximizing displacement in (97) is unique, so differentiation gives the clipped expression in the main text; its values agree at the two clipping transitions. For $K = 0$ choose the active endpoint as a subgradient of $\max\{v\ell, vr\}$, and either endpoint or any convex combination when $v = 0$. The chain rule for affine arguments supplies a valid subgradient of each support constraint. All these choices can be rational. Constraint values, subgradients, the box residual and the final bound are then exact rational operations. No feasibility or stationarity assumption on z_0 has entered the proof. *Q.E.D.*

For completeness, $A = 0$ is an identity problem: the only feasible action is zero and every action contrast is exactly zero. One checks the paid value intervals and the supports at that point; neither derivative interpolation nor an action query is required. Repeated anchors do not create zero-length intervals. An empty feasible set indicates inconsistent supplied data and must be reported; constraints cannot be discarded to manufacture a gain.

27.1. *Interval slopes and a global robust lower obstacle*

The shared slope variables are an exact convex lifting, not additional information relative to enforcing the entire robust pointwise obstacle. To see this, let a convex function h satisfy $h(a) \geq W$, and suppose

$$h(x) \geq W + \min_{p \in [p^-, p^+]} p(x - a) \qquad (0 \leq x \leq A).$$

Define

$$L_a = \sup_{x<a} \frac{h(x) - W}{x - a}, \qquad R_a = \inf_{x>a} \frac{h(x) - W}{x - a},$$

with the empty-side conventions $L_a = -\infty$ and $R_a = +\infty$. For $x < a < y$, convexity and $W \leq h(a)$ give $(h(x) - W)/(x - a) \leq (h(y) - W)/(y - a)$, hence $L_a \leq R_a$. The robust obstacle gives $L_a \leq p^+$ and $R_a \geq p^-$. Consequently $[L_a, R_a] \cap [p^-, p^+]$ is nonempty. Any p in that intersection satisfies $h(x) \geq W + p(x - a)$ everywhere. The converse follows by taking the minimum over the slope interval. This also proves

the endpoint cases with the empty-side conventions. Adding back the common quadratic establishes the assertion for (95).

27.2. *An exact action contrast that uses the omitted support*

PROPOSITION S.41—A strict economic effect of retaining a paid support: *Take* $A = 1$, $\mu = 1$, $L = 2$, $M = 3$, *and disclose*

$$q(0) = 0, \quad q(1/2) \in [0,1], \quad q'(1/2) \in [-5/4, 3].$$

The paid support at $a_1 = 1/2$ *has* $W_1 = 0$ *and exact slope* $p_1 = 1$, *so*

$$S_1(x) = x^2/2 + x/2 - 3/8.$$

Let $\mathcal{Q}_{\text{red}}$ *be the class retaining the displayed levels and derivatives and the curvature and derivative-cap assumptions, but omitting* $q \geq S_1$. *Let* $\mathcal{Q}_{\text{full}}$ *also retain that support. For the incumbent* $a = 1$ *and candidate* $b = 0$,

$$\max_{q \in \mathcal{Q}_{\text{red}}} [q(0) - q(1)] = -1/4, \qquad \max_{q \in \mathcal{Q}_{\text{full}}} [q(0) - q(1)] = -5/8.$$

Both maxima are attained. With arbitrary Borel $\eta : [0,1] \to [0,\delta]$, *the corresponding maximum original-policy contrasts are* $-1/4 + \delta$ *and* $-5/8 + \delta$.

PROOF: Strong convexity at the midpoint gives $q(1/2) \leq [q(0) + q(1)]/2 - 1/8$, hence $q(1) \geq 1/4$. The function $q_{\text{red}}(x) = x^2/2 - x/4$ attains this bound. It has curvature one, derivative between $-1/4$ and $3/4$, $q_{\text{red}}(1/2) = 0$, and $q'_{\text{red}}(1/2) = 1/4$; it therefore satisfies every reduced-class restriction. In the full class the support gives $q(1) \geq S_1(1) = 5/8$. The function $q_{\text{full}}(x) = x^2/2 + x/8$ attains this bound. Its midpoint value is $3/16$, its midpoint derivative is $5/8$, and its derivative remains within the cap. Moreover

$$q_{\text{full}}(x) - S_1(x) = 3(1 - x)/8 \geq 0.$$

Thus it satisfies the entire global support. Both functions vanish at zero, proving the two contrast maxima. Taking $\eta(0) = \delta$ and $\eta(1) = 0$ attains the upper future-error

difference for each witness, while the uniform bound gives the reverse inequality. *Q.E.D.*

The derivative interval in this example does not hide a stronger free derivative query. Converting the support with its paid upper level $U_1 = 1$ uses gap $e = 1$, $K = 1$ and both feasible radii $1/2$. The exact bound is $\Psi(1, 1/2, 1/2) = 9/4$, giving $[-5/4, 13/4]$, and intersection with $[-3, 3]$ gives the stated interval. The improvement is thus over the level–derivative summary of that payload. Directly evaluating $S_1(1)$ already obtains the stronger answer here; no exclusive benefit of a conic solver or neural representation is asserted. If $1/4 \le \delta < 5/8$, this omitted information changes whether the proposed policy change is uniformly certified at all.

27.3. *Why nodewise obstacle checks are insufficient*

Take $K = 1$ and $h(x) = x^2/2 - x/2$ on $[0, 1]$, with nodes $0, 1/4, 1$. Its jets satisfy every adjacent interpolation condition. The affine obstacle $\ell(x) = -7/64$ lies below the function at these nodes: their gaps are $7/64, 1/64, 7/64$. Nevertheless $h(1/2) = -1/8 < -7/64$. At the middle node $g = -1/4$, so $v = 1/4$ and

$$\Phi_1(1/4; -1/4, 3/4) = 1/32 > 1/64 = f - \ell.$$

The new constraint rejects this impossible global-support payload. Equivalently, with $\mu = 1$, the proposed objective is $q(x) = x^2 - x/2$ and the proposed support is $S(x) = x^2/2 - 7/64$. It can be written at anchor $1/4$ with $W = -5/64$ and slope $p = 1/4$. This is a negative algebraic check of inconsistent data, not a claimed admissible economic observation.

28. PROOF OF THE COMPLETE ACQUISITION ACCOUNT

INITIAL BRACKET AND THEOREM S.40: Let a_q^* be the constrained minimizer. Its variational inequality and strong convexity give $q(a_0) - q(a_q^*) \ge \mu |a_0 - a_q^*|^2/2$. Since $q(a_0) \le J(a_0)$, the original-policy regret bound implies (102), including either capacity boundary. An outward radius and intersection with another valid optimizer interval cannot exclude a_q^*.

Write w for the actual interval width at one planned step. If $w \le t$, no query is necessary. Otherwise the actual midpoint's distance from each domain boundary is at least $(1/2 - \xi)w > (1/2 - \xi)t \ge h$. Thus both trial displacements $\pm h$ in the support conversion are feasible. Smoothness and the support inequality imply

$$p^- - e/h - Kh/2 \le q'(a) \le p^+ + e/h + Kh/2.$$

The interval width is at most $v + 2e/h + Kh$. Under (103) its three terms are at most $\mu t/4, \mu t/2, \mu t/4$, respectively. For $K = 0$ the last term is zero. The derivative interval therefore has width at most μt. The inherited rounded-midpoint result now gives new optimizer width at most

$$\max\{\gamma w, \mu t/\mu\} \le \max\{\gamma w_k, t\} = t.$$

This argument also covers a sign-resolved derivative interval, which may contract at a coarser realized precision than requested. It uses a valid original derivative enclosure; differentiating the fitted critic does not establish the premise.

Minimality of N gives $\gamma w_0 \le w_1 < w_0$, with the upper inequality weak if a redundant boundary case is retained. Subsequent planned widths satisfy $w_{j+1} = \gamma w_j$. Induction using the preceding step proves the asserted final width. Summing the per-probe work bounds gives

$$\sum_{j=1}^{N} (C_0 + C_1 w_j^{-p}) = NC_0 + C_1 w_*^{-p} \sum_{j=0}^{N-1} \gamma^{pj},$$

which is (104). Skipped steps can only reduce this nonnegative bound. For the certificate-planning optimality claim, any N-step plan reaching $w_N \le w_*$ under $w_{j+1} \ge \gamma w_j$ must have $w_j \le w_* \gamma^{-(N-j)}$. The displayed schedule takes the largest feasible width at every step, minimizing each nonincreasing cost term simultaneously. Fewer than N steps cannot attain the target under the guaranteed contraction floor. A longer plan has the same upper constraints on its final N widths; its additional nonnegative cost terms cannot improve the minimum. This proves optimality for this declared bound and planning interface, without asserting that an actual oracle must cost the upper bound or that observations cannot produce better contraction.

Finally a_q^* lies in the certified interval, giving $|b - a_q^*| \le r_b$. Integrating the derivative cap gives $q(b) - q(a_q^*) \le Mr_b$. Adding $0 \le \eta(b) \le \delta$ proves the regret bound in (105) directly, without an absolute-value width penalty. Also $q(a_q^*) \ge q(b) - Mr_b \ge W_b - Mr_b$ supplies a lower optimum bound when one is needed. Combining only this latter lower bound with $U_b + \delta$ would give the weaker bound $e_b + \delta + Mr_b$; an implementation can use the minimum of all valid regret certificates instead. If $q'(a_q^*) = 0$, the smooth Taylor upper inequality instead gives $q(b) - q(a_q^*) \le Lr_b^2/2$ and the corresponding replacement. An interval strictly inside the feasible domain contains an interior minimizer and therefore certifies that stationary condition. At a capacity minimum its derivative can be nonzero, which is why the quadratic bound is not asserted there without further evidence. The absolute interval for J has width $e_b + \delta$, a separate condition from optimizer localization. The payment for its value query and for all other service operations is not present in the sum over the N localization probes and must be added explicitly. *Q.E.D.*

28.1. *Sharp terminal information requirements*

The following exact reduced-interface statement separates action resolution from the precision of an absolute policy value. It does not claim a lower bound after raw supports, additional value data or the original investment formulas have also been supplied.

PROPOSITION S.42—Matching terminal regret at a stationary bracket interface: *Suppose the only action information is $0 \le \ell \le r \le A$, $0 < \mu \le L$, and the fact that the unique minimizer z belongs to $[\ell, r]$ and satisfies $q'(z) = 0$. Let $\mathcal{Q}_B$ contain all differentiable μ-strongly convex functions with L-Lipschitz derivative and these properties. A derivative cap, if imposed for this statement, is at least LA. For every Borel law ν of $B \in [0, A]$,*

$$\sup_{q \in \mathcal{Q}_B} \mathbb{E}_\nu[q(B) - \min q] = \frac{L}{2}\left\{\mathrm{Var}_\nu(B) + \max\left[(\mathbb{E}_\nu B - \ell)^2, (\mathbb{E}_\nu B - r)^2\right]\right\}.$$

Hence

$$\inf_\nu \sup_{\substack{q \in \mathcal{Q}_B \\ 0 \le \eta \le \delta}} \mathbb{E}_\nu[q(B) + \eta(B) - \min q] = \delta + L(r - \ell)^2/8.$$

The deterministic midpoint attains the bound; randomization cannot improve it. Independently, if the only absolute-level information at a chosen action is $q(b) \in [W, U]$, with $0 \le \eta(b) \le \delta$ and arbitrary additive shifts otherwise allowed, the exact compatible range of $J(b)$ is $[W, U + \delta]$.

PROOF: Smoothness at the stationary minimizer gives $q(B) - q(z) \le L(B - z)^2/2$. Taking expectations and maximizing over $z \in [\ell, r]$ bounds the left side by the stated expression, because a convex quadratic in z has its interval maximum at an endpoint. For either endpoint z the single function $q_z(x) = L(x - z)^2/2$ belongs to the class, is also μ-strongly convex, satisfies the permitted cap, and attains its corresponding expression. Choosing the maximizing endpoint proves equality; the adverse function knows the law before the random action is drawn. The expression is minimized by zero variance and mean $(\ell + r)/2$, giving $L(r - \ell)^2/8$. The constant error $\eta \equiv \delta$ attains the additional worst-case allowance. For the absolute-level statement, an arbitrary compatible shape can be shifted to give either endpoint W, U at b. Taking the policy error respectively zero or δ attains both endpoints of $[W, U + \delta]$; the assumed bounds give its containment. No optimizer localization can resolve this free level shift without further value information. *Q.E.D.*

29. DIRECT STOPPING FOR THE ORIGINAL ECONOMIC DECISION

This section establishes the stopping and replacement rules used by the new direct service. It assembles the inherited original-support results into a complete service guarantee; it does not claim new convexity or root-finding principles. All statements condition on valid observations of the same original Bellman objective. The critic supplies a feasible proposal and never supplies an unverified derivative of that objective.

Fix one state and the original risk law. Let $A > 0$, and let $q : [0, A] \to \mathbb{R}$ be continuously differentiable, μ-strongly convex and have L-Lipschitz derivative, with $0 < \mu \le L$ and $|q'| \le M$. Its unique constrained minimizer is a^* and its value is $q^* = q(a^*)$. The complete policy with the unchanged future controller has value

$$J(a) = q(a) + \eta(a), \qquad 0 \le \eta(a) \le \delta. \tag{106}$$

No continuity or differentiability of η is assumed. The comparison objective is q^*, the original optimum with optimal future decisions; we do not replace it by $\min J$.

Let the retained original observation collection be finite and nonempty, as it is after the direct service has bought its first support. Every observation j has an action a_j, an interval $W_j \le q(a_j) \le U_j$, and a derivative interval $g_j^- \le q'(a_j) \le g_j^+$. Its global support supplies a single $p_j \in [p_j^-, p_j^+]$ such that

$$q(t) \ge W_j + p_j(t - a_j) + \frac{\mu}{2}(t - a_j)^2 \quad (0 \le t \le A).$$

The support and derivative intervals are obtained from the original continuous-risk calculation and the finite-radius conversion of Theorem 6. Neither an approximate actor nor a fitted continuation derivative establishes these premises. We also retain any independently valid original-optimum bound $B_0 \le q^*$.

29.1. *Transported derivatives and the constrained optimizer*

For each observation define the continuous piecewise-affine functions

$$\lambda_j(t) = g_j^- + \begin{cases} L(t - a_j), & t < a_j, \\ \mu(t - a_j), & t \ge a_j, \end{cases}$$

$$v_j(t) = g_j^+ + \begin{cases} \mu(t - a_j), & t < a_j, \\ L(t - a_j), & t \ge a_j. \end{cases}$$

The direct service uses $\lambda(t) = \max_j \lambda_j(t)$ and $v(t) = \min_j v_j(t)$. The global cap M remains available for the separate distance bound below. For any continuous strictly increasing f define its constrained zero by

$$Z(f) = \begin{cases} 0, & f(0) \ge 0, \\ A, & f(A) \le 0, \\ \text{the unique zero of } f \text{ in } (0, A), & \text{otherwise.} \end{cases}$$

LEMMA S.43—A nested optimizer enclosure: *The inequalities $\lambda \le q' \le v$ hold on $[0, A]$. Both envelopes are continuous and satisfy $f(t) - f(s) \ge \mu(t - s)$ for $t > s$.*

Consequently

$$\ell = Z(v) \le a^* \le Z(\lambda) = r. \tag{107}$$

Retaining all old observations when adding a valid new observation can only increase ℓ and decrease r. If

$$v(0) \le 0 \le \lambda(A), \tag{108}$$

then $q'(a^*) = 0$. This condition allows a boundary optimum whose one-sided derivative is zero; a general boundary optimum need not be stationary.

PROOF: For $t \ge a_j$, strong convexity and Lipschitz continuity of q' give $\mu(t - a_j) \le q'(t) - q'(a_j) \le L(t - a_j)$. For $t < a_j$ the inequalities reverse the two curvature coefficients when multiplied by the negative displacement. This proves $\lambda_j(t) \le q'(t) \le v_j(t)$, hence the asserted envelope inequalities. Each transported function is continuous and its slopes are μ or L. If every f_j satisfies $f_j(t) \ge f_j(s) + \mu(t - s)$, its finite maximum and finite minimum satisfy the same inequality. Thus both envelopes are continuous and strictly increasing.

For such functions, pointwise order reverses the order of constrained zeros: $f \le g$ implies $Z(g) \le Z(f)$. To verify this, an interior zero of g has $f \le 0$, so it cannot lie to the right of $Z(f)$; if the constrained zero is an endpoint, the relevant endpoint sign gives the same conclusion. The first-order variational inequality for minimizing q on $[0, A]$ gives $a^* = Z(q')$, proving (107). Additional observations increase λ and decrease v pointwise, so their constrained zeros give the stated nesting. Finally (108) implies $q'(0) \le 0 \le q'(A)$. By continuity and strict monotonicity, q' has exactly one zero in $[0, A]$, which is its constrained minimizer. Without these signs a constrained minimum can have a nonzero outward derivative. *Q.E.D.*

29.2. *Projected gaps and separate stopping gates*

For a feasible realized action a and a derivative interval $g_- \le q'(a) \le g_+$, put

$$d(z) = \operatorname{proj}_{[-a, A-a]}(-z/\mu),$$

$$D_\mu(a; g_-, g_+) = \max_{z \in \{g_-, g_+\}} \left\{ -z\, d(z) - \frac{\mu}{2} d(z)^2 \right\}. \tag{109}$$

The maximum is nonnegative because the feasible displacement set contains zero. This is the projected strong-convexity calculation already used by the inherited action service. In particular, it recognizes a constrained optimum without requiring its unconstrained gradient to vanish.

For completeness, write $C^-(u,v) \le q(v) - q(u) \le C^+(u,v)$ for the integrals of the appropriate transported derivative envelopes, with their order reversed when $v < u$ and both equal to zero when $u = v$. Define

$$S_j^-(t) = W_j + \min\{p_j^-(t-a_j), p_j^+(t-a_j)\} + \frac{\mu}{2}(t-a_j)^2,$$

$$L_q(a) = \max_j \{S_j^-(a),\ W_j + C^-(a_j,a)\},$$

$$U_q(a) = \min_j \{U_j + C^+(a_j,a)\},$$

$$B = \max\left\{B_0,\ \min_{t\in[0,A]} \max_j S_j^-(t)\right\}.$$

The finite piecewise-quadratic minimum is evaluated by retaining all breakpoints and feasible stationary points. It does not assume that the interval-slope support envelope is convex. These are the original-level quantities used by the direct implementation.

PROPOSITION S.44—A complete direct action certificate: *Let $[\ell, r]$ be the bracket in Lemma S.43, let $\rho(a) = \max\{|a-\ell|, |a-r|\}$, and use $g_- = \lambda(a)$ and $g_+ = v(a)$ in (109). Each of the following is an upper bound on $J(a) - q^*$:*

$$U_q(a) + \delta - B, \qquad \delta + D_\mu(a; g_-, g_+), \qquad \delta + M\rho(a). \tag{110}$$

When (108) is certified, $\delta + L\rho(a)^2/2$ is a further upper bound. Their minimum, $G(a)$, is therefore a valid nonnegative original-policy regret bound. Separately,

$$L_q(a) \le J(a) \le U_q(a) + \delta, \qquad W_J(a) = U_q(a) + \delta - L_q(a). \tag{111}$$

Hence the two tests $G(a) \le \tau$ and $W_J(a) \le \omega$ establish the declared original-regret and absolute-value-width requirements. A smaller prescribed target $\tau_* \le \tau$ is an additional regret outcome and requires $G(a) \le \tau_*$ separately.

For a fixed action, retaining valid old observations makes every available bound and the exact width nonincreasing. A stationary certificate, once established, remains established. Thus an already qualified unchanged action stays qualified under exact accumulation of valid information. Moreover $|a - a^*| \le \sqrt{2G(a)/\mu}$*; this quantity is a bound, not a measurement of realized action error.*

PROOF: Strong convexity gives

$$q(a+d) \ge q(a) + q'(a)d + \frac{\mu}{2}d^2 \quad (-a \le d \le A - a).$$

Consequently

$$q(a) - q^* \le \max_{-a \le d \le A-a} \{-q'(a)d - \mu d^2/2\}.$$

For a fixed z the maximizer is exactly $d(z)$. The maximum over displacements is a convex function of z, being a supremum of affine functions of z. Its maximum on $[g_-, g_+]$ therefore occurs at an endpoint, proving the projected bound. The argument includes $a = 0, A$. For example, if $a = A$ and $g_+ \le 0$, each projected displacement is zero and the projected gap vanishes.

The bracket gives $|a - a^*| \le \rho(a)$. Integrating the derivative cap yields $q(a) - q^* \le M\rho(a)$. If $q'(a^*) = 0$, the smooth upper Taylor inequality gives $q(a) - q^* \le L|a - a^*|^2/2 \le L\rho(a)^2/2$ instead. Adding $0 \le \eta(a) \le \delta$ proves these direct regret routes.

Every raw support S_j^- is below q, since its minimum over the reported slope interval is no larger than the value at the true shared slope. Integrating valid derivative envelopes also gives valid contrasts. It follows that $L_q(a) \le q(a) \le U_q(a)$ and $B \le q^*$. Equation (106) now proves the level gap and (111). Taking the minimum of valid upper bounds preserves validity. No error is differentiated and the future allowance is added once.

Adding observations tightens both contrast endpoints, increases each maximum defining a lower bound, and decreases each minimum defining an upper bound. It increases the maximum of the raw supports pointwise, and hence cannot decrease its minimum. Thus B and $L_q(a)$ cannot decrease, and $U_q(a)$ cannot increase. The derivative interval narrows, so its robust projected gap cannot increase. The bracket

nests, so $\rho(a)$ cannot increase. The endpoint signs in (108) are preserved because v decreases and λ increases. These observations prove monotonicity of the bounds, their minimum and the exact width. Finally the constrained minimizer satisfies $q'(a^*)(a - a^*) \geq 0$. Strong convexity then gives $q(a) - q^* \geq \mu|a - a^*|^2/2$. Since $q(a) \leq J(a)$, the claimed distance bound follows. *Q.E.D.*

The implementation converts L_q downwards, $U_q + \delta$ and G upwards to finite binary64 endpoints, and rounds their reported width upwards. It checks both the exact rational tests and the reported outward tests. A claimed pass therefore requires an outward regret no larger than the stated tolerance and an outward width no larger than its separate tolerance. The mathematical guarantee concerns these outward enclosures; a nonfinite or inconsistent numerical conversion is not a successful certificate.

PROPOSITION S.45—Why the two gates cannot be merged: *An action-regret certificate alone need not determine an absolute policy-value interval or an original-optimum lower bound. In particular, $J^+(a) - G(a)$ need not be a lower bound for q^* when $J^+(a)$ is merely an upper enclosure and $G(a)$ is a direct regret bound. Conversely, exact knowledge of $J(a)$ does not certify a small decision regret without information about alternatives.*

PROOF: On $[0,1]$, consider the compatible family $q_C(t) = C + (t - 1/2)^2$, $0 \leq C \leq V$, for a rational $V > 0$. Its derivative and unique optimizer $a = 1/2$ are known exactly. The common support $(t - 1/2)^2$ is valid for every member, while the absolute-level interval at the optimizer is $[0, V]$. Under $0 \leq \eta \leq \delta$, the direct regret bound is $G(a) = \delta$ and the upper policy endpoint is $J^+(a) = V + \delta$. For $C = 0$ and $\eta \equiv 0$, the true optimum is zero whereas $J^+(a) - G(a) = V > 0$. An arbitrarily accurate optimizer certificate therefore cannot remove the unresolved additive level. It follows only that $q^* \geq J(a) - G(a) \geq J^-(a) - G(a)$, using a lower, not an upper, policy endpoint. For the converse, take $q(t) = (t - 1)^2$, $\eta = 0$, and $a = 0$. Its value at the chosen action is exactly one, with zero value uncertainty, but its regret is one. Both examples satisfy the stated smooth strong-convexity assumptions. *Q.E.D.*

29.3. *Strict policy changes and rounded localization*

PROPOSITION S.46—Same-objective replacement: *Let a be a completed qualified action and let a distinct realized candidate b satisfy both qualification gates. Set*

$$u(a,b) = \min\{C^+(a,b),\ U_q(b) - L_q(a)\}.$$

If $u(a,b) + \delta < 0$, replacing a by b decreases the original complete-policy objective by at least $-u(a,b) - \delta > 0$. No claim of such improvement follows merely from a smaller regret bound. For $b = a$, the actual complete-policy contrast is exactly zero under the unchanged future controller.

PROOF: Both entries in the displayed minimum bound $q(b) - q(a)$ from above. Hence

$$J(b) - J(a) = q(b) - q(a) + \eta(b) - \eta(a) \le u(a,b) + \delta.$$

The result follows. The single allowance is sufficient because $\eta(b) \le \delta$ and $\eta(a) \ge 0$. Without further regularity or coupling of the future errors it cannot in general be removed: the admissible assignments $\eta(b) = \delta$, $\eta(a) = 0$ attain that allowance. Identical actions use the identical complete future policy, so their actual values are equal. The first qualified action requires no comparison with an earlier completed policy; its selection establishes qualification only. *Q.E.D.*

LEMMA S.47—The actual rounded-midpoint condition: *Let a valid old optimizer interval be $I = [\ell, r]$, of width w, and let a feasible queried action a have $\epsilon = |a - (\ell + r)/2|$. Suppose its derivative interval is $[g_-, g_+]$. Intersect I with $[0, a]$ if $g_- > 0$, with $[a, A]$ if $g_+ < 0$, and otherwise with $[a - g_+/\mu, a - g_-/\mu]$. The resulting interval remains valid and has width at most*

$$\max\{w/2 + \epsilon,\ (g_+ - g_-)/\mu\}. \tag{112}$$

When I is the bracket of the retained old envelopes, the new full transported-envelope bracket is at least as informative as this update.

For lattice spacing $h > 0$, the implemented feasible downward rounding $a = h\lfloor((\ell + r)/2)/h\rfloor$ satisfies $0 \le (\ell + r)/2 - a < h$. Whenever the actual updated bracket passes

the checked inequality $w_{\text{new}} \le w/2 + \epsilon$, a genuine localization improvement with that rounding allowance has been certified. After m consecutive such checked updates from width w_0,

$$w_m \le 2^{-m} w_0 + 2h(1 - 2^{-m}).$$

These statements do not assert that a finite prescribed precision ladder must produce a passing update.

PROOF: If $q'(a) > 0$, monotonicity and the constrained optimality condition imply $a^* \le a$; if $q'(a) < 0$, they imply $a^* \ge a$. This proves the two sign-resolved intersections, whose widths are at most $w/2 + \epsilon$. If $g_- \le 0 \le g_+$ and $a^* < a$, strong monotonicity and $q'(a^*) \ge 0$ when the minimizer is the left boundary give $q'(a) \ge \mu(a - a^*)$. Thus $a^* \ge a - g_+/\mu$, while the proposed upper endpoint is at least a. If $a^* > a$, the analogous inequality, including $q'(A) \le 0$ at a right boundary optimum, gives $a^* \le a - g_-/\mu$, while the proposed lower endpoint is at most a. Equality $a = a^*$ is immediate. The unresolved-sign interval has width $(g_+ - g_-)/\mu$, proving (112). The transported cones of the observation impose these same restrictions on their constrained zeros; retaining further valid observations can only tighten them.

The floor of a nonnegative feasible midpoint is feasible, including when A is not a lattice point, and has the asserted error. The checked inequality therefore implies $w_{k+1} \le w_k/2 + h$. Iterating this scalar inequality gives the stated expression. No contraction is inferred for a failed check, for the first arbitrary actor center, or for a query that stops before obtaining a valid derivative interval. A lattice floor, a terminal accuracy floor or the resource limit can leave the stricter regret target unresolved. *Q.E.D.*

29.4. *What a finite direct-service return guarantees*

The frozen service takes $h = 2^{-30}$, $M = 7$, $\mu = 2$ and $\delta = 15/2^{24}$, with the proved original L appropriate to the risk level. Its ordinary tolerances are $\tau = 2^{-8}$ and $\omega = 2^{-13}$; its separately reported stronger target is $\tau_* = 2^{-16}$. It starts with the same paid actor or conventional proposal as its comparator and buys its own first original support. It does not inherit a completed comparison service without paying for it.

The first actor-center departure is allowed without a halving assertion; subsequent departures require the actual bracket check in Lemma S.47. Every candidate is the realized feasible action, including its rounding, rather than an unimplemented ideal midpoint.

PROPOSITION S.48—Finite service qualification and the cost boundary: *Assume all completed support records satisfy the premises above and all returned floating endpoints enclose their exact quantities. Every successful direct-service return satisfies both ordinary gates, either through Proposition S.44 or through its separately paid, independently executed qualified fallback. Each recorded distinct-action replacement has the strict same-objective guarantee in Proposition S.46. Requalification of an unchanged action is not recorded as a policy gain. Reaching a resource or precision limit with an ordinarily qualified incumbent does not imply attainment of* τ_*.

The direct route starts no more than 8×7 *queries, never purchases the same action at the same precision twice, and checks its cap of 32,768 requested terminal contexts before starting each query. A query at* n *bins charges* $2n + 1$ *requested contexts, with* $n \in \{32, 64, 128, 256, 512, 1024, 2048\}$. *These are finite budget statements, not a bound on all arithmetic operations or elapsed time. All failed or discarded direct work, actor acquisition and inference, certificate work, and any completed fallback belong to the complete service cost.*

PROOF: Before selection the service has no completed incumbent to improve. Its first selection is restricted to candidates passing both exact and outward ordinary gates. After selection it reconsiders the same action using all retained information. Proposition S.44 preserves exact qualification under valid information accumulation. A different action is selected only if it also passes both gates and the strict replacement test; Proposition S.46 proves the claimed gain. Induction establishes qualification of every completed selected policy. The stronger stopping condition explicitly checks both $G \leq \tau_*$ and width. Ending at a different declared limit returns the qualified incumbent with its actual stronger-target status. A recoverable numerical failure returns a retained completed qualified policy when one is available; otherwise the service pays the entire qualified fallback. If the fallback or an infrastructure check aborts, there is no successful-return claim.

There are eight center rounds and seven declared bin levels per round. The visited-pair record prevents duplicate purchases. Before beginning a new query the service adds its $2n+1$ requested contexts only if doing so remains within the declared cap, and retains the charge even if the query subsequently fails. The requested-context count is conservative about a failed partial query and is distinct from the saved actual operation ledger. The actor, sorting, terminal solving, support algebra, serialization, and complete fallback can require additional work. No part of the budget argument licenses omitting those costs or assigning a zero cost to a refused candidate. The certificate and accounting rules apply equally to neural, matched hinge, polynomial, spline, grid and online proposals supplied with the same information. *Q.E.D.*

30. EXACT REUSE OF BELLMAN CERTIFICATE GEOMETRY

The following results concern the separately frozen within-service cache. They apply to a finite nonempty collection of rational observations with $A > 0$ and $0 < \mu \le L$. Every observation anchor, requested action and integration endpoint is feasible. A case and its payload entries are immutable throughout its lifetime inside one service. The normalized cases have a fixed number of scalar fields outside their finite payload list. The implementation retains a strong reference to every case used for an object-identity binding; thus a discarded temporary case cannot have its identifier reassigned to a different case while the binding exists. Mutating a bound case is outside these assumptions.

Let s be the number of retained derivative observations, including repeated anchors. The two transported functions of observation j are

$$\lambda_j(t) = g_j^- + \begin{cases} L(t-a_j), & t < a_j, \\ \mu(t-a_j), & t \ge a_j, \end{cases}$$

$$v_j(t) = g_j^+ + \begin{cases} \mu(t-a_j), & t < a_j, \\ L(t-a_j), & t \ge a_j. \end{cases}$$

Write $e_-(t) = \max_j \lambda_j(t)$ and $e_+(t) = \min_j v_j(t)$. These are precisely the inherited transported envelopes, without adding an unrecorded information constraint. Their

geometry key is

$$\kappa_D = (A, \mu, L, ((a_j, g_j^-, g_j^+))_{j=1}^{s}).$$

The observation order is retained. A subset request uses the corresponding ordered subset in this key. The inherited algorithm first partitions the domain at all observation anchors. On each resulting open interval it retains every strict interior intersection of nonparallel transported lines, even when those lines do not form the active envelope. On each subinterval it chooses the extremizing line at the midpoint, with the original fixed lexicographic rule for ties. No segments are coalesced by the cache.

For either envelope, write its full ordered partition as

$$(x_i, x_{i+1}, m_i, c_i)_{i=0}^{P-1}, \qquad 0 = x_0 < \cdots < x_P = A, \qquad e(t) = m_i t + c_i \quad (x_i < t < x_{i+1}).$$

At endpoints use the continuous extension. Define exact prefixes

$$I_0 = 0, \qquad I_{i+1} = I_i + \frac{m_i}{2}(x_{i+1}^2 - x_i^2) + c_i(x_{i+1} - x_i).$$

For $t \in [x_i, x_{i+1}]$ the stored primitive evaluates

$$F(t) = I_i + \frac{m_i}{2}(t^2 - x_i^2) + c_i(t - x_i). \tag{113}$$

The values at 0 and A are explicitly initialized. Binary search on the ordered starts chooses the interval for every other new endpoint; a boundary can be assigned to either adjacent interval because the two formulas agree.

PROPOSITION S.49—Exact geometry and service identity: *Under the stated assumptions, the cache returns exactly the same rational result as the inherited implementation for each of the following operations: full or partial envelope partition; both directed contrast endpoints, including the best-single-record mode; derivative-envelope value; constrained envelope zero; and raw-support minimum with its selected minimizing action.*

The raw-support minimum is keyed separately by

$$\kappa_S = (A, \mu, ((a_j, W_j, p_j^-, p_j^+))_{j=1}^{s}).$$

The original incumbent, future allowance, original-optimum floor and upper value levels are not arguments of that raw minimum. They remain arguments of the separately recalculated policy certificate. In particular, a cached raw minimum is not a cached value of the final original-optimum lower bound.

Suppose the original oracle and actor return the same outputs on the same inputs, and the frozen service's control decisions depend only on scientific values and declared discrete budgets, not on elapsed time. Replacing the listed geometry operations by their cached versions then leaves the complete sequence of economic calls, candidate decisions and scientific certificates unchanged whenever both executions complete their required arithmetic. The new cache counters and clocks may differ. Resetting the cache at each paid decision preserves this conclusion without transferring acquired information between services.

PROOF: First consider a partial interval $[u, v] \subseteq [0, A]$ with $u < v$. Its old partition has the endpoints u, v, every observation anchor in its interior, and every pairwise line crossing in the interior of each resulting anchor interval. These are exactly the full-domain breakpoints strictly between u and v, together with u, v. To see this, restrict any full anchor interval to $[u, v]$. Each observation still has the same transported line on that restriction. Its intersection with any other nonparallel line is the same rational number; it is retained precisely when it lies inside the restriction. A crossing at an anchor or at u, v is already an endpoint. Parallel equal lines create no extra crossing in either calculation.

The relative order of distinct affine lines is constant on each open subinterval between all crossings. Thus its active envelope line cannot change when a subinterval is truncated at u or v. If several lines agree there, their slope and intercept agree throughout that open interval, so the fixed tie rule also gives the same coefficients. Clipping the full list therefore gives exactly the old partial list, including the retained inactive-line crossings and coefficient pairs. If $u = v$, both routines return the empty list. The same argument applies to a declared subset.

Induction on i shows $I_i = \int_0^{x_i} e(t)\, dt$. Integrating the selected affine line on the final partial segment proves (113). Consequently its differences are the exact integrals returned by summing the old partial partition. Rational addition introduces no

order-dependent rounding. For $a < b$, the lower contrast uses $F_-(b) - F_-(a)$ and the upper one uses $F_+(b) - F_+(a)$. For $a > b$, reversing the integration direction reverses the two bounds. For $a = b$, both routines return zero. The best-single-record mode applies this identity to each singleton record and then takes the same maximum of lower endpoints and minimum of upper endpoints.

Each envelope is continuous, because it is a finite maximum or minimum of continuous transported functions. Evaluating the selected affine piece at an interior point, a crossing, an anchor or either domain endpoint therefore gives the same value as direct evaluation of every observation. All slopes are at least $\mu > 0$, so the envelope is strictly increasing. The two endpoint sign tests and, when needed, the first feasible zero $-c_i/m_i$ therefore return its identical constrained zero. Saving an already computed value or zero cannot change it.

The raw-support routine minimizes

$$S(t) = \max_j \left\{ W_j + \min[p_j^-(t - a_j), p_j^+(t - a_j)] + \frac{\mu}{2}(t - a_j)^2 \right\} \quad (0 \le t \le A).$$

Every argument of this function is contained in κ_S; the omitted fields do not enter it. The cache executes the unchanged exact routine on the first occurrence of this complete key, retains its value and original tie-selected action, and returns that same pair thereafter. Equality of κ_D likewise implies equality of every argument of the transported geometry. The object-identity binding is sound by immutability and its strong references. This proves all individual identities.

For the service conclusion, induct on the ordered external calls and scientific control decisions. At the first call the frozen inputs agree. An original economic or actor call with the same inputs has the same output by the stated premise. A substituted geometry call has the same output by the identities just proved. A scientific branch or discrete-budget update therefore takes the same branch in both executions. The induction preserves all subsequent oracle arguments, certificates, candidate refusals, accepted changes and stopping decisions. Cache-table accesses are certificate work; they neither call an economic primitive nor replace a paid economic record. Each complete service starts with an empty cache and the context restores the original functions and clears its bindings on exit, including exceptional

exit. A failure caused by unavailable machine resources is not ruled out by an algebraic identity, and a failed execution is not asserted to complete. *Q.E.D.*

LEMMA S.50—Size and construction cost of the inherited partitions: *For one sign of a derivative ledger with $s \geq 1$ observations, the number of retained full-domain pieces obeys*

$$P \leq 1 + s + \binom{s}{2}. \tag{114}$$

The inherited all-crossing construction, followed by its prefix calculation, uses $O(s^3)$ exact scalar arithmetic, ordering and item-inspection operations and stores $O(s^2)$ piece data. These are operation counts, not bounds on the bit cost of rational numerators and denominators.

A first exact raw-support minimum for a key with s supports also has an $O(s^3)$ bound in this operation model. It need not be a convex function, and its minimum is not obtained by assuming convexity of its slope-interval envelope.

PROOF: There are at most s distinct internal observation anchors. Consider two anchors $a_j \leq a_k$ and the difference of their lower transported functions. It is constant on each exterior region $t < a_j$ and $t > a_k$, because both functions have the same slope there. On (a_j, a_k) its slope is $\mu - L \leq 0$. For the upper transported functions the middle slope is $L - \mu \geq 0$. Thus a pair of either kind has at most one isolated interior crossing not already an anchor. If $L = \mu$ it has none. Coincident lines on an interval contribute no isolated crossing. Counting all pairs proves (114), even though the algorithm retains crossings of inactive lines and repeated anchors.

There are at most $s + 1$ anchor intervals. Forming their s lines and checking every pair costs $O(s^3)$ operations. Let P be the actual number of pieces. Sorting the retained crossings costs at most $O((P + s)\log(s + 2))$, and selecting the extremizing line on each piece costs $O(sP)$. With (114), both are $O(s^3)$. The prefixes add $O(P)$ operations and storage. These statements describe the inherited enumeration; no optimal lower-envelope construction algorithm is being substituted in the complexity calculation.

For the support minimum subtract the common quadratic $\mu t^2/2$. Each remaining support is a continuous piecewise-affine function with one break at its own anchor.

A pair has at most three affine regions, hence at most three isolated crossings. There are consequently at most $1 + s + 3\binom{s}{2}$ resulting pieces before selecting their maximum. The inherited routine enumerates all pairwise crossings on every anchor interval and selects the maximal line on each piece. It retains all piece endpoints and the feasible stationary point of the common quadratic plus its selected affine line. These candidates include a minimizer on every closed piece. Evaluating the original support maximum at the candidates and choosing its minimum requires $O(s)$ work per candidate. There are $O(s^2)$ candidates and $O(s^3)$ pair-enumeration work, yielding the asserted bound. Downward slope-interval kinks cause no gap in this piecewise argument. *Q.E.D.*

PROPOSITION S.51—Finite operation and storage account for geometry reuse: *Within one paid service let $\mathcal{D}$ be the distinct derivative keys and $\mathcal{S}$ the distinct raw-support keys. Write s_κ for the support count at a key and P_κ for the total number of its derivative pieces actually built across both signs. For a derivative key let e_κ count previously unseen primitive arguments, v_κ previously unseen gradient arguments, and l_κ nondegenerate requests for a clipped partition. Let Q count all requests to the cache routines, including calls made by another cache routine, and let C count the total clipped pieces returned. Let B count formations of a derivative key, and R count requests for a raw-support minimum; write s_b, s_r for their respective record lengths.*

There are at most $2|\mathcal{D}|$ full derivative-partition builds, $2|\mathcal{D}|$ initial constrained-root calculations, and $|\mathcal{S}|$ executions of the original raw-support-minimum routine. The explicit geometry arithmetic, comparisons and key-entry inspections are bounded by

$$O\Bigg(\sum_{\kappa\in\mathcal{D}}\Big[s_\kappa^3 + P_\kappa + (e_\kappa + v_\kappa + l_\kappa)\log(2+P_\kappa)\Big] + \sum_{\sigma\in\mathcal{S}} s_\sigma^3 + Q + C + \sum_{b=1}^{B} s_b + \sum_{r=1}^{R} s_r\Bigg). \tag{115}$$

In particular, if every record has at most S entries and each built geometry at most P pieces, this is

$$O\big((|\mathcal{D}| + |\mathcal{S}|)S^3 + Q[S + \log(2+P)] + C\big).$$

The routines also make $O(Q)$ cache-table operations. This counts the operations requested of the tables and does not assume a constant bound on their internal hashing, collision handling or allocation time. Rational bit arithmetic and those table costs are additional machine work, included in the measured complete service.

Cached geometry, memoized arguments, exact keys and retained immutable case references require at most $O(\sum_{\kappa\in\mathcal{D}} P_\kappa + QS)$ stored scalar or reference entries under the same bound S. This memory is confined to one service. The result establishes neither a lower bound for the uncached computation nor a uniform complete-time advantage for the cached one.

PROOF: Each derivative key has two possible sign entries and each entry is built only on its first request. Lemma S.50 pays for those builds and the prefix sums. The first gradient or primitive evaluation at an argument uses binary search on ordered rational starts, requiring $O(\log(2 + P_\kappa))$ comparisons, followed by a constant number of exact scalar operations. Repeated arguments retrieve the exact saved value. A clipped partition pays for a binary search and one iteration per returned piece. Each initial root scans at most its stored partition; there are only two roots per key, so these scans are covered by the P_κ term. The four primitive evaluations and endpoint reversal in a fused contrast are included among the requests. Recursive best-single contrasts also enter the request count rather than being charged as one constant-cost operation.

The raw-support routine runs once per distinct complete support key. Importantly, its current implementation forms that key on every request, including a cache hit. Its s_r entry inspections therefore remain in (115); they are not replaced by an unproved constant cost. A previously bound immutable case avoids rebuilding its derivative key, whereas each fresh binding or explicit subset forms all s_b entries. Every request performs only a fixed number of additional table operations, apart from recursive requests already counted in Q. Summing these costs proves the displayed bound. Since key formations and minimum requests are included among the calls, replacing their lengths by S gives the simpler bound.

Each built piece stores a fixed number of rational entries and a prefix. The number of memoized arguments, roots, bindings and support keys is at most a constant multiple of the request count. A mathematical key or a retained case contains at most

$O(S)$ scalar or reference entries. These facts prove the storage bound. Clearing the caches on context exit prevents these retained cases or evaluations from becoming an uncharged initial state for the next service.

The upper bound counts the stated calculations. An uncached partial interval can contain fewer pieces than the full domain, or a service may never request the same geometry twice. Its work therefore has no matching lower bound merely because the cached algorithm has the displayed upper bound. Growing rational bit lengths, table operations, extra retained objects and cache setup or cleanup can also affect actual cost. The paired experiment measures all those costs while requiring identical original economic work and policies. It supplies any empirical speed comparison; the algebra alone supplies none. *Q.E.D.*

31. FINITE CUT CERTIFICATES AND ACTION REGRET

31.1. *Exact constraints and their rational tangents*

Use the notation $K = L - \mu$, $h = q - \mu x^2/2$, $f_i = h(x_i)$, and $g_i = h'(x_i)$ from Theorem S.39. Every item of information claimed for the compatibility class must be encoded before computing a cut. Additional original-model bounds are kept separately as described below. In particular, an uncertain supporting slope has one variable p_j shared by every node. A different slope chosen for each node would describe a different information class.

Suppose first $K > 0$. On an adjacent interval, let $d = x_{i+1} - x_i$, $s = g_{i+1} - g_i$, and $w = f_{i+1} - f_i - g_i d$. In addition to the affine restrictions $-s \le 0$ and $s - Kd \le 0$, the two nonlinear restrictions are

$$c_3 = s^2 - 2Kw \le 0, \qquad c_4 = s^2 - 2K(sd - w) \le 0.$$

In the coordinate order $(f_i, f_{i+1}, g_i, g_{i+1})$, their gradients are

$$\nabla c_3 = (2K, -2K, -2s + 2Kd, 2s),$$
$$\nabla c_4 = (-2K, 2K, -2s, 2s - 2Kd).$$

The Hessian of each is twice the outer product of the linear coefficient vector of s with itself. The remaining terms are affine. Thus these are convex quadratics and their displayed tangents are global affine minorants.

For a support at a_j with level W_j, its node restriction is

$$c_{ij} = \ell_j(x_i; p_j) + \Phi_K(v; -x_i, A - x_i) - f_i \le 0,$$
$$v = p_j - \mu a_j - g_i, \qquad t = \operatorname{proj}_{[-x_i, A-x_i]}(v/K).$$

Its only nonzero gradient coordinates are

$$\partial_{f_i} c_{ij} = -1, \qquad \partial_{g_i} c_{ij} = -t, \qquad \partial_{p_j} c_{ij} = x_i - a_j + t.$$

The formula follows by differentiating $\Phi_K(v; \ell, r) = vt - Kt^2/2$. The maximizer is unique when $K > 0$, including at the clipping thresholds, and the derivative with respect to v is t. Every evaluation is rational for rational inputs. The function is convex because Φ_K is a pointwise maximum of affine functions of v; subtracting f_i and adding ℓ_j preserves convexity.

When $K = 0$, impose $s = w = 0$ as affine equalities. The support function is $\Phi_0(v; \ell, r) = \max\{v\ell, vr\}$. Choose either maximizing endpoint $t \in \{\ell, r\}$, including either one in a tie. The same three displayed coordinates are a valid rational subgradient. This handles the constant-curvature case without division by K or an unstated limiting argument.

Observed value and derivative intervals, slope intervals, the cap $|g_i + \mu x_i| \le M$, and the integrated value boxes are affine restrictions. They can be retained as exact rows from the outset. No fitted derivative, altered preference, or support from another state may enter this set without its own declared transport theorem and complete charge.

31.2. *Proof of Theorem 8*

For every convex constraint, the subgradient inequality gives

$$c_{i_r}(z) \ge c_{i_r}(v_r) + s_r^\top (z - v_r) = A_r z - b_r.$$

If $z \in P$, its right side is nonpositive. Therefore, for nonnegative y, $H(z) + y^\top (Az - b) \le H(z)$ on P. The expression on the left is affine in z, and its exact minimum on the finite product box is

$$h_0 - y^\top b + \sum_k \min\{(h + A^\top y)_k \underline{z}_k, (h + A^\top y)_k \bar{z}_k\} = B.$$

Thus $B \le \inf_{z \in P} H(z)$. The exact interpolation theorem identifies the latter infimum as $-U_{\text{full}}(a, b)$. This proves the first assertion without any optimization assumption on the cut points or weights.

If $z^{\text{f}} \in P$, then $B \le \min_P H \le H(z^{\text{f}})$. Reversing signs gives (29). In particular, the only accuracy assertion licensed by the two witnesses is the actual checked gap. A favorable floating-point objective, a small printed stationarity residual, or an approximate primal feasibility flag does not replace either witness.

An upper bound remains valid after taking its minimum with other valid upper bounds on the same number. The common node set represents the same function class for each requested contrast; hence the same valid cuts may be reused with a different linear H. Their generation, storage, rational checks, and all numerical optimization calls remain part of the service's work. If common acquisition is supplied to different services, its allocation must be fixed and charged equally. This proves all remaining assertions. □

The assertion about the exact compatibility extremum applies to the raw bound $-B$ and to intersections with bounds valid throughout that same compatibility class. A separately licensed original-model bound may describe a smaller class. In particular, the saved number $B_{\text{old}} \le \min q$ implies $q(a) \ge B_{\text{old}}$ for the actual original objective. The implemented cheap comparator uses this fact even when the global restriction $q(x) \ge B_{\text{old}}$ has not been encoded in P. Its intersection with $-B$ certifies the actual original action contrast, but need not dominate the supremum over the larger P. Its economic validity does not by itself prove the sharpness or numerical accuracy of the complete-support program. The full-class regret sandwich uses full-class upper contrasts; an original-policy regret upper bound may instead use any bounds valid for that same actual q.

Numerical dual proposals. For a minimization linear program with rows $Az \le b$, a library may report nonpositive objective derivatives with respect to the right-hand sides. Under that convention the proposed nonnegative weights are the negatives of those reported values. This is an implementation convention to be verified, not a premise of the theorem. An independent checker reconstructs A, b from the original constraint identifiers and rational cut points, rejects negative weights, and recomputes B. Any rationalized nonnegative weights are valid, whether or not they are close to a numerical optimum. Exact box minimization pays for all stationarity error introduced by that rationalization. Retaining the largest certified B prevents a poorer later proposal from weakening the certificate. Numerical linear-program infeasibility is never treated as proof that the original economic record is impossible.

31.3. *The limiting calculation and its precise assumptions*

PROPOSITION S.52—Convergence of exact nested outer relaxations: *Let $\mathcal{B}$ be a compact convex box, $P \neq \varnothing$, and let each convex c_i be finite and continuous on a neighborhood of $\mathcal{B}$, with bounded selected subgradients there. Start with any finite valid cut set. At iteration n, minimize the affine H exactly on the current outer polytope $P_n \supseteq P$, obtaining z_n. Add a tangent cut for every c_i at z_n, without discarding earlier cuts. Then*

$$\min_{P_n} H \uparrow \min_{P} H,$$

and every cluster point of z_n belongs to P and attains its minimum. The conclusion also holds when cuts are taken at points v_n with $\|v_n - z_n\| \to 0$, under the same bounded-subgradient condition.

PROOF: Each tangent is valid on P, so the nonempty compact outer polytopes are nested. Their minima increase and are bounded above by $\min_P H$; write their limit as ℓ. Take any convergent subsequence $z_{n_k} \to z_*$. For every fixed iteration r, all sufficiently late points belong to the closed P_r, so $z_* \in P_r$. In particular, the cut added at any v_{n_k} is satisfied at z_*. Hence

$$c_i(v_{n_k}) + s_{i,n_k}^{\top}(z_* - v_{n_k}) \le 0.$$

Continuity, $v_{n_k} \to z_*$, and bounded subgradients imply $c_i(z_*) \le 0$ for every i. Thus $z_* \in P$. Continuity of H gives $H(z_*) = \ell$, which, combined with $\ell \le \min_P H$, proves equality and optimality. The same argument applies to every convergent subsequence. *Q.E.D.*

No interior point or constraint qualification is required for this limiting statement. It gives no finite iteration count and no convergence certificate for an arbitrary capped numerical solve. Deleting cuts, adding cuts for only a selected subset of constraints, or accepting a numerical subproblem solution without its required control changes its hypotheses. The implemented economic guarantee must therefore remain the finite rational bound of Theorem 8. When no exact primal witness is available, the distance to the sharp information bound remains uncertified.

31.4. *Proof of Theorem 9*

For any fixed original objective in the information class,

$$q(a) - \min_x q(x) = \max_x \{q(a) - q(x)\}.$$

Taking a supremum over the class and restricting the inner maximization to the finite grid gives

$$\Gamma(a) \ge \max_j \sup_q \{q(a) - q(x_j)\} = \max_j u_j.$$

The maximum and supremum commute because the grid is finite. There is no interchange of an infimum and a supremum in this argument.

For the upper bound, let x_q^* minimize q. If it is a boundary point, it is already a grid node. Otherwise $q'(x_q^*) = 0$. There is a grid node x_j at distance at most $h/2$, and smoothness gives

$$q(x_j) - q(x_q^*) \le \frac{L}{2}|x_j - x_q^*|^2 \le \frac{Lh^2}{8}.$$

It follows that $q(a) - q(x_q^*) \le \max_j u_j + Lh^2/8$, uniformly in q. Taking the supremum proves (31).

To verify sharpness, take the domain $[0, h]$, the grid $\{0, h\}$, $a = 0$, and the singleton class $q(x) = L(x - h/2)^2/2$. All parameters can be positive rational numbers and

$0 < \mu \le L$. Exact endpoint values and endpoint derivatives with total derivative increment Lh identify this quadratic within the L-smooth class: for every intermediate x, the upper Lipschitz bound from zero and the lower bound from h coincide. Then $u_0 = u_1 = 0$ and $\Gamma(a) = Lh^2/8$.

For the cell calculation, the function $x \mapsto q(x) - Lx^2/2$ is concave on $[0, A]$. For $x = (1-t)x_j + tx_{j+1}$ this gives the lower secant bound

$$q(x) \ge (1-t)q(x_j) + tq(x_{j+1}) - \frac{Ld_j^2}{2}t(1-t).$$

After subtracting from $q(a)$ and applying the two verified contrast bounds, the right side of (32) dominates $q(a) - q(x)$ on the entire cell. Maximizing over all cells and then all compatible q proves $\Gamma(a) \le G(a)$. Because a is a grid node, one adjacent cell endpoint evaluates to zero; hence $G(a) \ge 0$.

For a cell, write $k_j = Ld_j^2/2 > 0$. Its objective is

$$\widehat{u}_j + (\widehat{u}_{j+1} - \widehat{u}_j + k_j)t - k_j t^2.$$

This concave quadratic is maximized at the clipped rational point in (33). Finally its value is at most $\max\{\widehat{u}_j, \widehat{u}_{j+1}\} + Ld_j^2/8$. Under the uniform accuracy premise it follows that

$$G(a) \le \max_j u_j + \epsilon + Lh^2/8 \le \Gamma(a) + \epsilon + Lh^2/8.$$

This proves the accuracy statement. None of these arguments uses a new value observation at a computational grid node. The exact interpolation theorem applies to any enlarged finite node set, so introducing the corresponding unknown values and derivatives preserves the same compatible original functions. □

31.5. *Robust action selection, original-policy gain and paid stopping*

For every compatible q, the function $a \mapsto q(a) - \min q$ is M-Lipschitz. Taking suprema proves the same Lipschitz bound for Γ. Also $q(a) - \min q - \mu a^2/2$ is convex; its supremum is convex. Thus Γ is μ-strongly convex. Compactness supplies a unique minimizing action a^{r}, which is the robust action for this information class,

not necessarily the actual optimum a_q^*. Choose a grid node a_j within $h/2$ of a^{r}. For a minimizer $\widehat{a}$ of the computed G over the grid,

$$\Gamma(\widehat{a}) \le G(\widehat{a}) \le G(a_j) \le \Gamma(a_j) + \epsilon + Lh^2/8 \le \Gamma(a^{\mathrm{r}}) + Mh/2 + \epsilon + Lh^2/8.$$

This proves (35). The accuracy premise requires a verified gap for every used pairwise calculation; a solver's nominal tolerance is insufficient. Without that premise, the selected action retains its finite regret certificate G, but the displayed near-minimax guarantee is not asserted.

For the actual unchanged future controller, $J(a) = q(a) + \eta(a)$ with $0 \le \eta(a) \le \delta$,

$$0 \le J(a) - \min q \le \Gamma(a) + \delta \le G(a) + \delta.$$

If there is a separately certified original-value interval $[W_a, U_a]$, the implemented policy interval is $[W_a, U_a + \delta]$. The separate width gate is therefore $U_a - W_a + \delta \le \omega$; it cannot be deduced from the level-invariant regret calculation.

For two distinct realized actions, $J(b) - J(a) \le U_{\mathrm{full}}(a, b) + \delta$. Replacing U_{full} by any of its verified upper bounds and requiring a strictly negative result proves actual gain for every compatible original objective and allowed future error. At identical actions, the unchanged implemented policy has exactly zero contrast; it must not be assigned a positive gain by charging two different future-error allowances. A finite sequence of strictly accepted replacements telescopes its actual gains.

Finally, every feasible alternative original policy has cost at least the original optimum. Starting from the qualified incumbent, its possible economic saving is at most $\min\{K_{\mathrm{old}}, G(a) + \delta\}$. Under the stated compulsory cost and no-reuse premise, its resource bill is at least λc_*. The stopping inequality therefore makes every allowed continuation weakly more expensive in total objective. Equality licenses weak dominance only. Reuse across future decisions, an additional uncharged oracle, or another preference changes that comparison and is excluded from the assertion.

32. INDEPENDENT FINITE-DESIGN ACQUISITION VERIFICATION

The independent acquisition audit operates on the four immutable R121 label files and all 56 stored actors. It changes neither the original Bellman objective nor

a deployed coefficient. Forty-eight numerical problems were originally fitted; the other eight representations are the exactly matched ordinary-hinge encodings of paid ReLU fits. The audit reconstructs the pre-label feature specification, every value row and required one-sided original-action derivative row, the coefficient box, rounding rule and stored serialization. It configures BVLS rather than TRF within the same SciPy library. Four actual VALUE solves execute its distinct active-set iterations; the other 20 return the feasible unconstrained solution through the shared initial numerical path. The four VALUE hinge readouts copy the paid ReLU solutions. Its independently formulated marginal program still uses the HiGHS library. Thus independence here concerns the implemented feature, problem-assembly, candidate-generation and exact-checking paths; it does not assert independent authorship or an independently written linear-programming solver.

The following familiar convex and weak-duality calculations make the checking path independent of floating solver status. They are verification lemmas, not claims that convex duality is a new neural result. Let $B = \prod_{j=1}^{p} [\underline{b}_j, \overline{b}_j]$ be a finite rational box, $a_i, d_i \in \mathbb{Q}^p$, and $y_i, \ell_i, u_i \in \mathbb{Q}$, with $\ell_i \le u_i$. The exact finite-design objectives are

$$F(\theta) = \sum_i (a_i^\top \theta - y_i)^2, \qquad R(\theta) = \max_i \{d_i^\top \theta - \ell_i, u_i - d_i^\top \theta\}. \tag{116}$$

In particular, $R(\theta) \ge 0$. These objectives are conditional on the original acquired interval labels. They are not whole-domain approximation errors or original-policy regret.

PROPOSITION S.53—Exact finite-box checking bounds: *For any rational feasible readout $\widehat{\theta} \in B$, set $g = \nabla F(\widehat{\theta})$ and*

$$L_F = \max\left\{0, F(\widehat{\theta}) - g^\top \widehat{\theta} + \sum_{j=1}^{p} \min\{g_j \underline{b}_j, g_j \overline{b}_j\}\right\}.$$

Then $L_F \le \min_{\theta \in B} F(\theta) \le F(\widehat{\theta})$. For arbitrary rational weights $w_i^+, w_i^- \ge 0$ satisfying $\sum_i (w_i^+ + w_i^-) = 1$, define

$$c = \sum_i (w_i^+ - w_i^-) d_i, \quad L_R = \max\left\{0, -\sum_i w_i^+ \ell_i + \sum_i w_i^- u_i + \sum_{j=1}^{p} \min\{c_j \underline{b}_j, c_j \overline{b}_j\}\right\}.$$

Then $L_R \le \min_{\theta\in B} R(\theta) \le R(\widehat{\theta})$. *The conclusions require no numerical stationarity, floating solver success, strict convexity or uniqueness of the minimizing readout.*

PROOF: Convexity gives $F(\theta) \ge F(\widehat{\theta}) + g^{\mathsf{T}}(\theta - \widehat{\theta})$. Minimizing this affine function independently over each coordinate of B gives the stated affine lower bound. The squared loss is nonnegative, so replacing the lower bound by its maximum with zero preserves validity. Evaluating the objective at the feasible readout gives the upper bound.

For the marginal problem, the maximum of the $2m$ displayed affine expressions is at least any convex combination of them. Therefore, for every $\theta \in B$,

$$R(\theta) \ge \sum_i w_i^+ (d_i^{\mathsf{T}}\theta - \ell_i) + \sum_i w_i^- (u_i - d_i^{\mathsf{T}}\theta) = c^{\mathsf{T}}\theta - \sum_i w_i^+ \ell_i + \sum_i w_i^- u_i.$$

Minimizing the affine expression over B yields its coordinatewise box minimum. Since at least one of $d_i^{\mathsf{T}}\theta - \ell_i$ and $u_i - d_i^{\mathsf{T}}\theta$ is nonnegative for every i, R is nonnegative. The maximum with zero is thus valid. Feasibility of $\widehat{\theta}$ gives the upper bound. Every operation uses finite rational arithmetic; approximate multipliers enter only as candidates for the subsequently checked nonnegative unit-mass weights. *Q.E.D.*

The audit converts a recorded binary64 multiplier to its exact binary rational value, takes the nonnegative part of its negative, and divides by the resulting total mass. If the mass is zero, it uses the predeclared uniform weights. Both constructions satisfy Proposition S.53 exactly. Weak witnesses remain weak and are reported without selecting a more favorable normalization after inspection. The exact difference between the two rounded readout objectives is reported separately from the two optimality-gap upper bounds.

PROPOSITION S.54—Rounding and what its bound controls: *Suppose every box endpoint belongs to the lattice* $h\mathbb{Z}$, *and let* P_B *denote coordinatewise projection onto* B. *Round a finite raw readout to the nearest lattice point, with ties to even, and project the result onto* B *to obtain* $\widehat{\theta}$. *Then* $\|\widehat{\theta} - P_B\theta^{\mathrm{raw}}\|_\infty \le h/2$. *For any two feasible readouts* $\theta, \theta + \delta \in B$,

$$|R(\theta + \delta) - R(\theta)| \le \max_i \|d_i\|_1 \, \|\delta\|_\infty,$$

and, writing A for the matrix with rows a_i^T,

$$|F(\theta+\delta)-F(\theta)| \le 2\|A\theta-y\|_2\|A\delta\|_2 + \|A\delta\|_2^2.$$

These are finite-design bounds. They do not supply a continuous-state or original-policy certificate.

PROOF: For a coordinate inside its box interval, nearest-lattice rounding has error at most $h/2$, and clipping to lattice-valued endpoints cannot increase the distance to that coordinate. For a coordinate outside the interval, monotonicity of nearest-lattice rounding and lattice-valued endpoints imply that subsequent clipping gives that endpoint. Its distance from the projected raw coordinate is zero. This proves the first assertion, including the case where a numerical solver slightly violates a box bound.

Each affine expression defining R changes by either $d_i^\mathsf{T}\delta$ or its negative. The difference between two finite maxima is bounded by the largest absolute change in their entries, which is at most $\max_i \|d_i\|_1\|\delta\|_\infty$. Finally,

$$F(\theta+\delta)-F(\theta) = 2(A\theta-y)^\mathsf{T}A\delta + \|A\delta\|_2^2.$$

Taking absolute values and applying Cauchy–Schwarz gives the second bound. Neither argument takes a supremum outside the declared design. *Q.E.D.*

The audit implementation uses $h = 2^{-30}$, records the actual exact rounded objectives, and checks their serialized numerator, denominator, binary64 and hexadecimal identities. A different coefficient vector can arise from a nonunique optimization problem; coefficient equality is therefore recorded, not imposed as a scientific success criterion. A successful finite-design audit does not independently reacquire the original labels, verify fitted error throughout the state–action cube, or create economic observations for the new readouts. These remaining limits stay attached to the response to R97 B8/M9.

32.1. *Policy identity and conditional adoption*

Separate value intervals need not exploit an exact identity between two implemented policies. The following observation gives a stronger adoption test when that identity is available. It applies to any two representations. The application below is a post-execution analysis of saved R125 records, with no new economic evaluation, timing run or experimental reclassification.

PROPOSITION S.55—Conditional cost dominance of an identical policy: *Fix a queried state, economic information, original transition law and finite-horizon recursive objective that is well-defined and finite for both policies. Suppose services n and c return the same realized first action and use the same implemented continuation policy on all future histories reachable from that state. The identity includes observation conventions, feasible actions, numerical rounding, and any policy randomization. For randomized policies, identity means the same conditional decision rules under a common adapted randomization and economic filtration. They then have identical original-policy objectives, $J_n = J_c$.*

Let B_n, B_c be their declared acquisition charges, $N > 0$ the common amortization volume, and $t_i \in [\underline{t}_i, \bar{t}_i]$ their complete per-decision service times. Suppose the recorded quantities establish

$$\varepsilon_N := B_n/N + \underline{t}_n - (B_c/N + \bar{t}_c) > 0. \tag{117}$$

For every pair of times in these ranges and every $\lambda > 0$,

$$[J_n + \lambda(B_n/N + t_n)] - [J_c + \lambda(B_c/N + t_c)] \geq \lambda\varepsilon_N > 0. \tag{118}$$

Thus service n cannot minimize the priced objective over any comparator set containing c at a strictly positive price. At $\lambda = 0$ it ties c; no optimality relative to a different policy is implied. The result is conditional on the stated cost ranges and accounting rule, and holds only for the specified queried state and its reachable continuation histories.

PROOF: Use the same future innovations and, where needed, the same coupled policy randomization for both services. Their initial states, economic information and realized first actions coincide. The common transition law therefore gives the same next state. The identical implemented continuation rule, including its

observations and rounding, then gives the same next action. Induction gives the same states, actions and realized costs at all future dates on the reachable histories. Their terminal costs coincide. Backward evaluation applies the same conditional risk or recursive-preference operator to identical continuation quantities at each date; it consequently gives the same value at each preceding history, including $J_n = J_c$ at the queried state. This argument does not replace the original recursive objective by an expectation or require a different information filtration.

For any admitted times,

$$B_n/N + t_n - (B_c/N + t_c) \geq B_n/N + \underline{t}_n - (B_c/N + \bar{t}_c) = \varepsilon_N.$$

Subtract the two priced objectives, cancel their equal original values, and multiply this inequality by $\lambda > 0$ to obtain (118). At zero price the entire computational term vanishes, so this pair ties. A policy outside the identical-policy pair can still have a lower original value. No independence of the two service times is needed: the inequality holds for every joint realization inside the stated ranges. *Q.E.D.*

What an interval-only outer region omits. If the earlier comparison retains only separate value boxes, its optimistic comparison with c uses

$$\underline{J}_n + \lambda(B_n/N + \underline{t}_n) \leq \bar{J}_c + \lambda(B_c/N + \bar{t}_c).$$

When (117) holds, this permits $\lambda \leq (\bar{J}_c - \underline{J}_n)/\varepsilon_N$. The right side may be positive even though the true difference $J_n - J_c$ is exactly zero. The separate boxes are a valid relaxation, but they do not encode the policy identity. Retaining them as the original reported outer region is consistent with deriving the stronger exclusion from the additional equality. Neither a narrower fitted residual nor a newly evaluated economic outcome is involved.

Saved first-state evidence and its limits. At the first declared state $(1/4, 1/4)$, all 116 learned or online services within each risk return the same action bits, `0x1.3fffffe000000p-3`. They use the same terminal controller with downward 40-bit state observations. The source-bound diagnostic verifies the state, risk, action and continuation identities. For each of the 128 neural-service/amortization cases,

it checks all 100 conventional services with that same policy and retains an exact rational witness of (117). The uniform-action DP control returns a different action and is not used in this identity comparison. Conditional on the recorded acquisition allocations and descriptive timing ranges, every strictly positive computation price is therefore excluded by a conventional identical-policy comparator. Zero price supplies a tie with that comparator, not a strict neural improvement or a conclusion about the different-policy grid control.

The recorded least-upper-cost blockers are online services in 64 cases and degree-two polynomial services in 64 cases. This is competition between complete services with their own charged acquisition routes; the blockers need not share features, training losses, information budgets or service modes with the neural service. It is not a matched-feature representation result or an architectural complexity separation. Equality of the original economic filtration and realized complete policy does not assert equality of the computations used to obtain that policy.

All 128 originally reported optimistic regions remain unchanged as nonempty interval-only outer sets, including their positive-price endpoints. All 128 originally reported sufficient regions remain empty. The new exclusion is a separately identified post-execution consequence of policy identity and saved cost evidence. The three observed timing ranges have no confidence-level or future-runtime interpretation; the conclusion does not extend to other states, workloads, hardware, future controllers or neural architectures. Its exact blocker witnesses and source hashes are retained in `ADOPTION_SUPPORT_DIAGNOSTIC125.json`.

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